

AI 500

Foundations of Artificial Intelligence

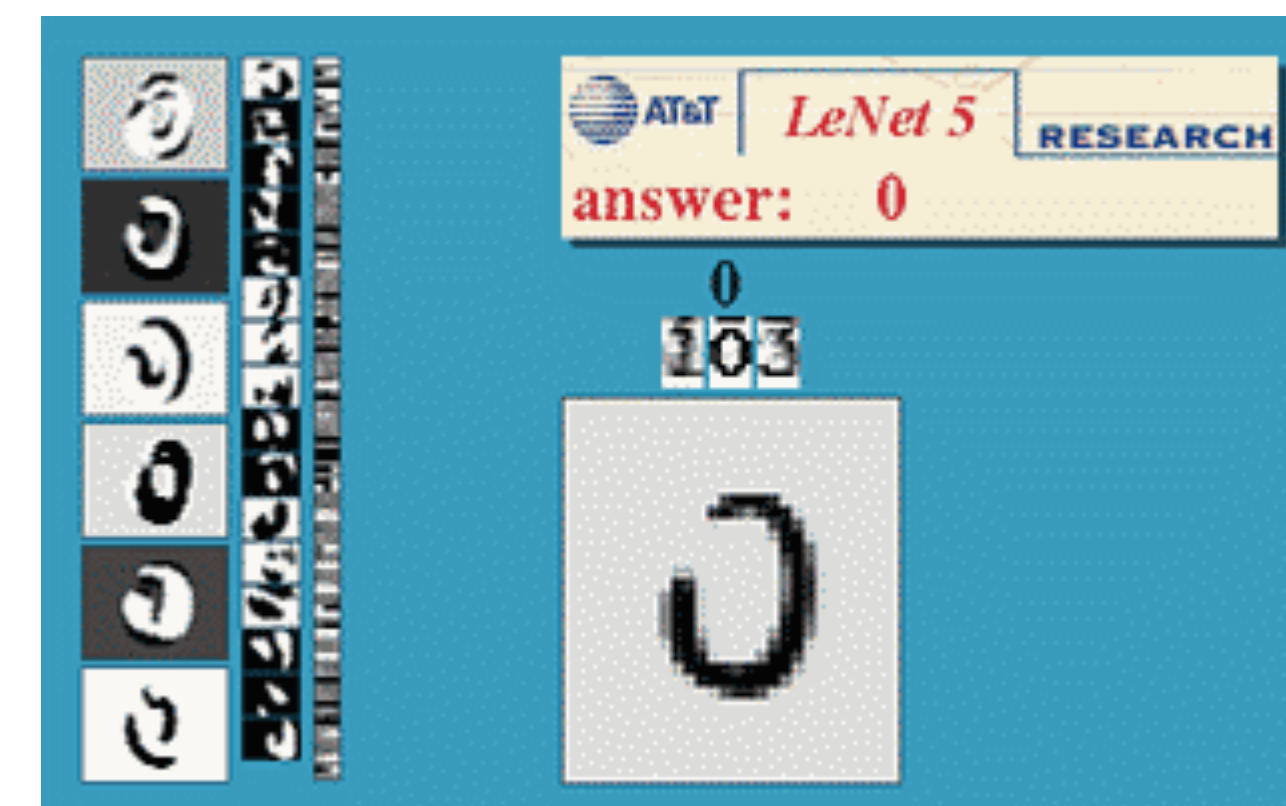
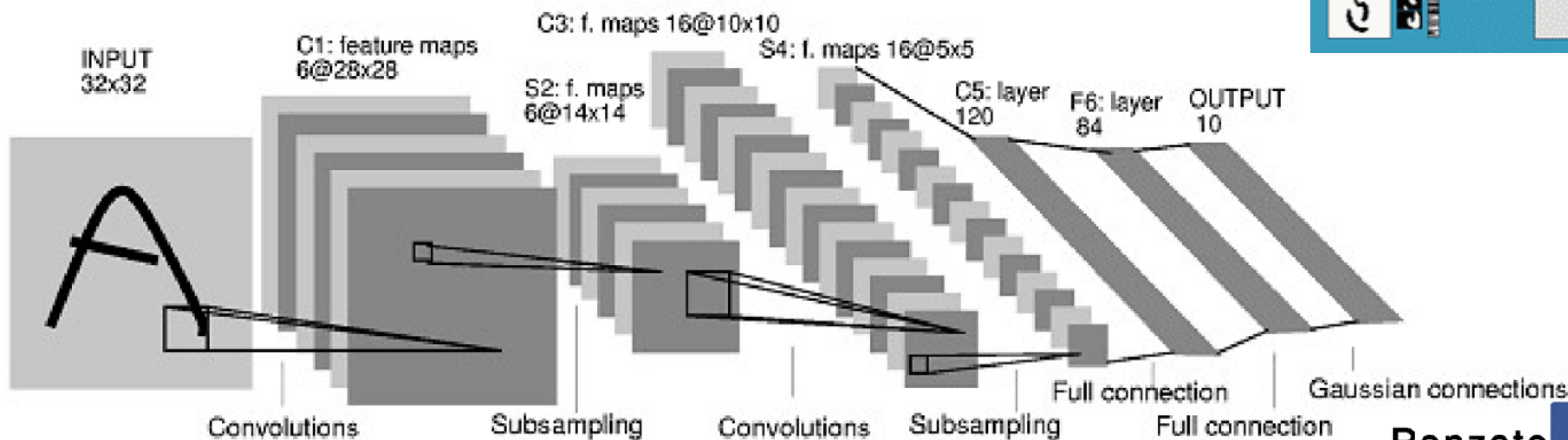
L13: Convolution Neural Network Part 2

By Murtaza Taj



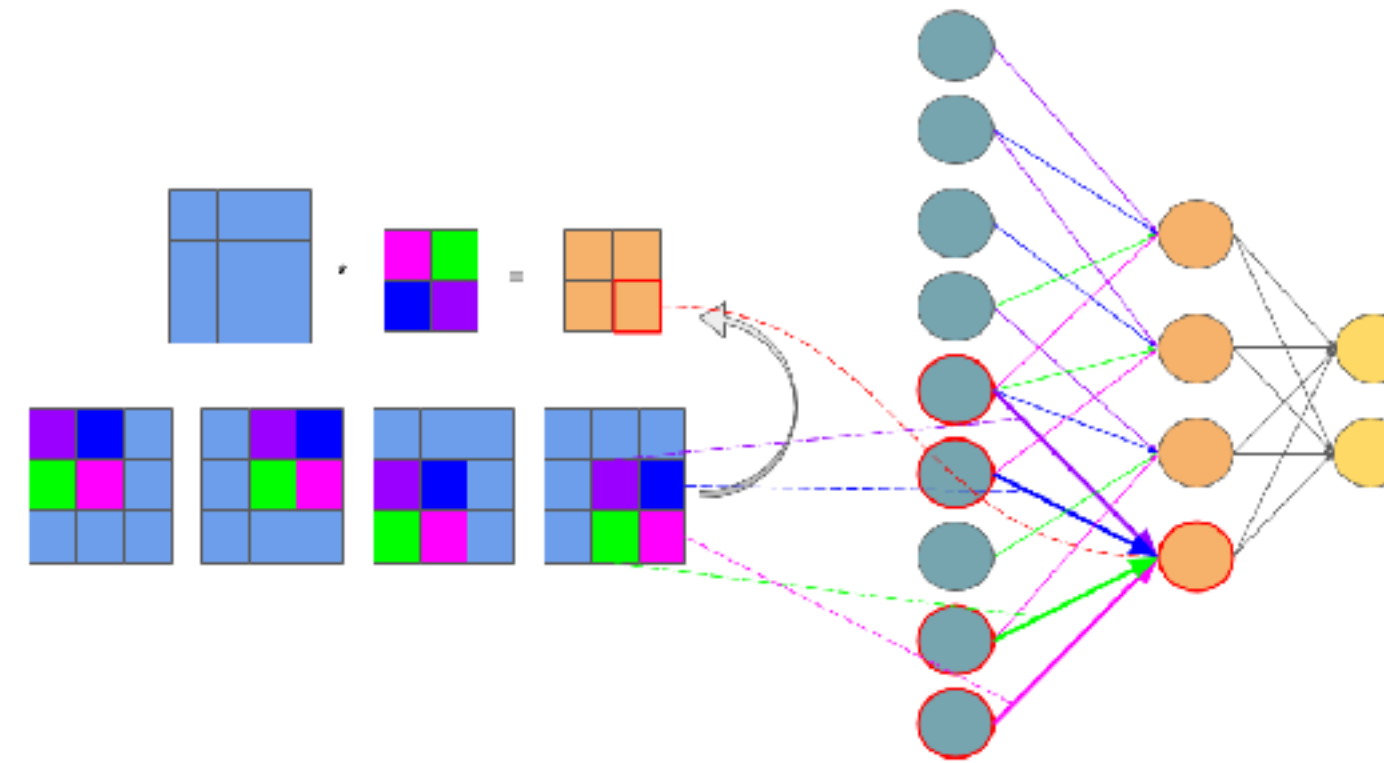
Convolutional Neural Network

- Le Cun 1998



Agenda

- Forward and Backward Propagation using Convolution operation



- Transforming Multilayer Perceptron to Convolutional Neural Network

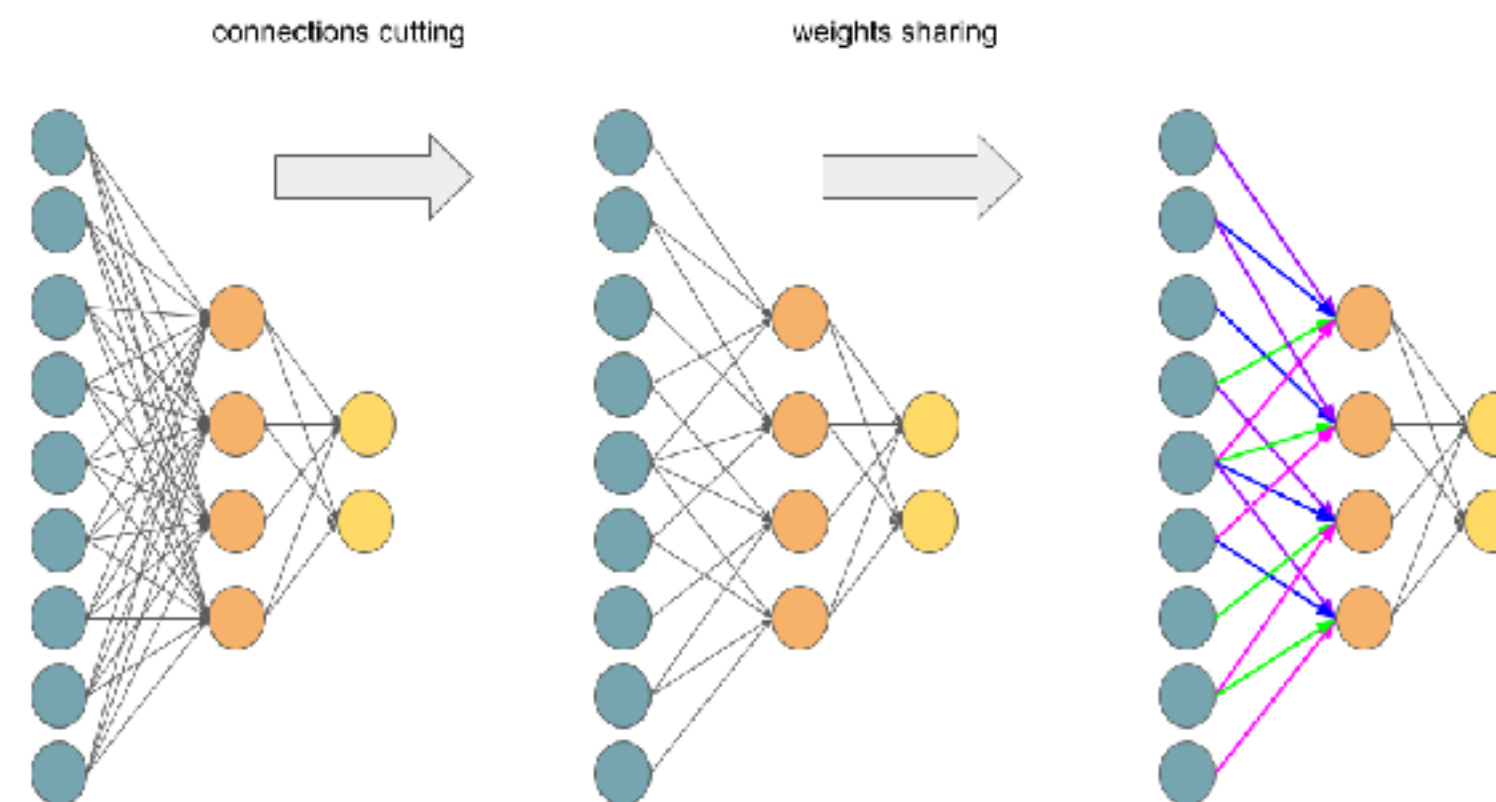


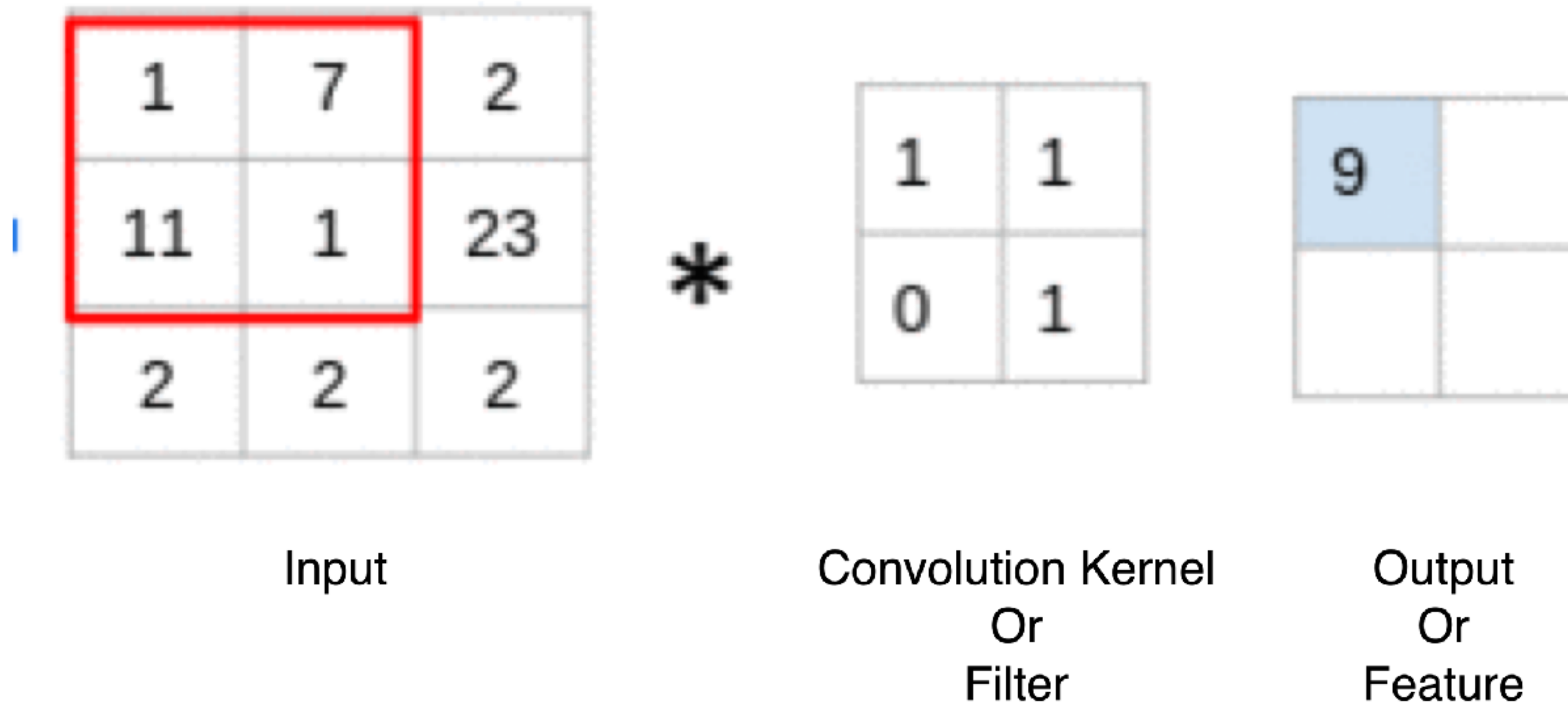
Image						Filter			Conv Output				
4	7	7	5	0	8	1	0	-1					
1	8	5	7	8	4	1	0	-1		-9	8	7	3
2	8	4	3	1	0	1	0	-1		-10	6	-4	10
0	7	4	7	8	3					-2	7	-7	-1
4	2	0	0	6	8	Ignoring flip				3	5	-11	-5
3	6	0	3	1	4								

Recap - Convolution - Smoothing Filter

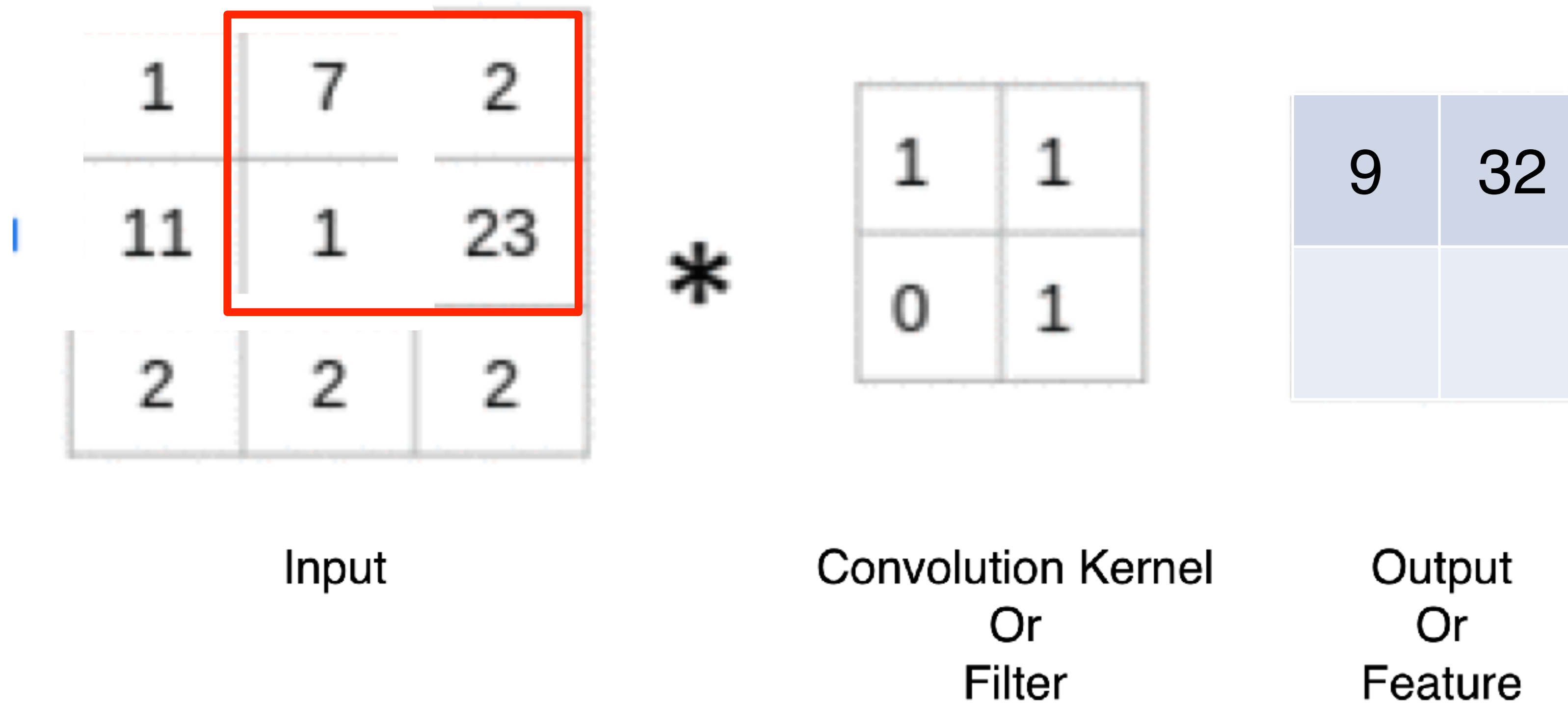
[illegible]

Transforming Multilayer Perceptron to Convolutional Neural Network

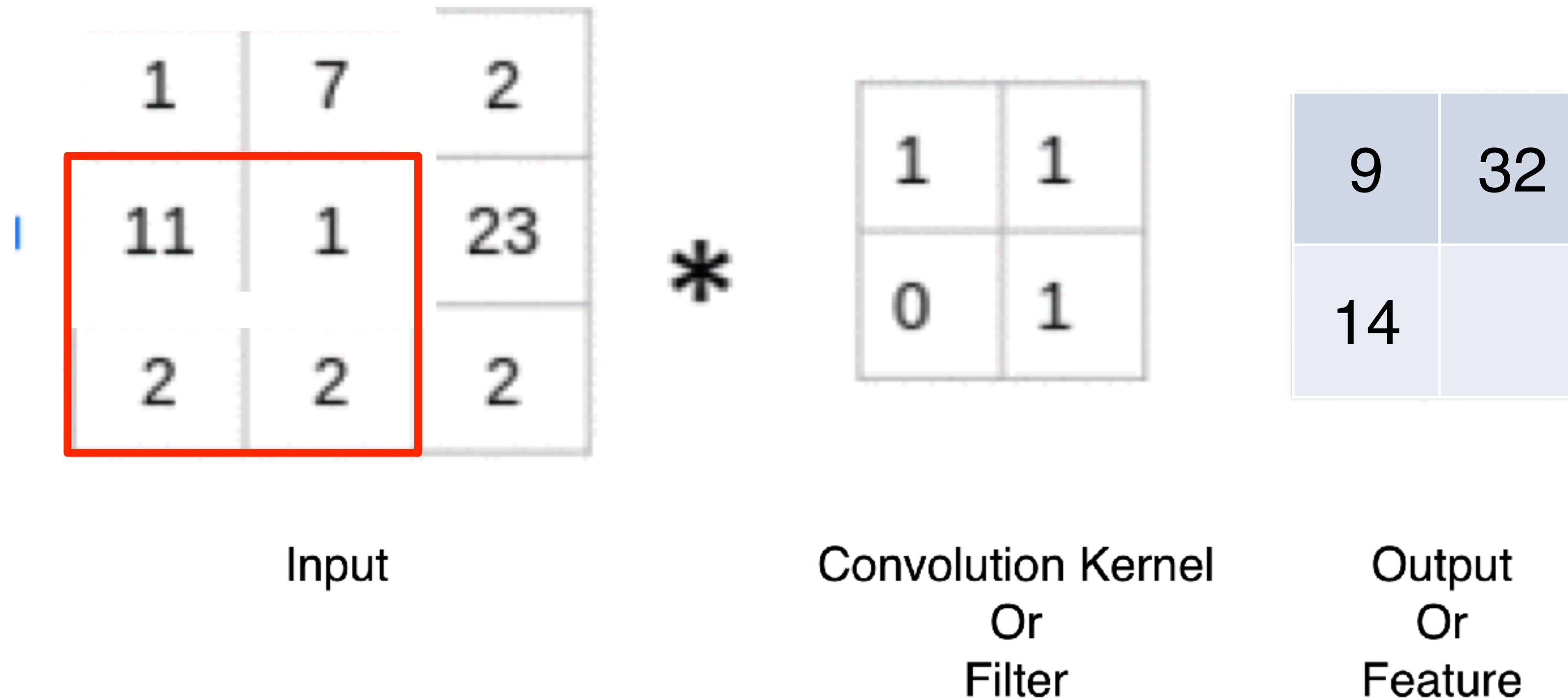
Recap: Convolution



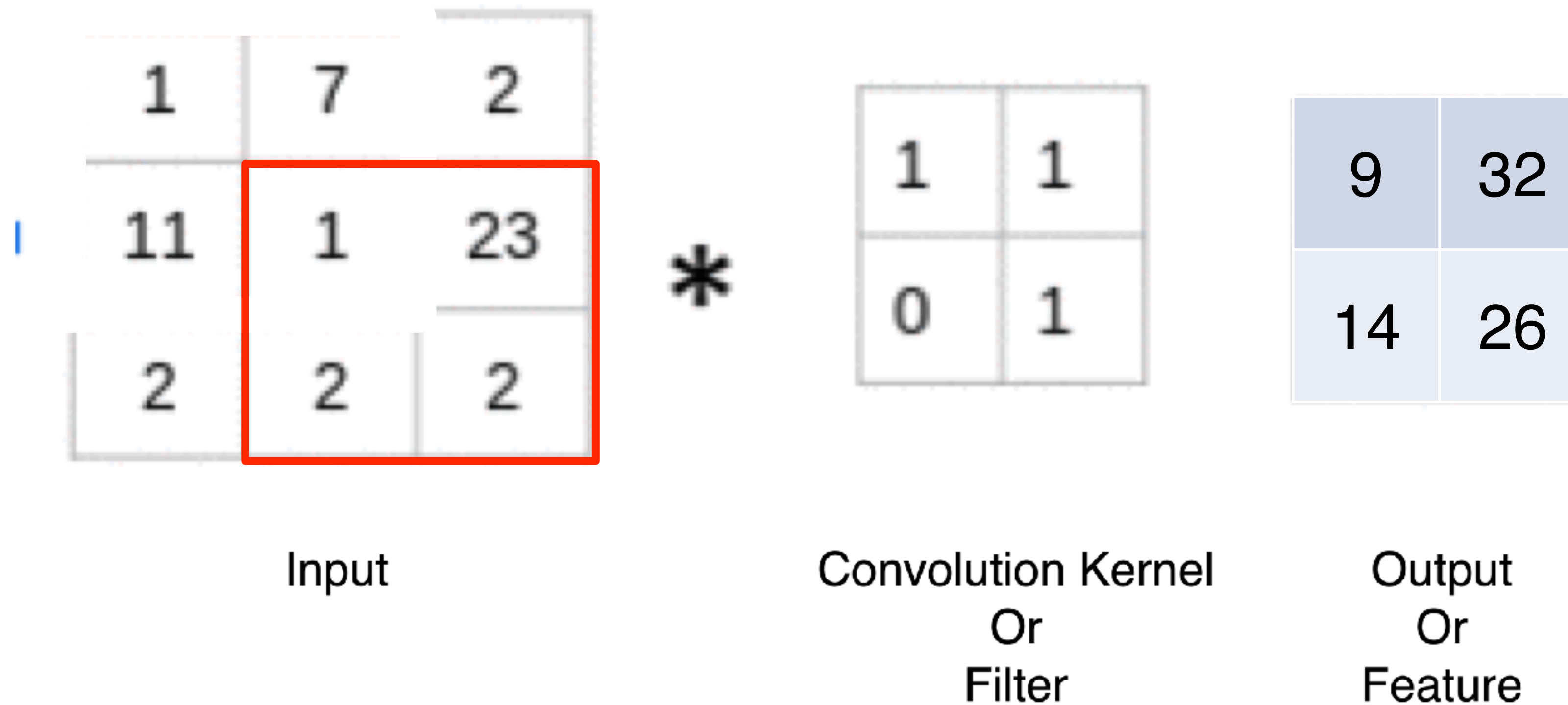
Recap: Convolution



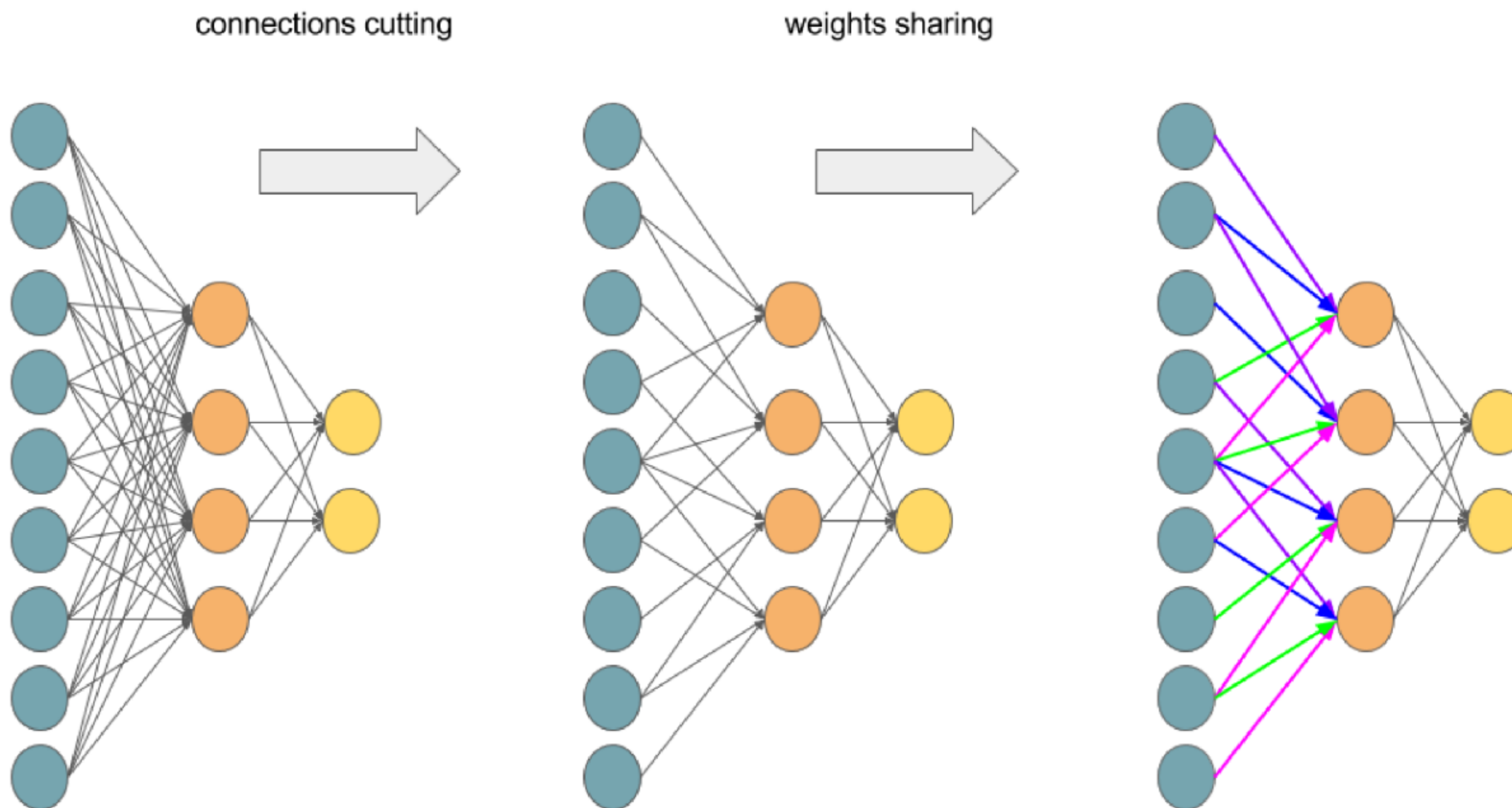
Recap: Convolution



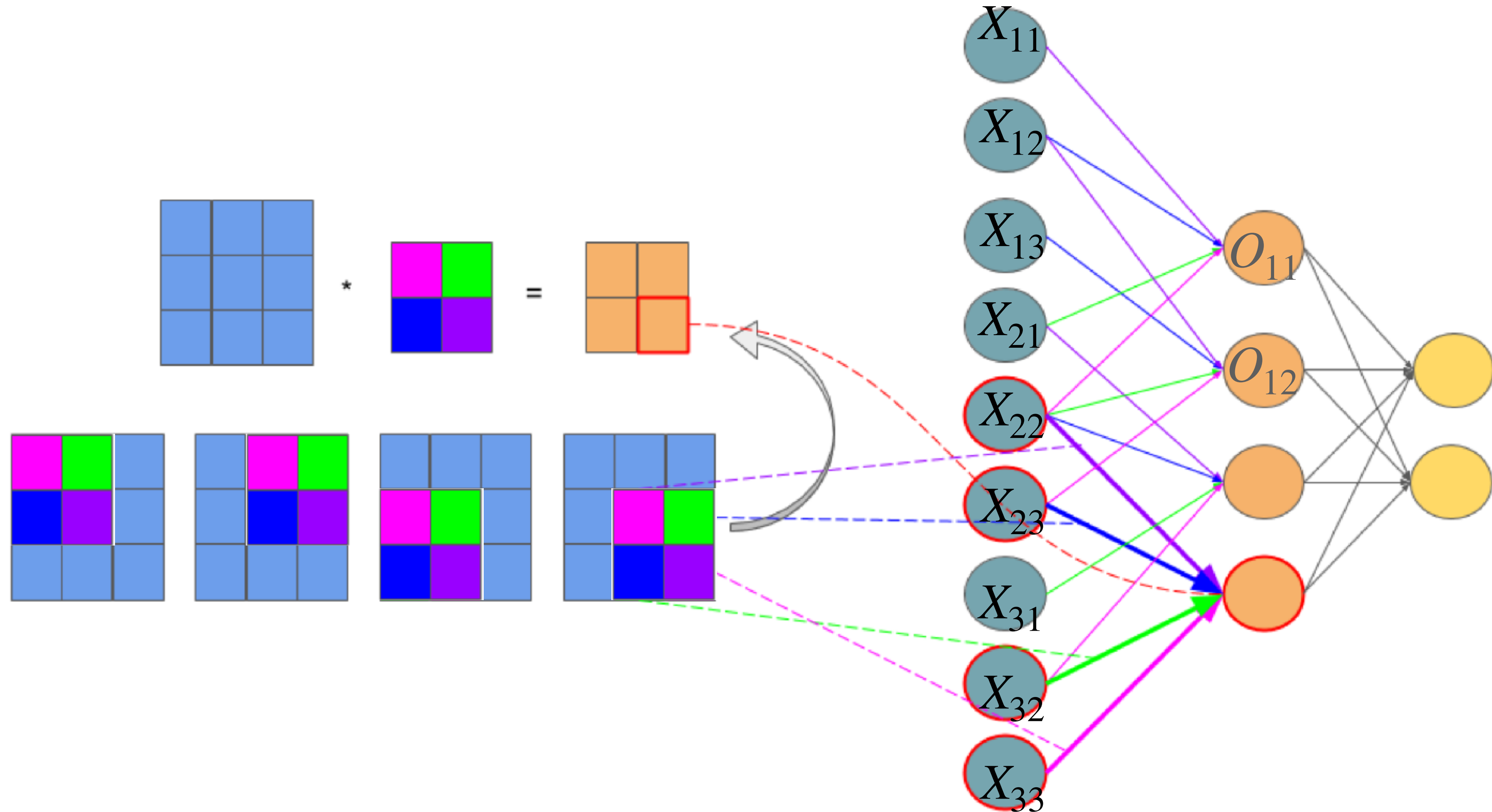
Recap: Convolution



Transforming Multilayer Perceptron to CNN



Feedforward in CNN is Identical with Convolution Operation



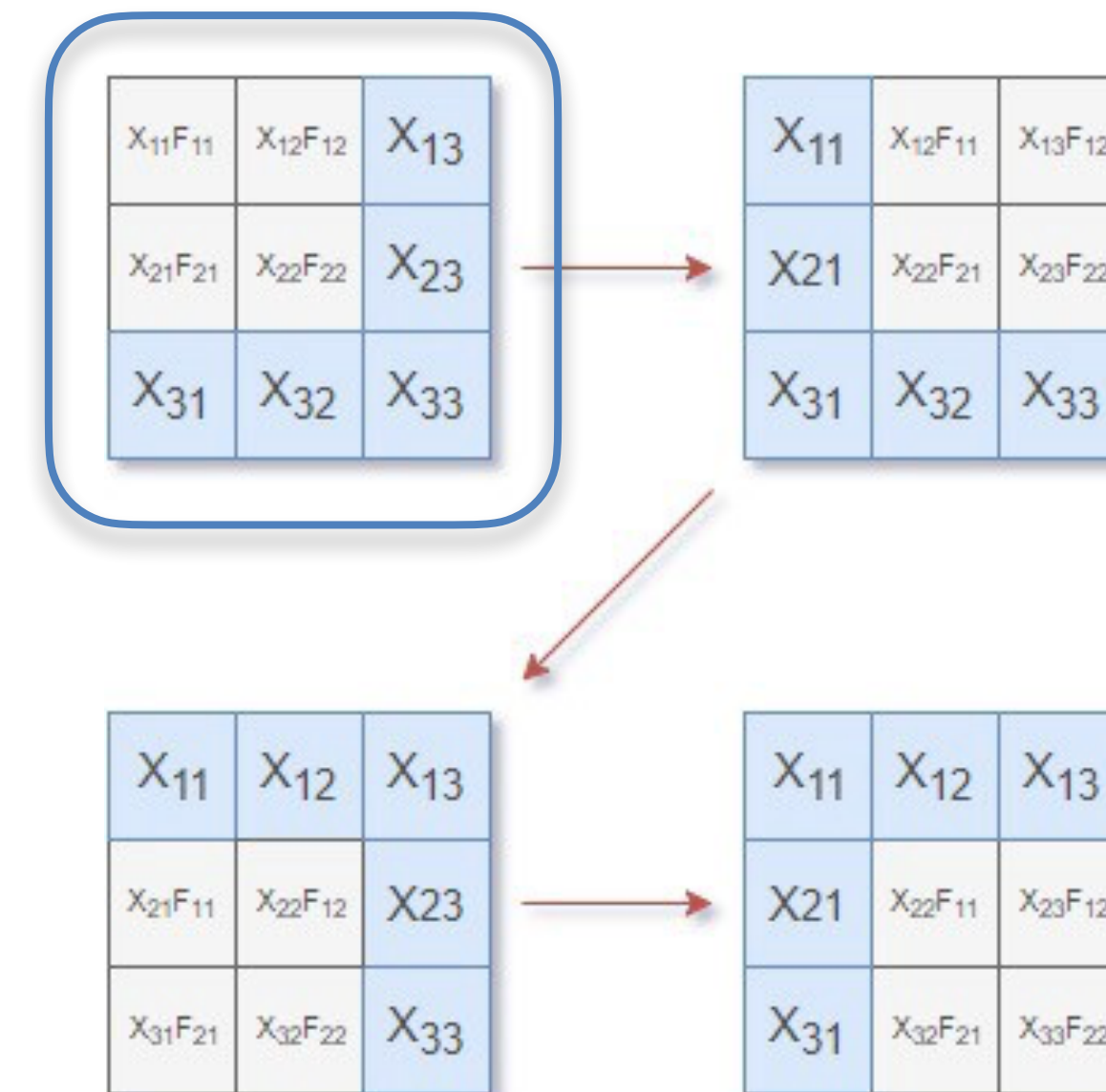
Forward and Backward Propagation using Convolution operation

Forward and Backward Propagation using Convolution operation

$$\begin{bmatrix} O_{11} & O_{12} \\ O_{21} & O_{22} \end{bmatrix} = \text{Convolution} \left(\begin{bmatrix} X_{11} & X_{12} & X_{13} \\ X_{21} & X_{22} & X_{23} \\ X_{31} & X_{32} & X_{33} \end{bmatrix}, \begin{bmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \end{bmatrix} \right)$$

$$O_{11} = F_{11}X_{11} + F_{12}X_{12} + F_{21}X_{21} + F_{22}X_{22}$$

- Forward Pass
 - Ignoring 180° rotation
 - Note: F instead of w



- It can be visualized as:

Forward and Backward Propagation using Convolution operation

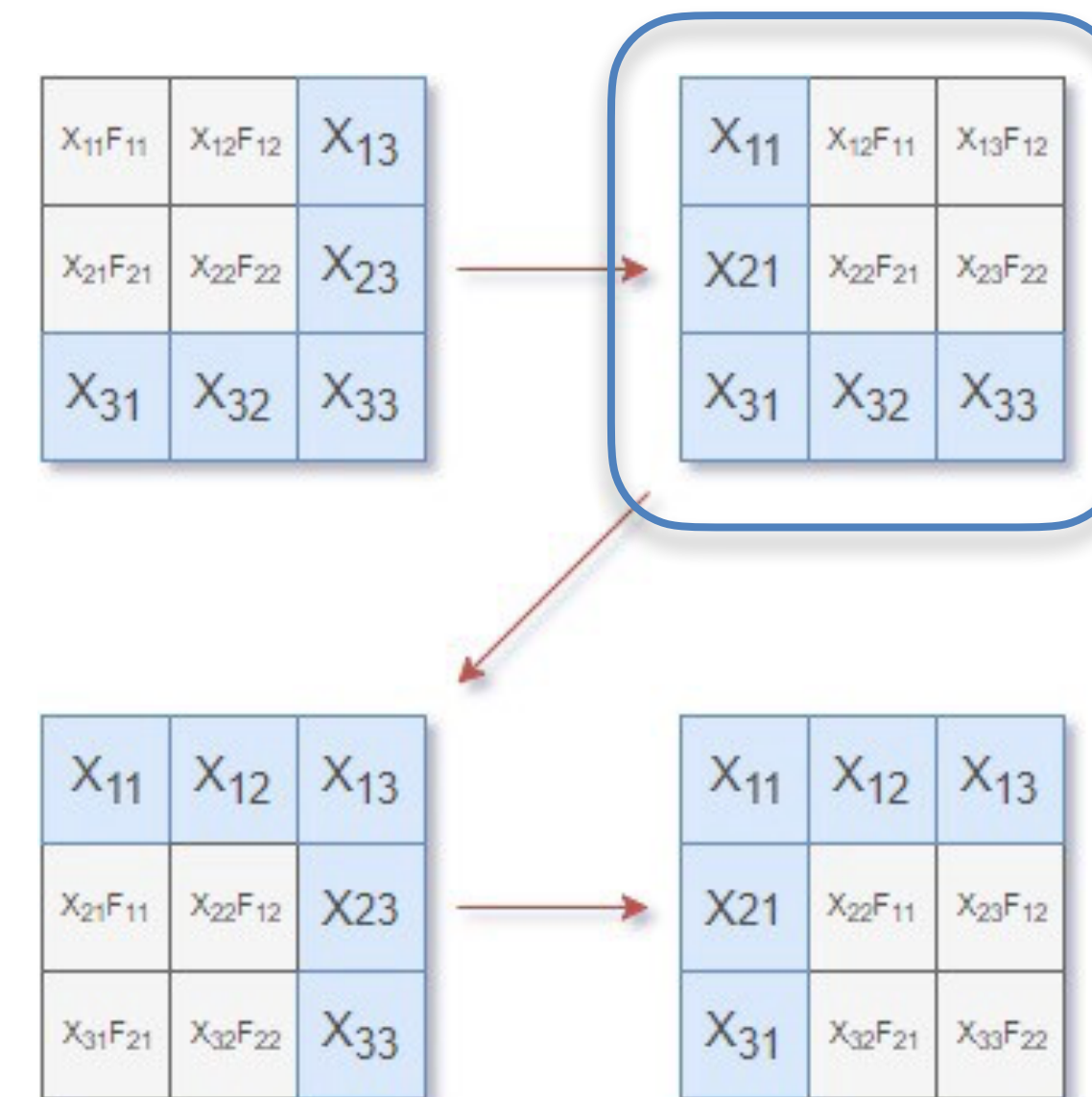
$$\begin{bmatrix} O_{11} & O_{12} \\ O_{21} & O_{22} \end{bmatrix} = \text{Convolution} \left(\begin{bmatrix} X_{11} & X_{12} & X_{13} \\ X_{21} & X_{22} & X_{23} \\ X_{31} & X_{32} & X_{33} \end{bmatrix}, \begin{bmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \end{bmatrix} \right)$$

- Forward Pass
 - Ignoring 180° rotation
 - Note: F instead of w

$$O_{11} = F_{11}X_{11} + F_{12}X_{12} + F_{21}X_{21} + F_{22}X_{22}$$

$$O_{12} = F_{11}X_{12} + F_{12}X_{13} + F_{21}X_{22} + F_{22}X_{23}$$

- It can be visualized as:



Forward and Backward Propagation using Convolution operation

$$\begin{bmatrix} O_{11} & O_{12} \\ O_{21} & O_{22} \end{bmatrix} = \text{Convolution} \left(\begin{bmatrix} X_{11} & X_{12} & X_{13} \\ X_{21} & X_{22} & X_{23} \\ X_{31} & X_{32} & X_{33} \end{bmatrix}, \begin{bmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \end{bmatrix} \right)$$

- Forward Pass

- Ignoring 180° rotation

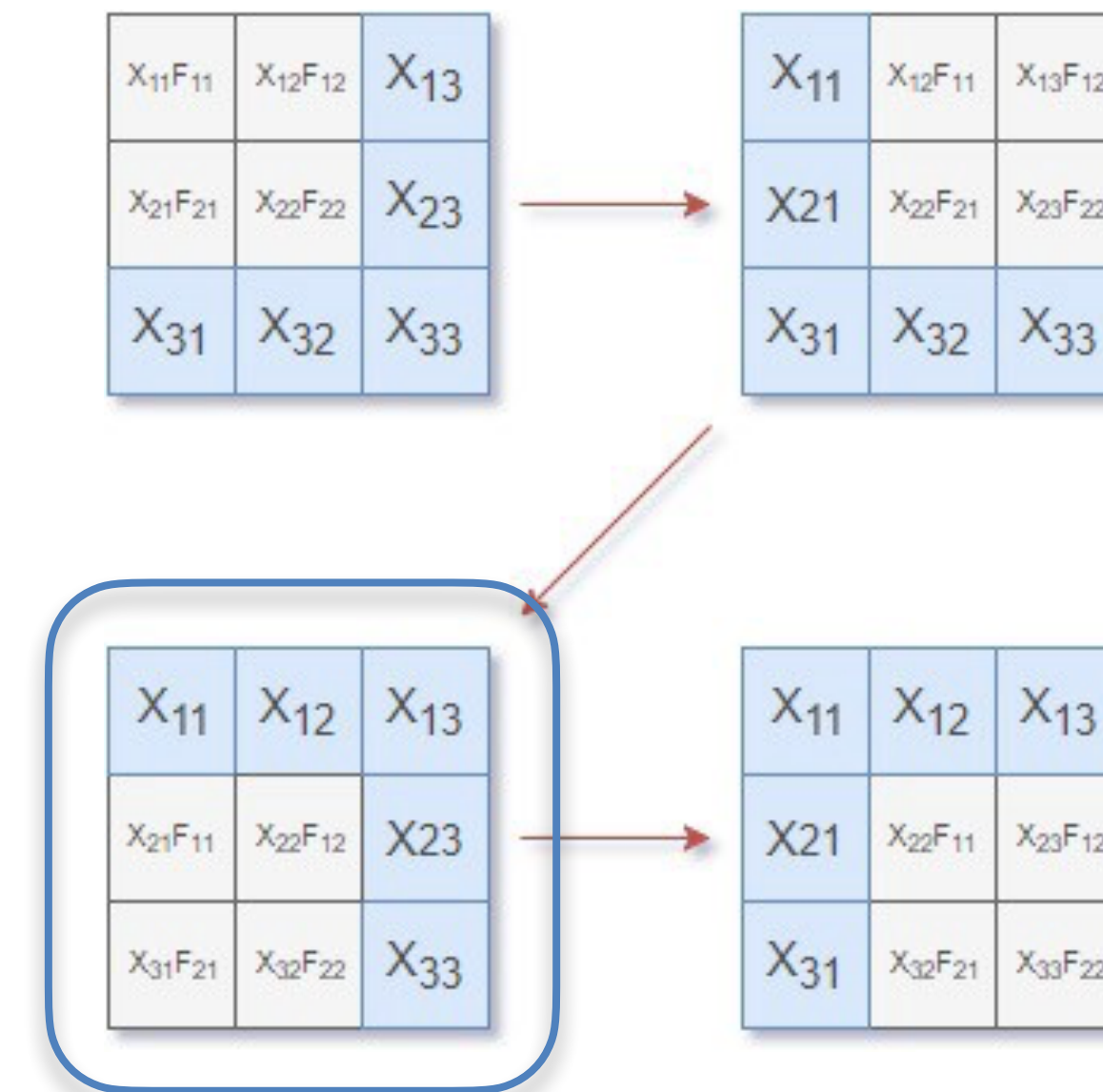
- Note: F instead of w

$$O_{11} = F_{11}X_{11} + F_{12}X_{12} + F_{21}X_{21} + F_{22}X_{22}$$

$$O_{12} = F_{11}X_{12} + F_{12}X_{13} + F_{21}X_{22} + F_{22}X_{23}$$

$$O_{21} = F_{11}X_{21} + F_{12}X_{22} + F_{21}X_{31} + F_{22}X_{32}$$

- It can be visualized as:



Forward and Backward Propagation using Convolution operation

$$\begin{bmatrix} O_{11} & O_{12} \\ O_{21} & O_{22} \end{bmatrix} = \text{Convolution} \left(\begin{bmatrix} X_{11} & X_{12} & X_{13} \\ X_{21} & X_{22} & X_{23} \\ X_{31} & X_{32} & X_{33} \end{bmatrix}, \begin{bmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \end{bmatrix} \right)$$

- Forward Pass

- Ignoring 180° rotation
- Note: F instead of w

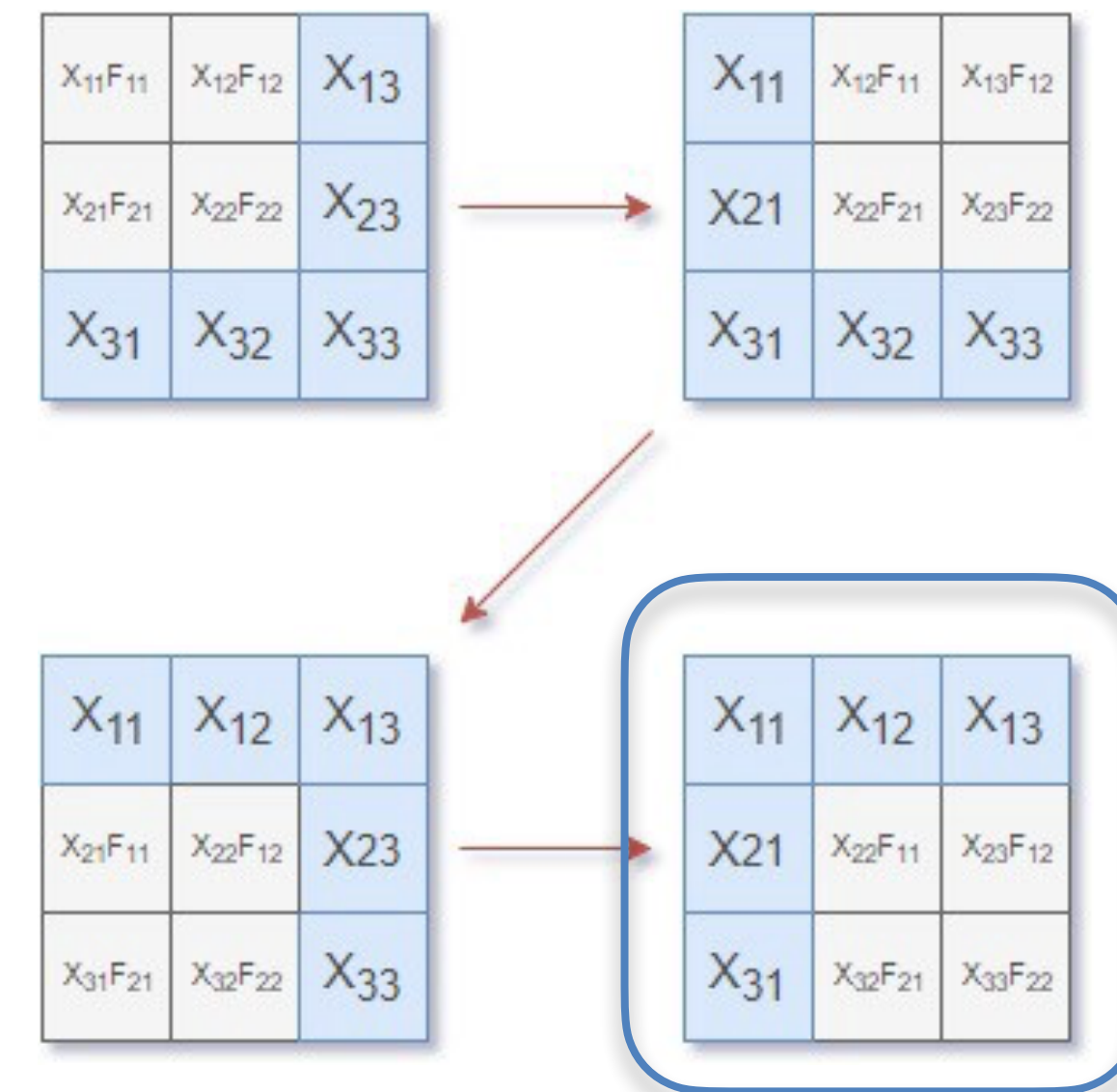
$$O_{11} = F_{11}X_{11} + F_{12}X_{12} + F_{21}X_{21} + F_{22}X_{22}$$

$$O_{12} = F_{11}X_{12} + F_{12}X_{13} + F_{21}X_{22} + F_{22}X_{23}$$

$$O_{21} = F_{11}X_{21} + F_{12}X_{22} + F_{21}X_{31} + F_{22}X_{32}$$

$$O_{22} = F_{11}X_{22} + F_{12}X_{23} + F_{21}X_{32} + F_{22}X_{33}$$

- It can be visualized as:



Forward and Backward Propagation using Convolution operation

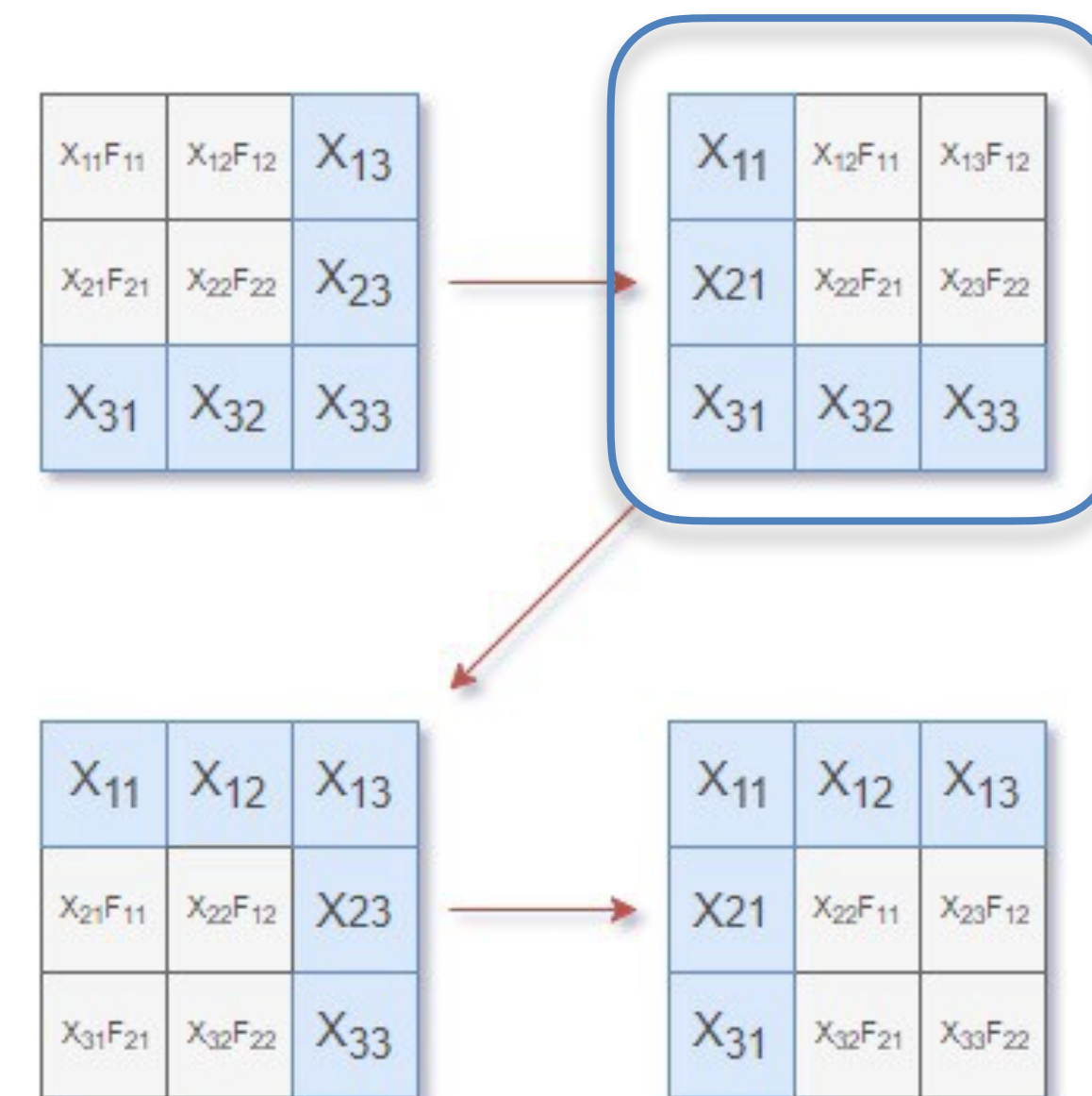
$$\begin{bmatrix} O_{11} & O_{12} \\ O_{21} & O_{22} \end{bmatrix} = \text{Convolution} \left(\begin{bmatrix} X_{11} & X_{12} & X_{13} \\ X_{21} & X_{22} & X_{23} \\ X_{31} & X_{32} & X_{33} \end{bmatrix}, \begin{bmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \end{bmatrix} \right)$$

- Forward Pass
 - Ignoring 180° rotation

$$O_{11} = F_{11}X_{11} + F_{12}X_{12} + F_{21}X_{21} + F_{22}X_{22}$$

$$O_{12} = F_{11}X_{12} + F_{12}X_{13} + F_{21}X_{22} + F_{22}X_{23}$$

- It can be visualized as:



Forward and Backward Propagation using Convolution operation

$$\begin{bmatrix} O_{11} & O_{12} \\ O_{21} & O_{22} \end{bmatrix} = \text{Convolution} \left(\begin{bmatrix} X_{11} & X_{12} & X_{13} \\ X_{21} & X_{22} & X_{23} \\ X_{31} & X_{32} & X_{33} \end{bmatrix}, \begin{bmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \end{bmatrix} \right)$$

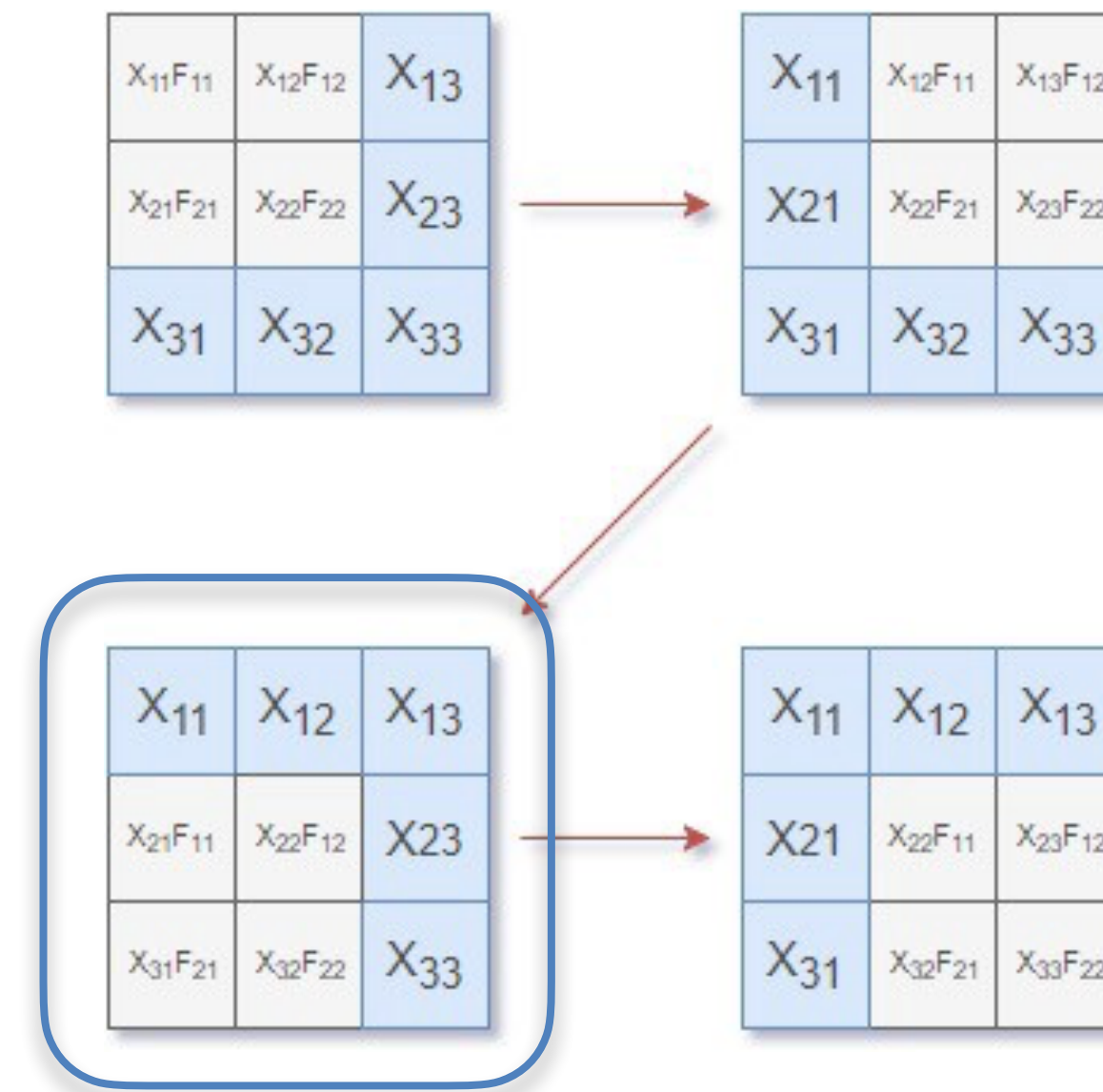
- Forward Pass
 - Ignoring 180° rotation

$$O_{11} = F_{11}X_{11} + F_{12}X_{12} + F_{21}X_{21} + F_{22}X_{22}$$

$$O_{12} = F_{11}X_{12} + F_{12}X_{13} + F_{21}X_{22} + F_{22}X_{23}$$

$$O_{21} = F_{11}X_{21} + F_{12}X_{22} + F_{21}X_{31} + F_{22}X_{32}$$

- It can be visualized as:



Forward and Backward Propagation using Convolution operation

$$\begin{bmatrix} O_{11} & O_{12} \\ O_{21} & O_{22} \end{bmatrix} = \text{Convolution} \left(\begin{bmatrix} X_{11} & X_{12} & X_{13} \\ X_{21} & X_{22} & X_{23} \\ X_{31} & X_{32} & X_{33} \end{bmatrix}, \begin{bmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \end{bmatrix} \right)$$

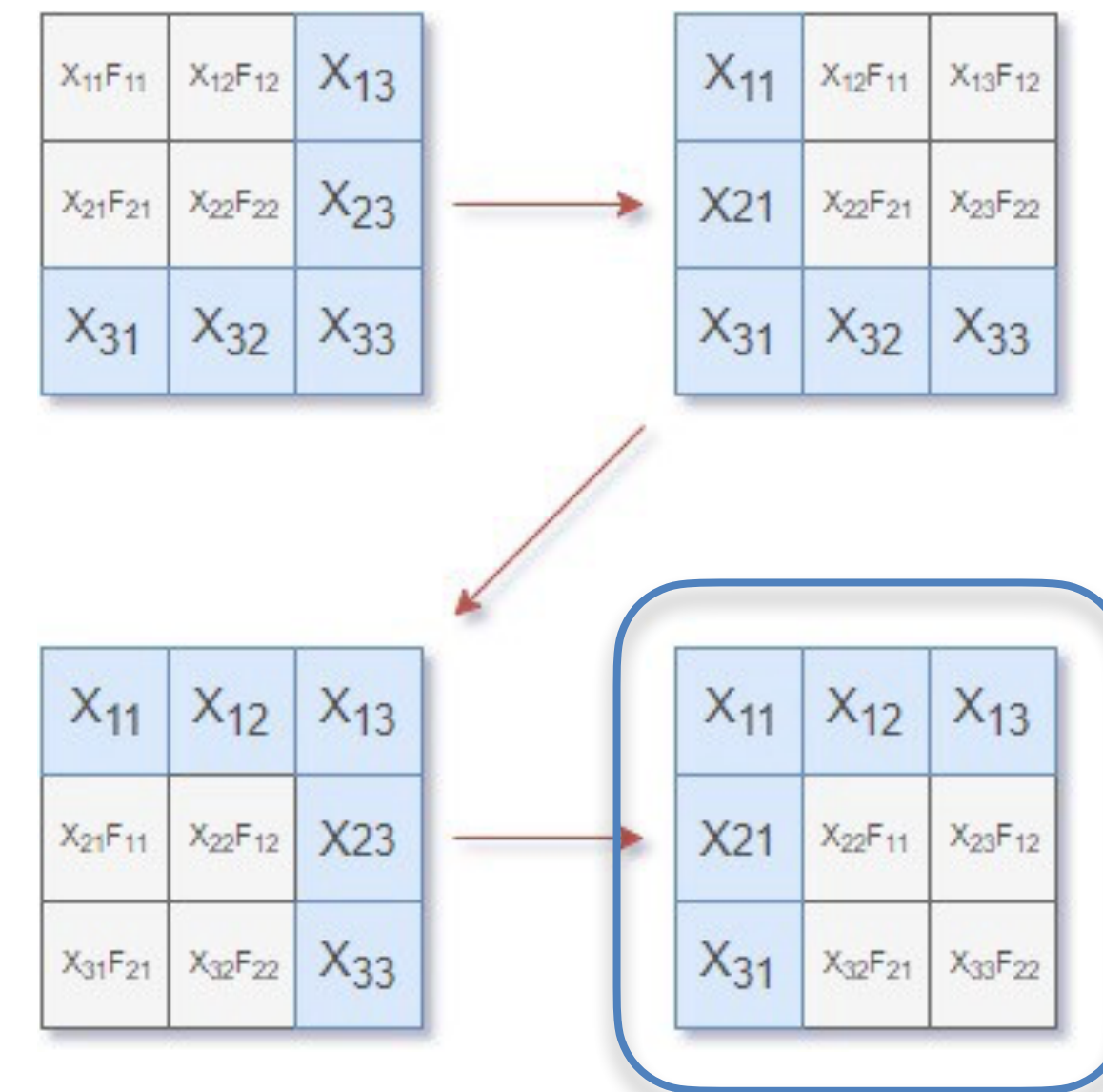
- Forward Pass
 - Ignoring 180° rotation

$$O_{11} = F_{11}X_{11} + F_{12}X_{12} + F_{21}X_{21} + F_{22}X_{22}$$

$$O_{12} = F_{11}X_{12} + F_{12}X_{13} + F_{21}X_{22} + F_{22}X_{23}$$

$$O_{21} = F_{11}X_{21} + F_{12}X_{22} + F_{21}X_{31} + F_{22}X_{32}$$

$$O_{22} = F_{11}X_{22} + F_{12}X_{23} + F_{21}X_{32} + F_{22}X_{33}$$



- It can be visualized as:

Feedforward in CNN is identical with convolution operation

$$\begin{bmatrix} O_{11} & O_{12} \\ O_{21} & O_{22} \end{bmatrix} = \text{Convolution} \left(\begin{bmatrix} X_{11} & X_{12} & X_{13} \\ X_{21} & X_{22} & X_{23} \\ X_{31} & X_{32} & X_{33} \end{bmatrix}, \begin{bmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \end{bmatrix} \right)$$

$$O_{11} = F_{11}X_{11} + F_{12}X_{12} + F_{21}X_{21} + F_{22}X_{22}$$

$$O_{12} = F_{11}X_{12} + F_{12}X_{13} + F_{21}X_{22} + F_{22}X_{23}$$

$$O_{21} = F_{11}X_{21} + F_{12}X_{22} + F_{21}X_{31} + F_{22}X_{32}$$

$$O_{22} = F_{11}X_{22} + F_{12}X_{23} + F_{21}X_{32} + F_{22}X_{33}$$

X_{11}

X_{12}

X_{13}

X_{21}

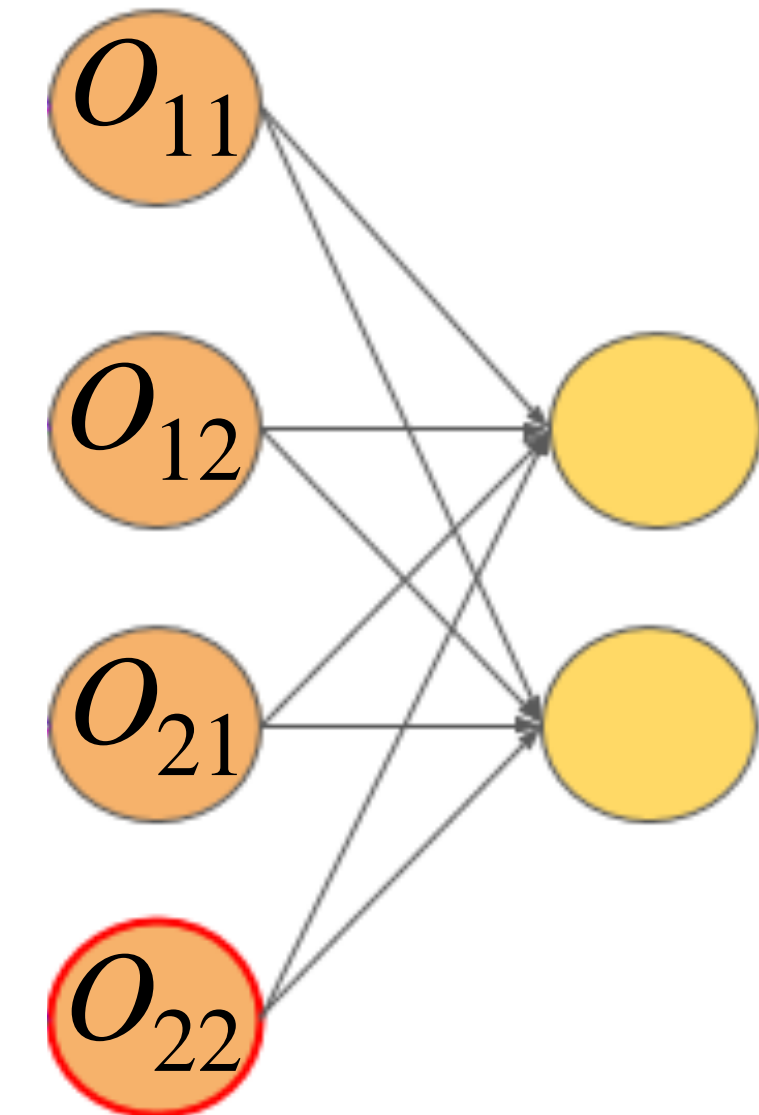
X_{22}

X_{23}

X_{31}

X_{32}

X_{33}



Forward and Backward Propagation using Convolution operation

- Gradients of filter 'F' with respect to the error 'E'

$$E = E_{O_{11}} + E_{O_{12}} + E_{O_{21}} + E_{O_{22}}$$

$$\frac{\partial E}{\partial F_{11}} = \frac{\partial E}{\partial O_{11}} \frac{\partial O_{11}}{\partial F_{11}} + \frac{\partial E}{\partial O_{12}} \frac{\partial O_{12}}{\partial F_{11}} + \frac{\partial E}{\partial O_{21}} \frac{\partial O_{21}}{\partial F_{11}} + \frac{\partial E}{\partial O_{22}} \frac{\partial O_{22}}{\partial F_{11}}$$

$$\frac{\partial E}{\partial F_{11}} = \frac{\partial E}{\partial O_{11}} X_{11} + \frac{\partial E}{\partial O_{12}} X_{12} + \frac{\partial E}{\partial O_{21}} X_{21} + \frac{\partial E}{\partial O_{22}} X_{22}$$

$$\frac{\partial E}{\partial F_{12}} = \frac{\partial E}{\partial O_{11}} \frac{\partial O_{11}}{\partial F_{12}} + \frac{\partial E}{\partial O_{12}} \frac{\partial O_{12}}{\partial F_{12}} + \frac{\partial E}{\partial O_{21}} \frac{\partial O_{21}}{\partial F_{12}} + \frac{\partial E}{\partial O_{22}} \frac{\partial O_{22}}{\partial F_{12}}$$

$$\frac{\partial E}{\partial F_{12}} = \frac{\partial E}{\partial O_{11}} X_{12} + \frac{\partial E}{\partial O_{12}} X_{13} + \frac{\partial E}{\partial O_{21}} X_{22} + \frac{\partial E}{\partial O_{22}} X_{23}$$

$$\frac{\partial E}{\partial F_{21}} = \frac{\partial E}{\partial O_{11}} \frac{\partial O_{11}}{\partial F_{21}} + \frac{\partial E}{\partial O_{12}} \frac{\partial O_{12}}{\partial F_{21}} + \frac{\partial E}{\partial O_{21}} \frac{\partial O_{21}}{\partial F_{21}} + \frac{\partial E}{\partial O_{22}} \frac{\partial O_{22}}{\partial F_{21}}$$

$$\frac{\partial E}{\partial F_{21}} = \frac{\partial E}{\partial O_{11}} X_{21} + \frac{\partial E}{\partial O_{12}} X_{22} + \frac{\partial E}{\partial O_{21}} X_{31} + \frac{\partial E}{\partial O_{22}} X_{32}$$

$$\frac{\partial E}{\partial F_{22}} = \frac{\partial E}{\partial O_{11}} \frac{\partial O_{11}}{\partial F_{22}} + \frac{\partial E}{\partial O_{12}} \frac{\partial O_{12}}{\partial F_{22}} + \frac{\partial E}{\partial O_{21}} \frac{\partial O_{21}}{\partial F_{22}} + \frac{\partial E}{\partial O_{22}} \frac{\partial O_{22}}{\partial F_{22}}$$

$$\frac{\partial E}{\partial F_{22}} = \frac{\partial E}{\partial O_{11}} X_{22} + \frac{\partial E}{\partial O_{12}} X_{23} + \frac{\partial E}{\partial O_{21}} X_{32} + \frac{\partial E}{\partial O_{22}} X_{33}$$

$$\begin{bmatrix} O_{11} & O_{12} \\ O_{21} & O_{22} \end{bmatrix} = \text{Convolution} \left(\begin{bmatrix} X_{11} & X_{12} & X_{13} \\ X_{21} & X_{22} & X_{23} \\ X_{31} & X_{32} & X_{33} \end{bmatrix}, \begin{bmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \end{bmatrix} \right)$$

$$O_{11} = F_{11}X_{11} + F_{12}X_{12} + F_{21}X_{21} + F_{22}X_{22}$$

$$O_{12} = F_{11}X_{12} + F_{12}X_{13} + F_{21}X_{22} + F_{22}X_{23}$$

$$O_{21} = F_{11}X_{21} + F_{12}X_{22} + F_{21}X_{31} + F_{22}X_{32}$$

$$O_{22} = F_{11}X_{22} + F_{12}X_{23} + F_{21}X_{32} + F_{22}X_{33}$$

$$\begin{bmatrix} \frac{\partial E}{\partial F_{11}} & \frac{\partial E}{\partial F_{12}} \\ \frac{\partial E}{\partial F_{21}} & \frac{\partial E}{\partial F_{22}} \end{bmatrix} = \text{Convolution} \left(\begin{bmatrix} X_{11} & X_{12} & X_{13} \\ X_{21} & X_{22} & X_{23} \\ X_{31} & X_{32} & X_{33} \end{bmatrix}, \begin{bmatrix} \frac{\partial E}{\partial O_{11}} & \frac{\partial E}{\partial O_{12}} \\ \frac{\partial E}{\partial O_{21}} & \frac{\partial E}{\partial O_{22}} \end{bmatrix} \right)$$

Forward and Backward Propagation using Convolution operation

- Gradients of filter 'F' with respect to the error 'E'

$$E = E_{O_{11}} + E_{O_{12}} + E_{O_{21}} + E_{O_{22}}$$

$$\begin{bmatrix} O_{11} & O_{12} \\ O_{21} & O_{22} \end{bmatrix} = \text{Convolution} \left(\begin{bmatrix} X_{11} & X_{12} & X_{13} \\ X_{21} & X_{22} & X_{23} \\ X_{31} & X_{32} & X_{33} \end{bmatrix}, \begin{bmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \end{bmatrix} \right)$$

$$\rightarrow \frac{\partial E}{\partial F_{11}} = \frac{\partial E}{\partial O_{11}} X_{11} + \frac{\partial E}{\partial O_{12}} X_{12} + \frac{\partial E}{\partial O_{21}} X_{21} + \frac{\partial E}{\partial O_{22}} X_{22}$$

$$\rightarrow \frac{\partial E}{\partial F_{12}} = \frac{\partial E}{\partial O_{11}} X_{12} + \frac{\partial E}{\partial O_{12}} X_{13} + \frac{\partial E}{\partial O_{21}} X_{22} + \frac{\partial E}{\partial O_{22}} X_{23}$$

$$\rightarrow \frac{\partial E}{\partial F_{21}} = \frac{\partial E}{\partial O_{11}} X_{21} + \frac{\partial E}{\partial O_{12}} X_{22} + \frac{\partial E}{\partial O_{21}} X_{31} + \frac{\partial E}{\partial O_{22}} X_{32}$$

$$\rightarrow \frac{\partial E}{\partial F_{22}} = \frac{\partial E}{\partial O_{11}} X_{22} + \frac{\partial E}{\partial O_{12}} X_{23} + \frac{\partial E}{\partial O_{21}} X_{32} + \frac{\partial E}{\partial O_{22}} X_{33}$$

$$\begin{bmatrix} \frac{\partial E}{\partial F_{11}} & \frac{\partial E}{\partial F_{12}} \\ \frac{\partial E}{\partial F_{21}} & \frac{\partial E}{\partial F_{22}} \end{bmatrix} = \text{Convolution} \left(\begin{bmatrix} X_{11} & X_{12} & X_{13} \\ X_{21} & X_{22} & X_{23} \\ X_{31} & X_{32} & X_{33} \end{bmatrix}, \begin{bmatrix} \frac{\partial E}{\partial O_{11}} & \frac{\partial E}{\partial O_{12}} \\ \frac{\partial E}{\partial O_{21}} & \frac{\partial E}{\partial O_{22}} \end{bmatrix} \right)$$

Forward and Backward Propagation using Convolution operation

- Gradients of the input matrix 'X' with respect to the error 'E'

$$E = E_{O_{11}} + E_{O_{12}} + E_{O_{21}} + E_{O_{22}}$$

O ₁₁	O ₁₂
O ₂₁	O ₂₂

= Convolution

X ₁₁	X ₁₂	X ₁₃
X ₂₁	X ₂₂	X ₂₃
X ₃₁	X ₃₂	X ₃₃

,

F ₁₁	F ₁₂
F ₂₁	F ₂₂

$$\frac{\partial E}{\partial X_{11}} = \frac{\partial E}{\partial O_{11}} \frac{\partial O_{11}}{\partial X_{11}} + \frac{\partial E}{\partial O_{12}} \frac{\partial O_{12}}{\partial X_{11}} + \frac{\partial E}{\partial O_{21}} \frac{\partial O_{21}}{\partial X_{11}} + \frac{\partial E}{\partial O_{22}} \frac{\partial O_{22}}{\partial X_{11}}$$

$$\frac{\partial E}{\partial X_{11}} = \frac{\partial E}{\partial O_{11}} F_{11} + \frac{\partial E}{\partial O_{12}} 0 + \frac{\partial E}{\partial O_{21}} 0 + \frac{\partial E}{\partial O_{22}} 0$$

$O_{11} = F_{11}X_{11} + F_{12}X_{12} + F_{21}X_{21} + F_{22}X_{22}$
 $O_{12} = F_{11}X_{12} + F_{12}X_{13} + F_{21}X_{22} + F_{22}X_{23}$
 $O_{21} = F_{11}X_{21} + F_{12}X_{22} + F_{21}X_{31} + F_{22}X_{32}$
 $O_{22} = F_{11}X_{22} + F_{12}X_{23} + F_{21}X_{32} + F_{22}X_{33}$

$$\frac{\partial E}{\partial X_{23}} = \frac{\partial E}{\partial O_{11}} \frac{\partial O_{11}}{\partial X_{23}} + \frac{\partial E}{\partial O_{12}} \frac{\partial O_{12}}{\partial X_{23}} + \frac{\partial E}{\partial O_{21}} \frac{\partial O_{21}}{\partial X_{23}} + \frac{\partial E}{\partial O_{22}} \frac{\partial O_{22}}{\partial X_{23}}$$

$$\frac{\partial E}{\partial X_{23}} = \frac{\partial E}{\partial O_{11}} 0 + \frac{\partial E}{\partial O_{12}} F_{22} + \frac{\partial E}{\partial O_{21}} 0 + \frac{\partial E}{\partial O_{22}} F_{12}$$

$$\frac{\partial E}{\partial X_{21}}$$

Forward and Backward Propagation using Convolution operation

- Gradients of the input matrix 'X' with respect to the error 'E'

$$\frac{\partial E}{\partial X_{11}} = \frac{\partial E}{\partial O_{11}} F_{11} + \frac{\partial E}{\partial O_{12}} 0 + \frac{\partial E}{\partial O_{21}} 0 + \frac{\partial E}{\partial O_{22}} 0$$

$$\frac{\partial E}{\partial X_{12}} = \frac{\partial E}{\partial O_{11}} F_{12} + \frac{\partial E}{\partial O_{12}} F_{11} + \frac{\partial E}{\partial O_{21}} 0 + \frac{\partial E}{\partial O_{22}} 0$$

$$\frac{\partial E}{\partial X_{13}} = \frac{\partial E}{\partial O_{11}} 0 + \frac{\partial E}{\partial O_{12}} F_{12} + \frac{\partial E}{\partial O_{21}} 0 + \frac{\partial E}{\partial O_{22}} 0$$

$$\frac{\partial E}{\partial X_{21}} = \frac{\partial E}{\partial O_{11}} F_{21} + \frac{\partial E}{\partial O_{12}} 0 + \frac{\partial E}{\partial O_{21}} F_{11} + \frac{\partial E}{\partial O_{22}} 0$$

$$\frac{\partial E}{\partial X_{22}} = \frac{\partial E}{\partial O_{11}} F_{22} + \frac{\partial E}{\partial O_{12}} F_{21} + \frac{\partial E}{\partial O_{21}} f_{12} + \frac{\partial E}{\partial O_{22}} F_{11}$$

$$\frac{\partial E}{\partial X_{23}} = \frac{\partial E}{\partial O_{11}} 0 + \frac{\partial E}{\partial O_{12}} F_{22} + \frac{\partial E}{\partial O_{21}} 0 + \frac{\partial E}{\partial O_{22}} F_{12}$$

$$\frac{\partial E}{\partial X_{31}} = \frac{\partial E}{\partial O_{11}} 0 + \frac{\partial E}{\partial O_{12}} 0 + \frac{\partial E}{\partial O_{21}} F_{21} + \frac{\partial E}{\partial O_{22}} 0$$

$$\frac{\partial E}{\partial X_{32}} = \frac{\partial E}{\partial O_{11}} 0 + \frac{\partial E}{\partial O_{12}} 0 + \frac{\partial E}{\partial O_{21}} F_{22} + \frac{\partial E}{\partial O_{22}} F_{21}$$

$$\frac{\partial E}{\partial X_{33}} = \frac{\partial E}{\partial O_{11}} 0 + \frac{\partial E}{\partial O_{12}} 0 + \frac{\partial E}{\partial O_{21}} 0 + \frac{\partial E}{\partial O_{22}} F_{22}$$

$$\begin{bmatrix} O_{11} & O_{12} \\ O_{21} & O_{22} \end{bmatrix} = \text{Convolution} \left(\begin{bmatrix} X_{11} & X_{12} & X_{13} \\ X_{21} & X_{22} & X_{23} \\ X_{31} & X_{32} & X_{33} \end{bmatrix}, \begin{bmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \end{bmatrix} \right)$$

$$O_{11} = F_{11}X_{11} + F_{12}X_{12} + F_{21}X_{21} + F_{22}X_{22}$$

$$O_{12} = F_{11}X_{12} + F_{12}X_{13} + F_{21}X_{22} + F_{22}X_{23}$$

$$O_{21} = F_{11}X_{21} + F_{12}X_{22} + F_{21}X_{31} + F_{22}X_{32}$$

$$O_{22} = F_{11}X_{22} + F_{12}X_{23} + F_{21}X_{32} + F_{22}X_{33}$$

$$\begin{bmatrix} \frac{\partial E}{\partial X_{11}} & \frac{\partial E}{\partial X_{12}} & \frac{\partial E}{\partial X_{13}} \\ \frac{\partial E}{\partial X_{21}} & \frac{\partial E}{\partial X_{22}} & \frac{\partial E}{\partial X_{23}} \\ \frac{\partial E}{\partial X_{31}} & \frac{\partial E}{\partial X_{32}} & \frac{\partial E}{\partial X_{33}} \end{bmatrix} = \text{Full_Convolution} \left(\begin{bmatrix} \frac{\partial E}{\partial O_{11}} & \frac{\partial E}{\partial O_{12}} \\ \frac{\partial E}{\partial O_{21}} & \frac{\partial E}{\partial O_{22}} \end{bmatrix}, \begin{bmatrix} F_{22} & F_{21} \\ F_{12} & F_{11} \end{bmatrix} \right)$$

$$\begin{bmatrix} F_{22} & F_{21} \\ F_{12} & F_{11} \end{bmatrix} \otimes \begin{bmatrix} \delta O_{11} & \delta O_{12} \\ \delta O_{21} & \delta O_{22} \end{bmatrix} = \begin{bmatrix} \delta X_{11} & \delta X_{12} \\ \delta X_{21} & \delta X_{22} \end{bmatrix}$$

$$\begin{bmatrix} F_{22} & F_{21} \\ F_{12} & F_{11} \end{bmatrix} \otimes \begin{bmatrix} \delta O_{11} & \delta O_{12} \\ \delta O_{21} & \delta O_{22} \end{bmatrix} = \begin{bmatrix} \delta X_{12} & \delta X_{13} \\ \delta X_{22} & \delta X_{23} \end{bmatrix}$$

$$\begin{bmatrix} F_{22} & F_{21} \\ F_{12} & F_{11} \end{bmatrix} \otimes \begin{bmatrix} \delta O_{11} & \delta O_{12} \\ \delta O_{21} & \delta O_{22} \end{bmatrix} = \begin{bmatrix} \delta X_{21} & \delta X_{22} \\ \delta X_{31} & \delta X_{32} \end{bmatrix}$$

$$\begin{pmatrix} \partial E / \partial X_{11} & \partial E / \partial X_{12} & \partial E / \partial X_{13} \\ \partial E / \partial X_{21} & \partial E / \partial X_{22} & \partial E / \partial X_{23} \\ \partial E / \partial X_{31} & \partial E / \partial X_{32} & \partial E / \partial X_{33} \end{pmatrix} = \text{Full_Convolution} \left(\begin{pmatrix} \partial E / \partial O_{11} & \partial E / \partial O_{12} \\ \partial E / \partial O_{21} & \partial E / \partial O_{22} \end{pmatrix}, \begin{pmatrix} F_{22} & F_{21} \\ F_{12} & F_{11} \end{pmatrix} \right)$$

$$\begin{array}{cc} F_{22} & F_{21} \\ F_{12} & F_{11} \delta O_{11} & \delta O_{12} \\ & \delta O_{21} & \delta O_{22} \end{array}$$

$$\delta X_{11} = F_{11} \delta O_{11}$$

$$\begin{array}{cc} F_{22} & F_{21} \\ F_{12} \delta O_{11} & F_{11} \delta O_{12} \\ \delta O_{21} & \delta O_{22} \end{array}$$

$$\delta X_{12} = F_{12} \delta O_{11} + F_{11} \delta O_{12}$$

$$\begin{array}{cc} & F_{22} & F_{21} \\ \delta O_{11} & F_{12} \delta O_{12} & F_{11} \\ \delta O_{21} & \delta O_{22} & \end{array}$$

$$\delta X_{13} = F_{12} \delta O_{12}$$

$$\begin{array}{cc} F_{22} & F_{21} \delta O_{11} & \delta O_{12} \\ F_{12} & F_{11} \delta O_{21} & \delta O_{22} \end{array}$$

$$\delta X_{21} = F_{21} \delta O_{11} + F_{11} \delta O_{21}$$

$$\begin{array}{cc} F_{22} \delta O_{11} & F_{21} \delta O_{12} \\ F_{12} \delta O_{21} & F_{11} \delta O_{22} \end{array}$$

$$\delta X_{22} = F_{22} \delta O_{11} + F_{21} \delta O_{12} + F_{12} \delta O_{21} + F_{11} \delta O_{22}$$

$$\begin{array}{cc} \delta O_{11} & F_{22} \delta O_{12} & F_{21} \\ \delta O_{21} & F_{12} \delta O_{22} & F_{11} \end{array}$$

$$\delta X_{23} = F_{22} \delta O_{12} + F_{12} \delta O_{22}$$

$$\begin{array}{cc} & \delta O_{11} & \delta O_{12} \\ F_{22} & F_{21} \delta O_{21} & \delta O_{22} \\ F_{12} & F_{11} & \end{array}$$

$$\delta X_{31} = F_{21} \delta O_{21}$$

$$\begin{array}{cc} \delta O_{11} & \delta O_{12} \\ F_{22} \delta O_{21} & F_{21} \delta O_{22} \\ F_{12} & F_{11} \end{array}$$

$$\delta X_{32} = F_{22} \delta O_{21} + F_{21} \delta O_{22}$$

$$\begin{array}{cc} \delta O_{11} & \delta O_{12} \\ \delta O_{21} & F_{22} \delta O_{22} & F_{21} \\ & F_{12} & F_{11} \end{array}$$

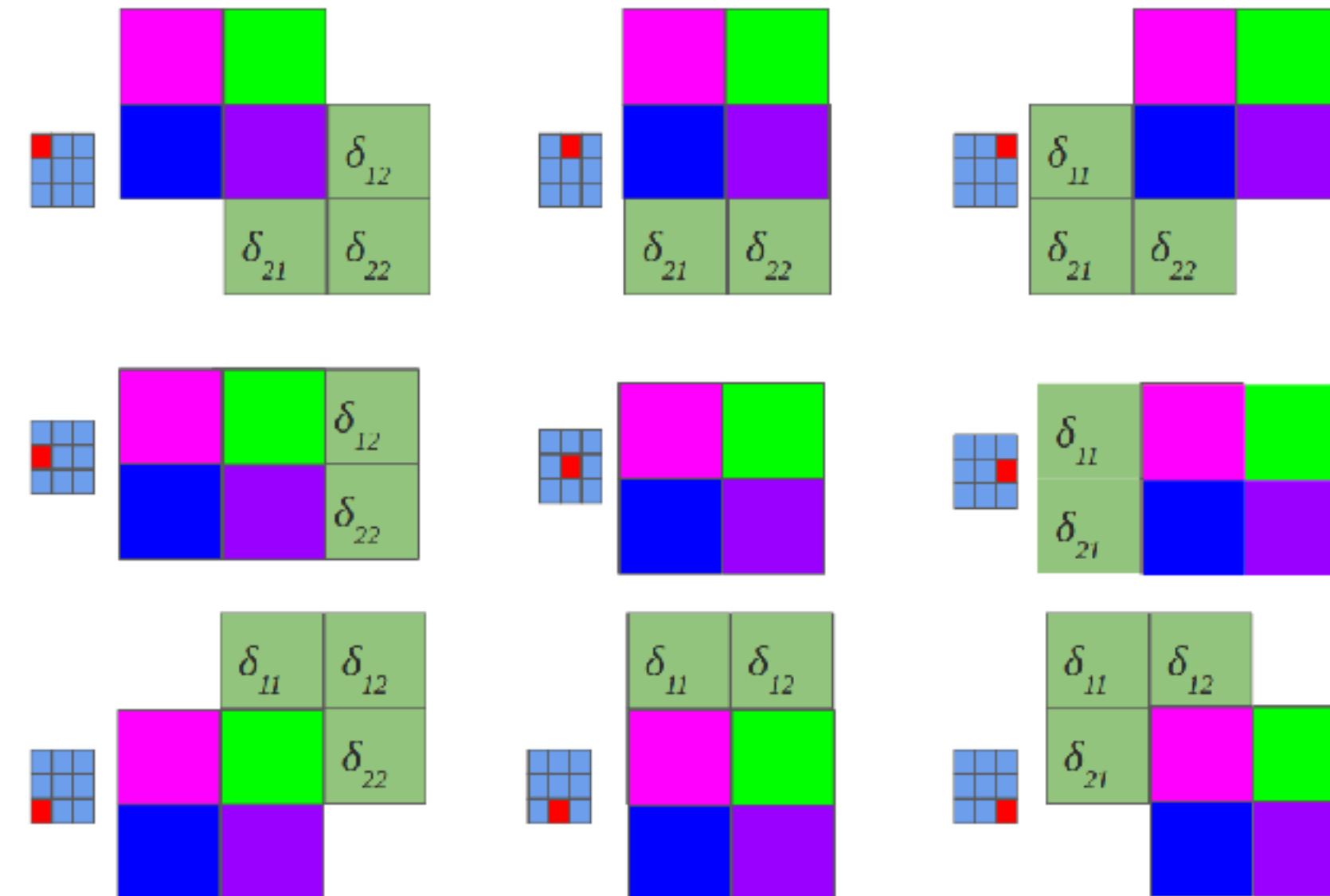
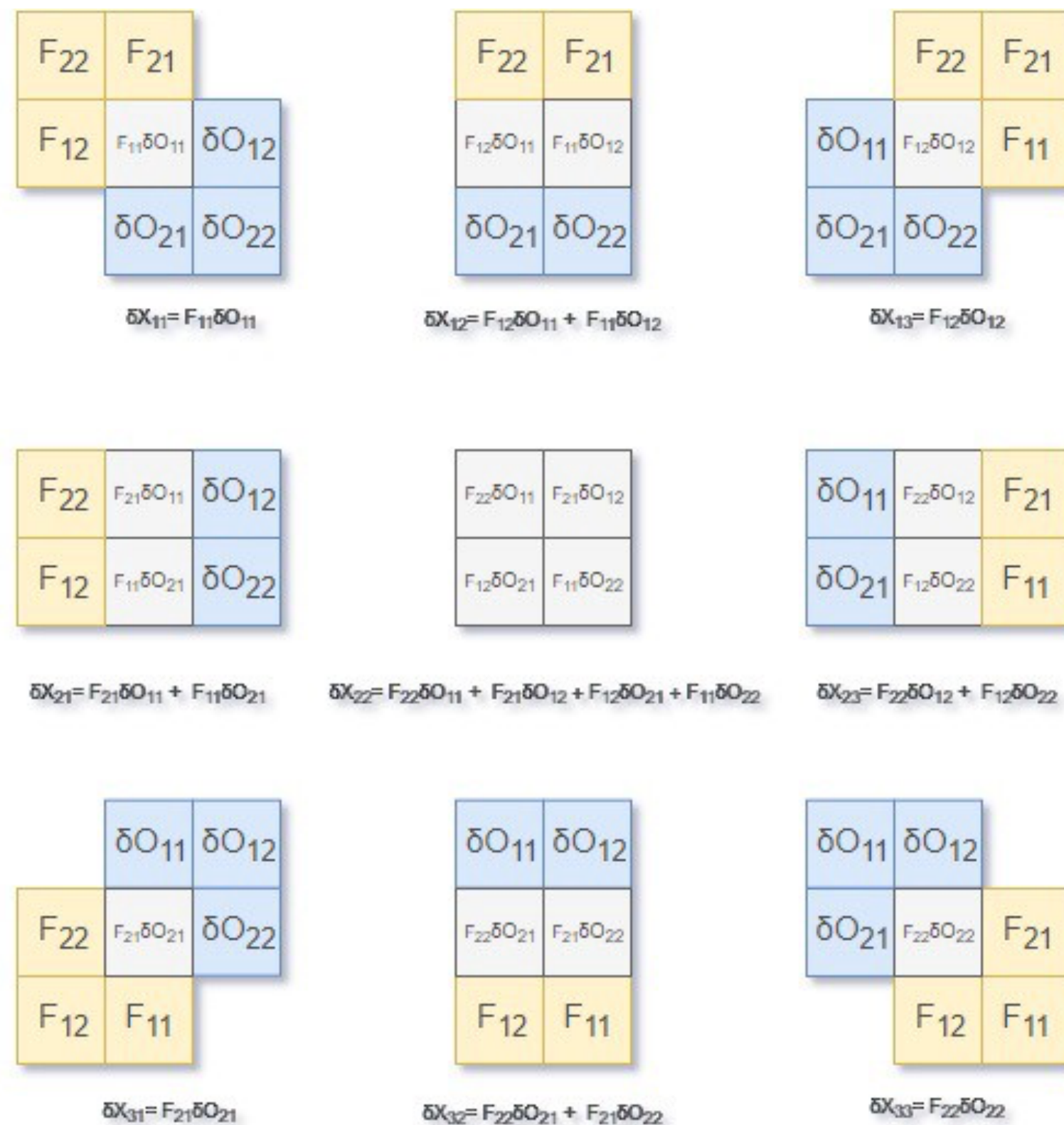
$$\delta X_{33} = F_{22} \delta O_{22}$$

Forward and Backward Propagation using Convolution operation

- Full Convolution

$$\begin{array}{|c|c|c|} \hline \partial E / \partial X_{11} & \partial E / \partial X_{12} & \partial E / \partial X_{13} \\ \hline \partial E / \partial X_{21} & \partial E / \partial X_{22} & \partial E / \partial X_{23} \\ \hline \partial E / \partial X_{31} & \partial E / \partial X_{32} & \partial E / \partial X_{33} \\ \hline \end{array} = \text{Full_Convolution} \left(\begin{array}{|c|c|} \hline \partial E / \partial O_{11} & \partial E / \partial O_{12} \\ \hline \partial E / \partial O_{21} & \partial E / \partial O_{22} \\ \hline \end{array}, \begin{array}{|c|c|} \hline F_{22} & F_{21} \\ \hline F_{12} & F_{11} \\ \hline \end{array} \right)$$

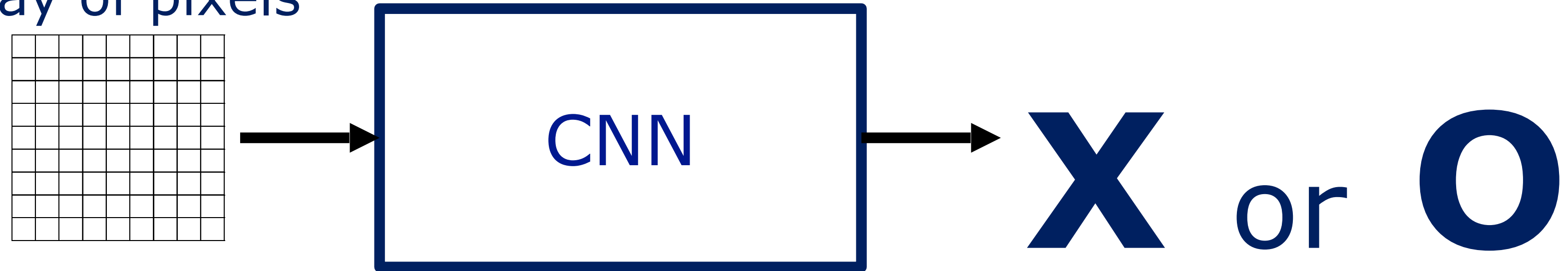
180° rotation



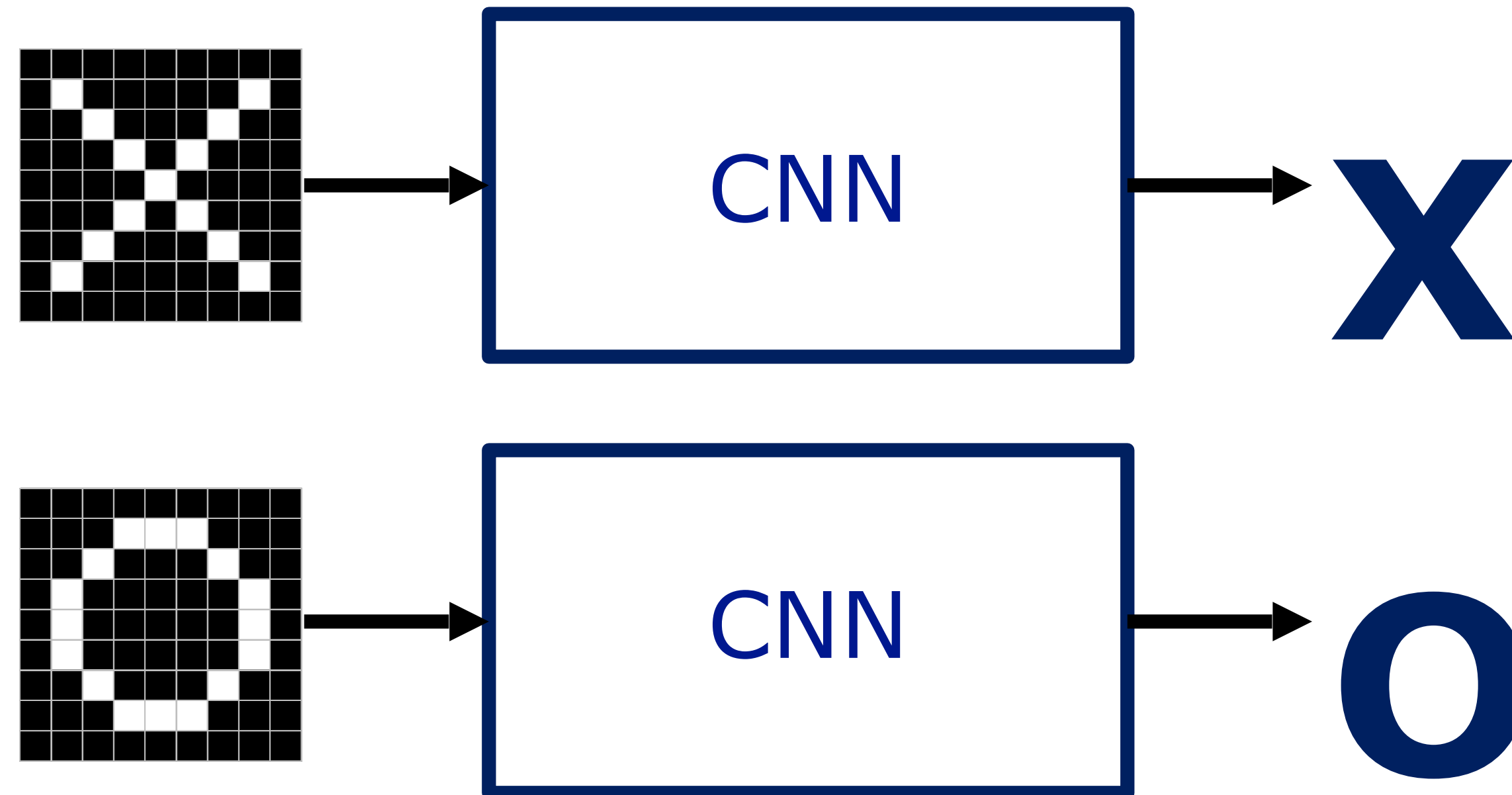
A toy ConvNet: X's and O's

Says whether a picture is of an X or an O

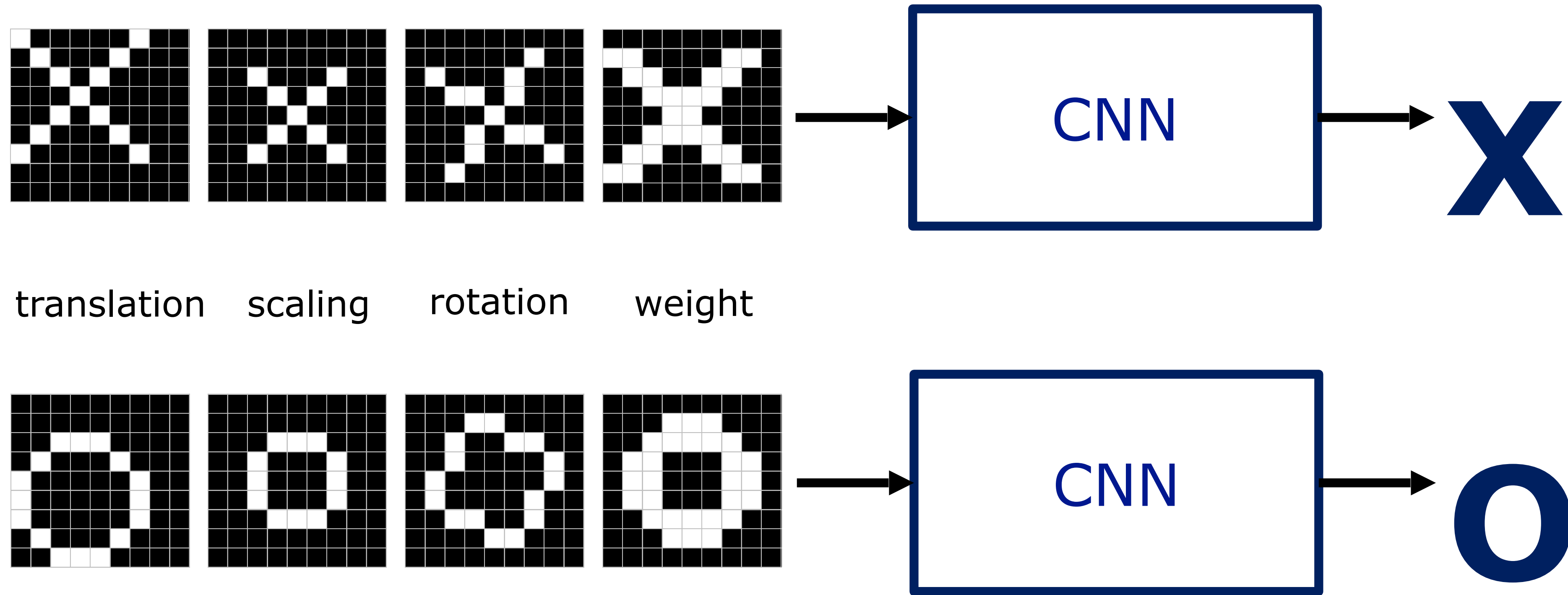
A two-dimensional
array of pixels



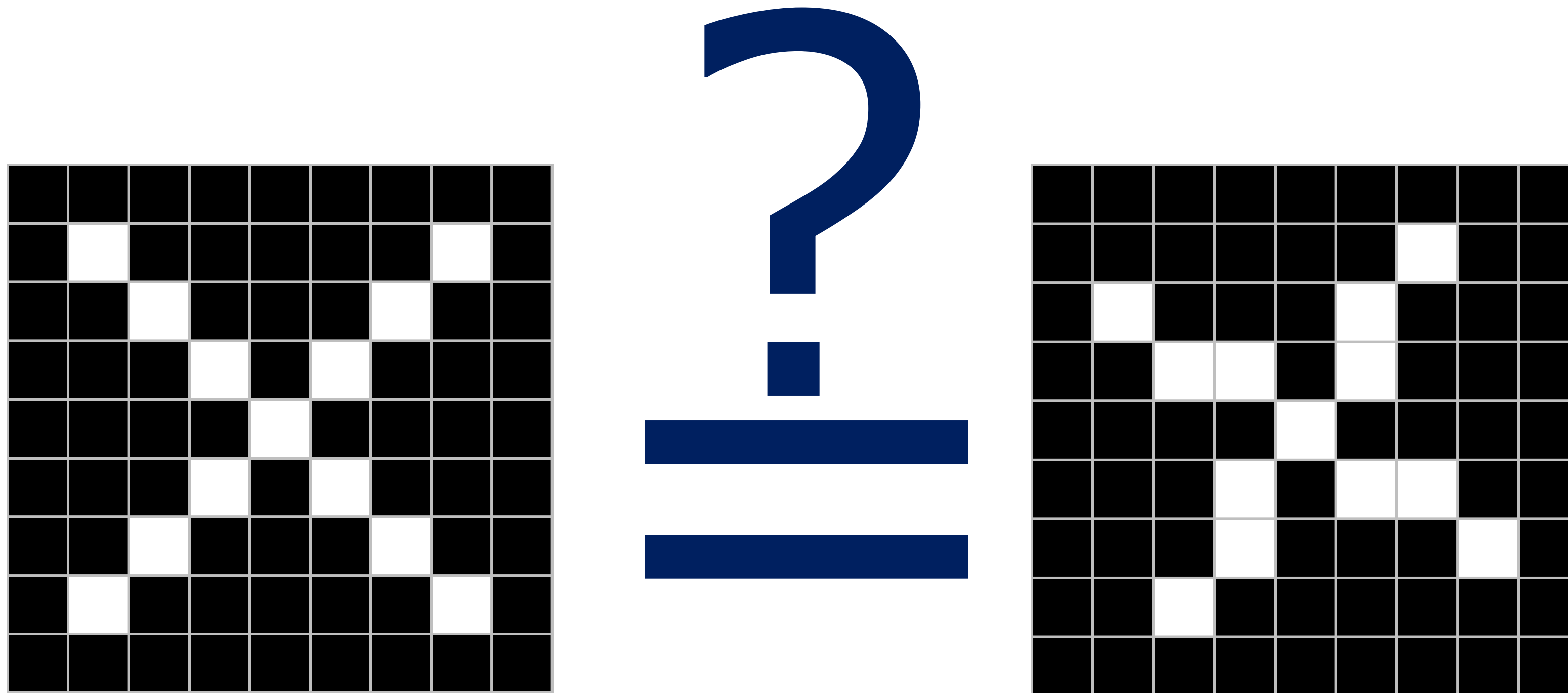
For example



Trickier cases



Deciding is hard



What computers see

-1	-1	-1	-1	-1	-1	-1	-1	-1
-1	1	-1	-1	-1	-1	-1	1	-1
-1	-1	1	-1	-1	-1	1	-1	-1
-1	-1	-1	1	-1	1	-1	-1	-1
-1	-1	-1	-1	1	-1	-1	-1	-1
-1	-1	-1	1	-1	1	-1	-1	-1
-1	-1	1	-1	-1	-1	1	-1	-1
-1	1	-1	-1	-1	-1	-1	1	-1
-1	-1	-1	-1	-1	-1	-1	-1	-1



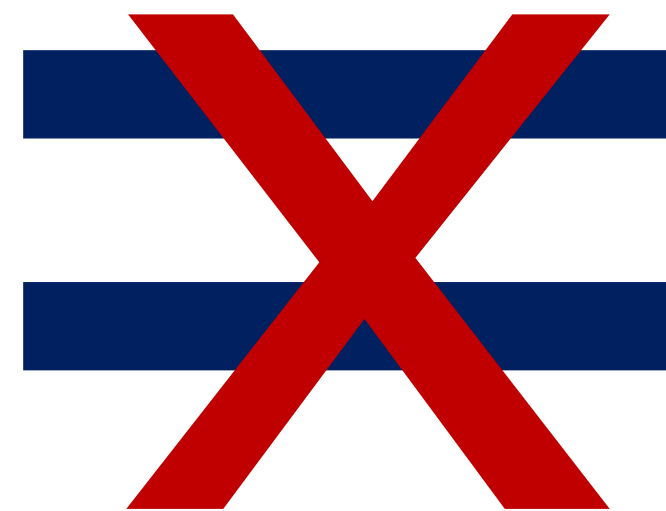
-1	-1	-1	-1	-1	-1	-1	-1	-1
-1	-1	-1	-1	-1	-1	1	-1	-1
-1	1	-1	-1	-1	1	-1	-1	-1
-1	-1	1	1	-1	1	-1	-1	-1
-1	-1	-1	-1	1	-1	-1	-1	-1
-1	-1	-1	1	-1	1	1	-1	-1
-1	-1	-1	1	-1	-1	-1	1	-1
-1	-1	1	-1	-1	-1	-1	-1	-1
-1	-1	-1	-1	-1	-1	-1	-1	-1

What computers see

-1	-1	-1	-1	-1	-1	-1	-1	-1
-1	X	-1	-1	-1	-1	X	X	-1
-1	X	X	-1	-1	X	X	-1	-1
-1	-1	X	1	-1	1	-1	-1	-1
-1	-1	-1	-1	1	-1	-1	-1	-1
-1	-1	-1	1	-1	1	X	-1	-1
-1	-1	X	X	-1	-1	X	X	-1
-1	X	X	-1	-1	-1	-1	X	-1
-1	-1	-1	-1	-1	-1	-1	-1	-1

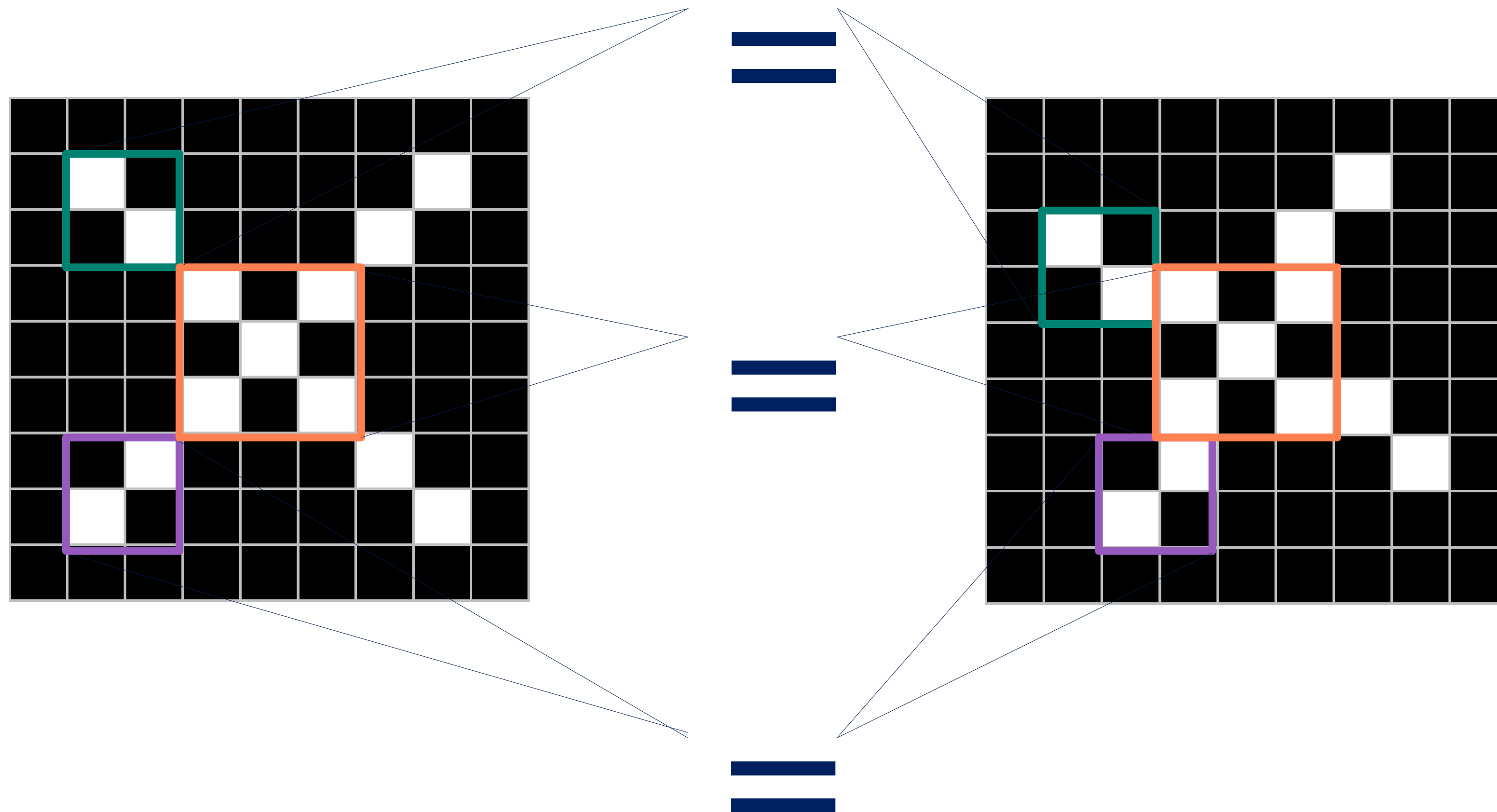
Computers are literal

-1	-1	-1	-1	-1	-1	-1	-1	-1
-1	1	-1	-1	-1	-1	-1	1	-1
-1	-1	1	-1	-1	-1	1	-1	-1
-1	-1	-1	1	-1	1	-1	-1	-1
-1	-1	-1	-1	1	-1	-1	-1	-1
-1	-1	-1	1	-1	1	-1	-1	-1
-1	-1	1	-1	-1	-1	1	-1	-1
-1	1	-1	-1	-1	-1	-1	1	-1
-1	-1	-1	-1	-1	-1	-1	-1	-1



-1	-1	-1	-1	-1	-1	-1	-1	-1
-1	-1	-1	-1	-1	-1	1	-1	-1
-1	1	-1	-1	-1	1	-1	-1	-1
-1	-1	1	1	-1	1	-1	-1	-1
-1	-1	-1	-1	1	-1	-1	-1	-1
-1	-1	-1	1	-1	1	1	-1	-1
-1	-1	-1	1	-1	-1	-1	1	-1
-1	-1	1	-1	-1	-1	-1	-1	-1
-1	-1	-1	-1	-1	-1	-1	-1	-1

ConvNets match pieces of the image

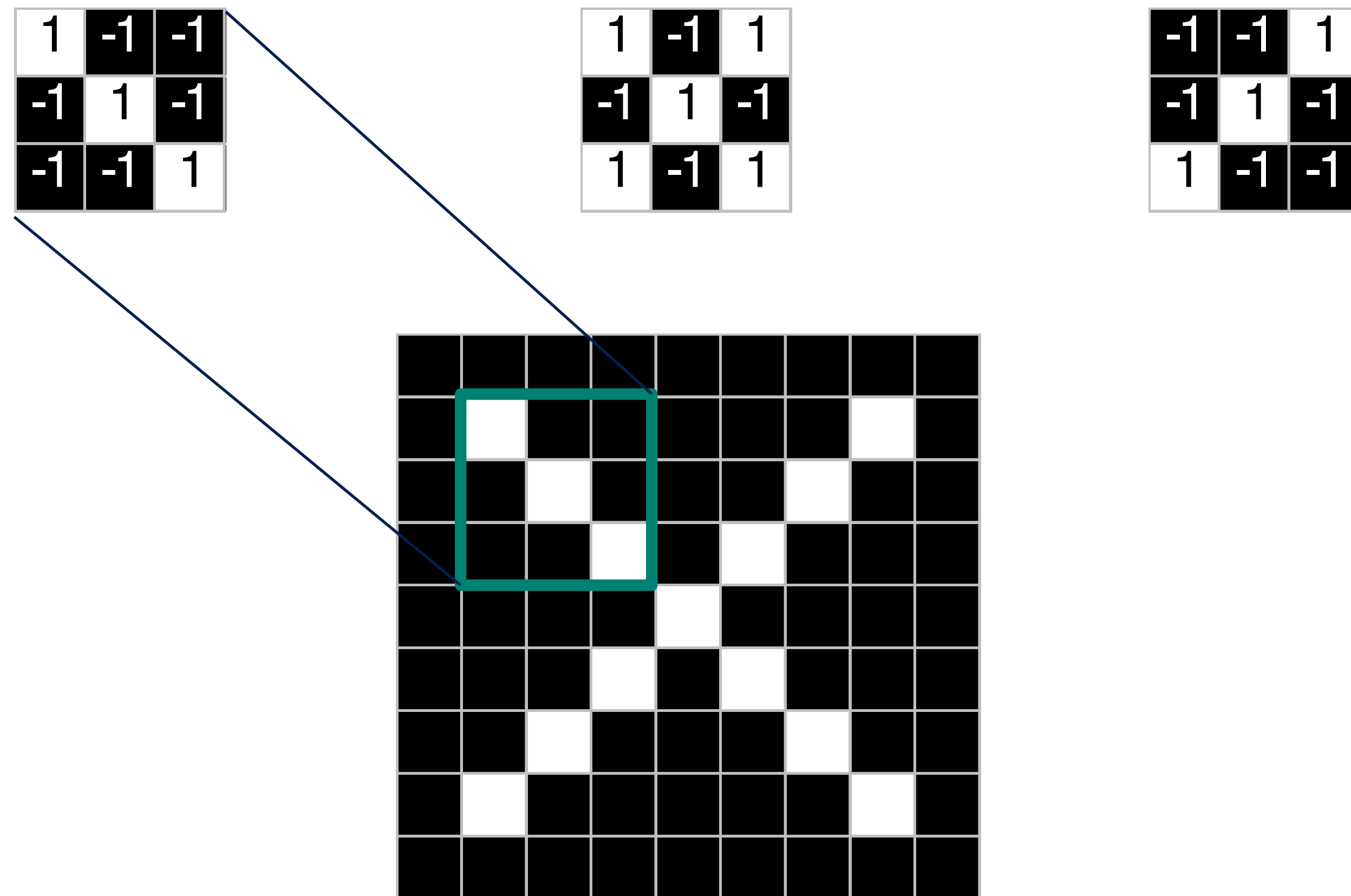


Features match pieces of the image

1	-1	-1
-1	1	-1
-1	-1	1

1	-1	1
-1	1	-1
1	-1	1

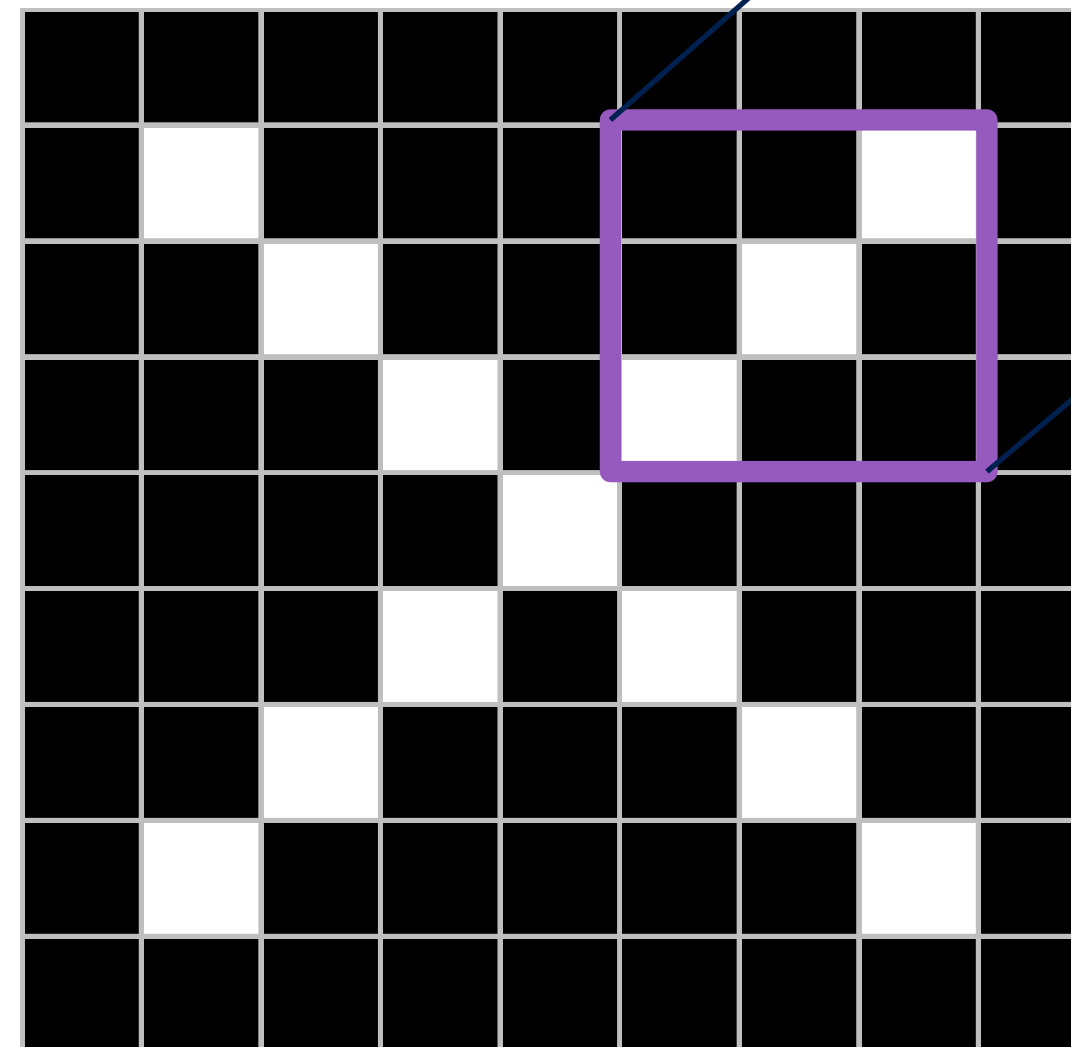
-1	-1	1
-1	1	-1
1	-1	-1



1	-1	-1
-1	1	-1
-1	-1	1

1	-1	1
-1	1	-1
1	-1	1

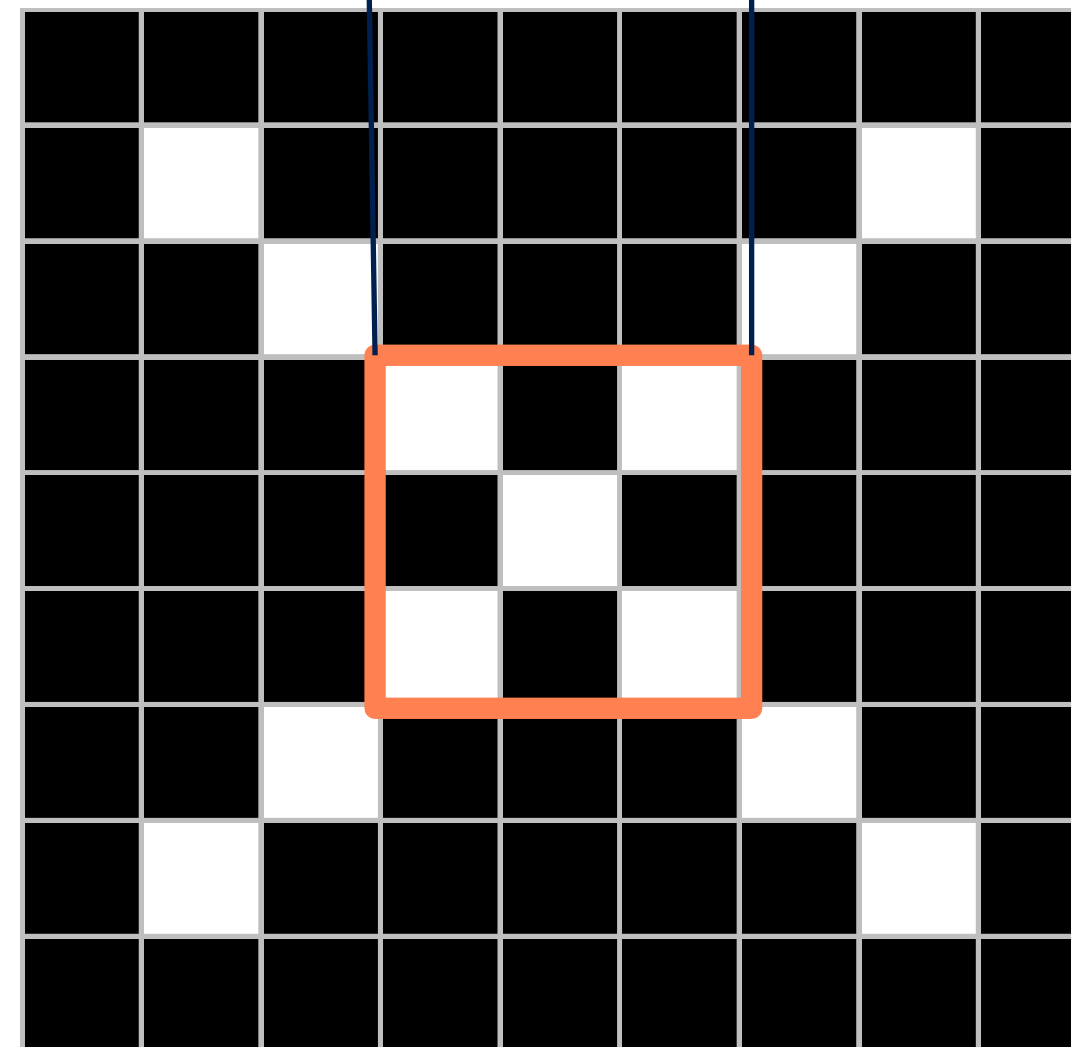
-1	-1	1
-1	1	-1
1	-1	-1

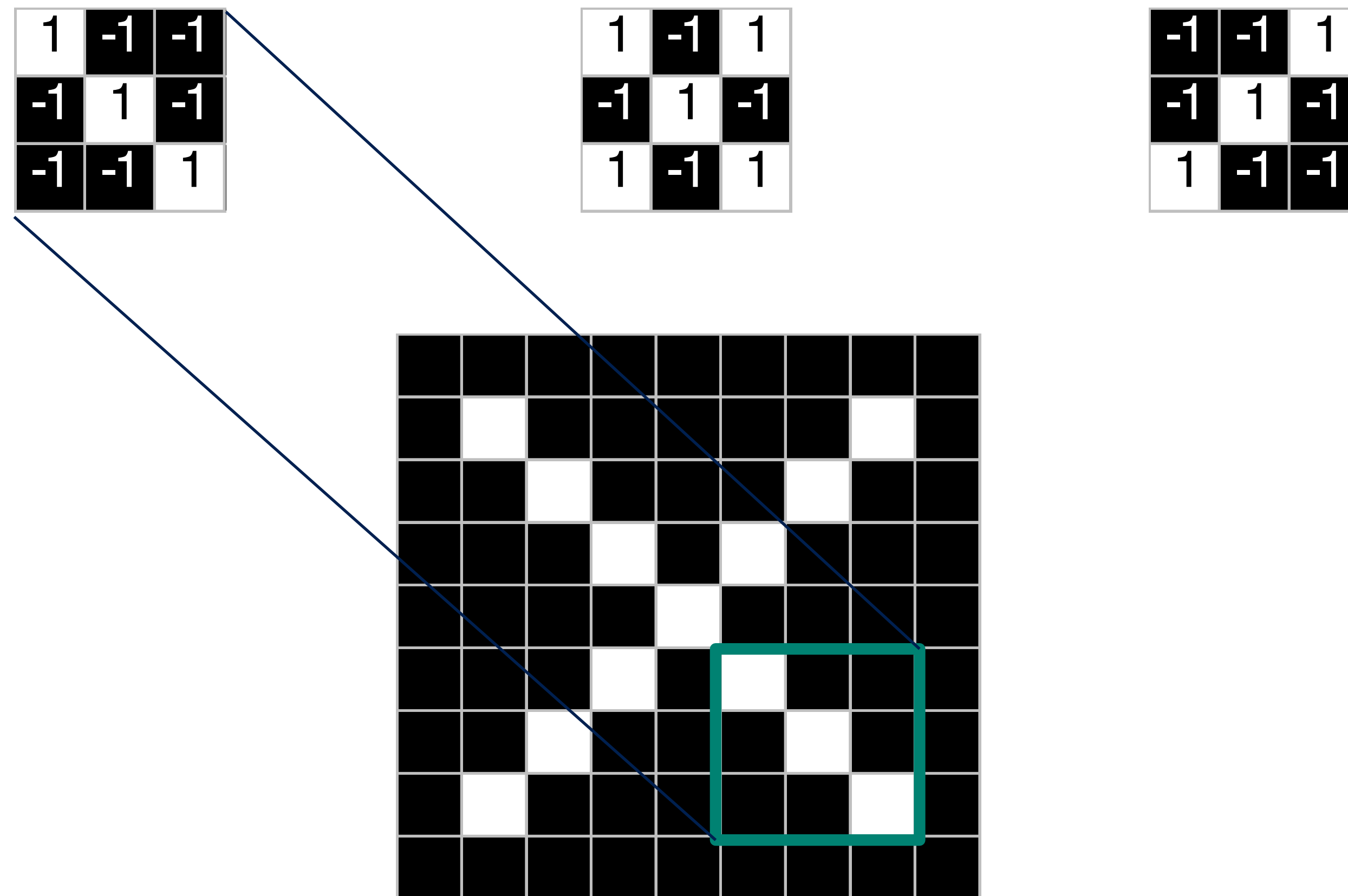


1	-1	-1
-1	1	-1
-1	-1	1

1	-1	1
-1	1	-1
1	-1	1

-1	-1	1
-1	1	-1
1	-1	-1

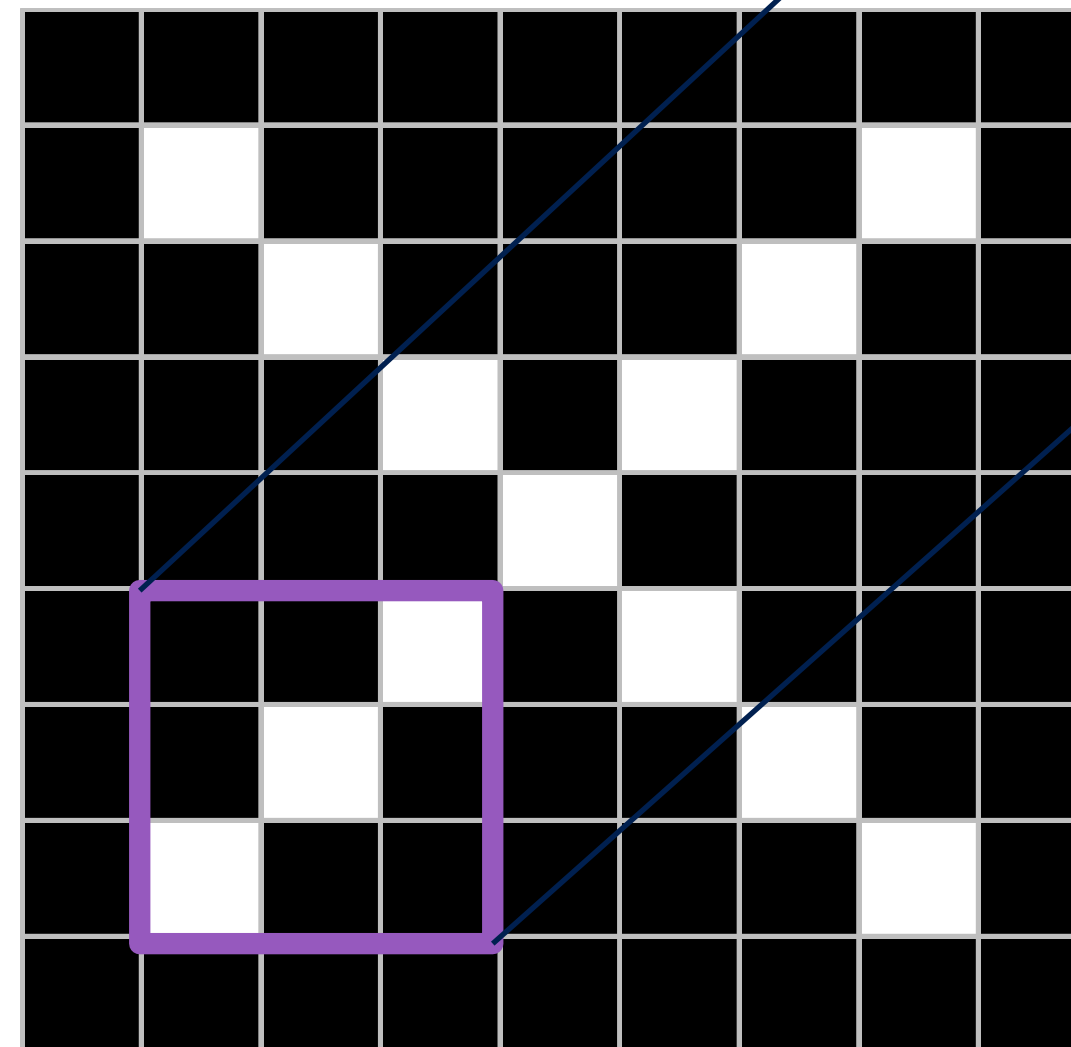




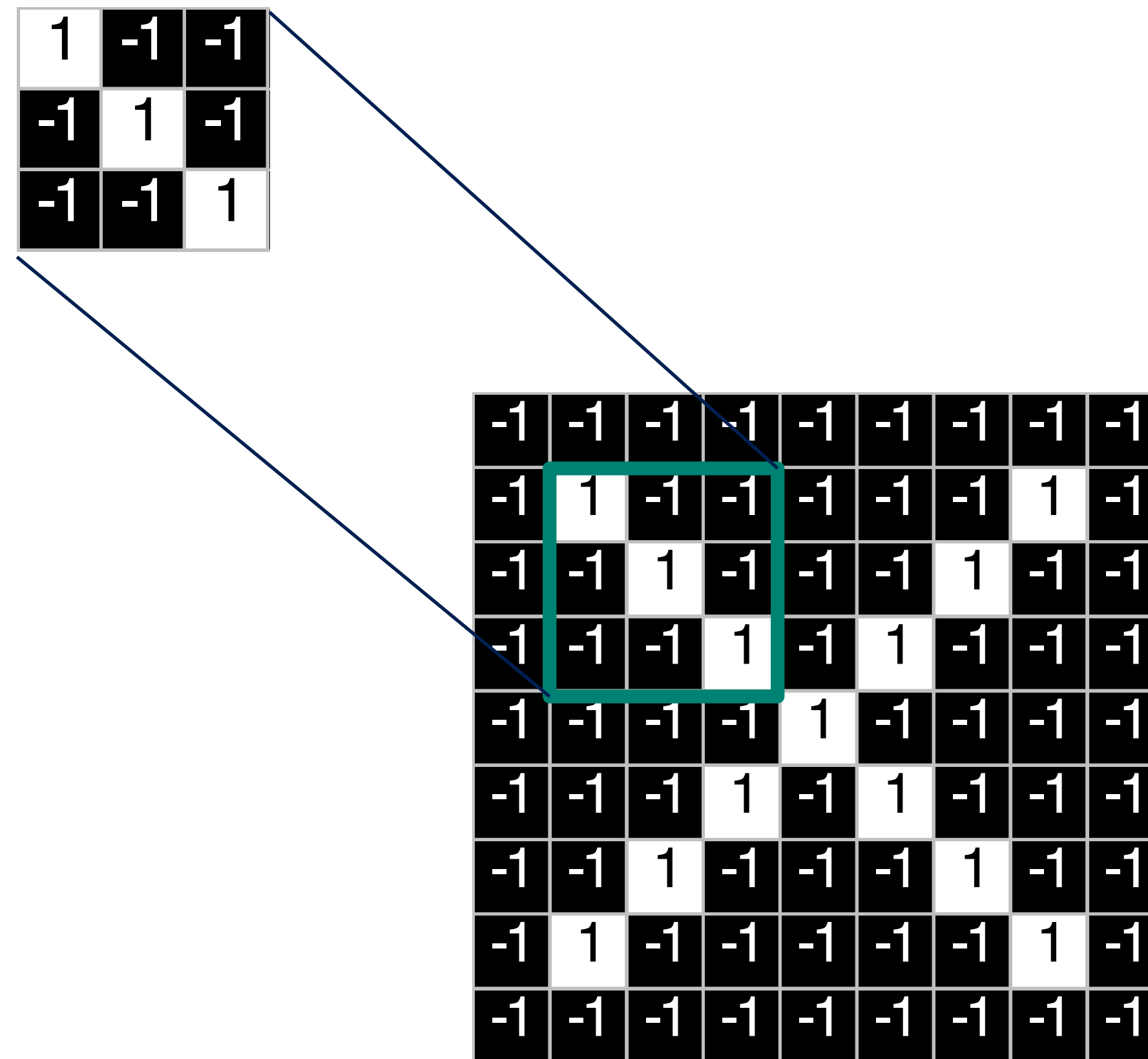
1	-1	-1
-1	1	-1
-1	-1	1

1	-1	1
-1	1	-1
1	-1	1

-1	-1	1
-1	1	-1
1	-1	-1



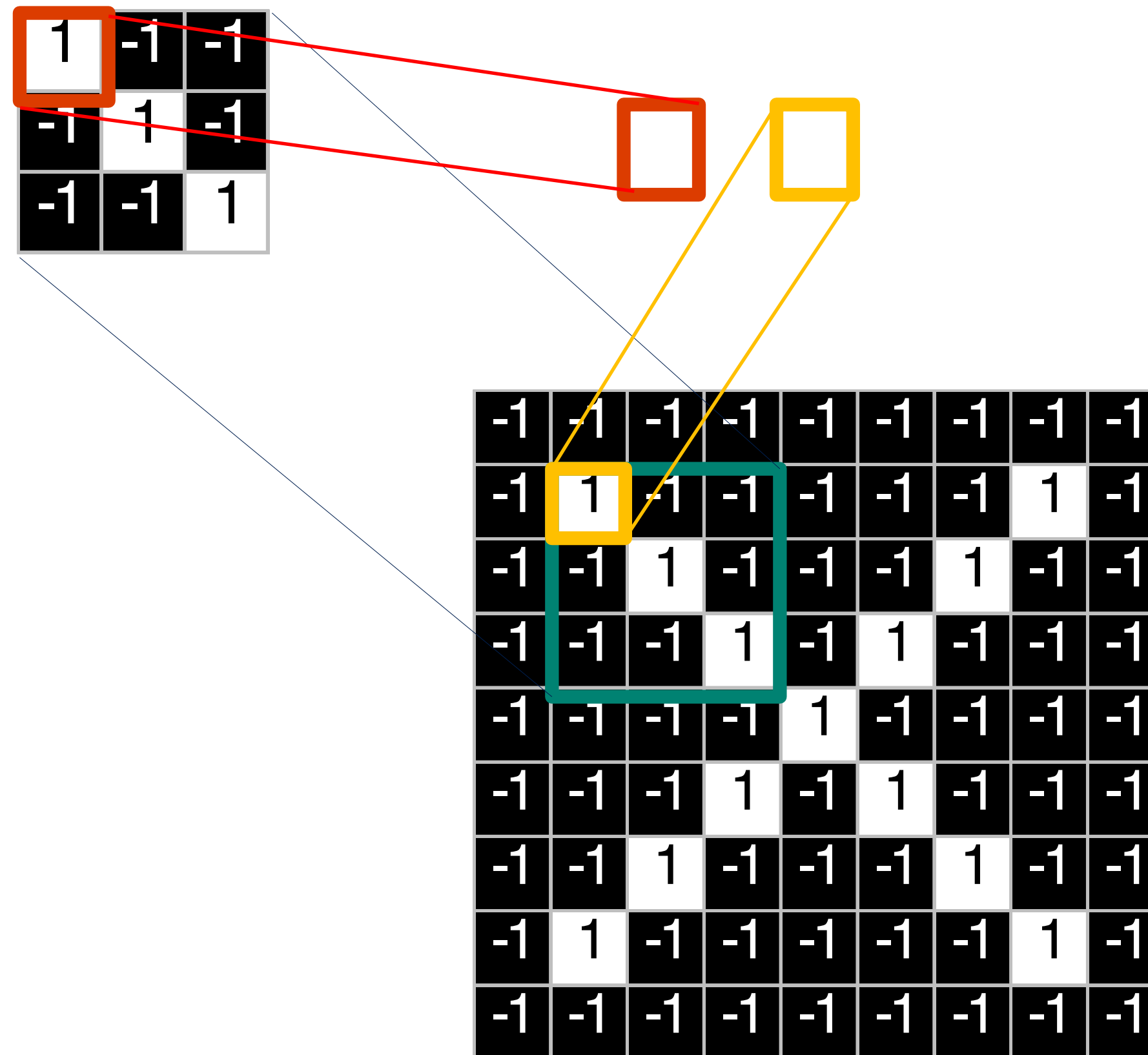
Filtering: The math behind the match



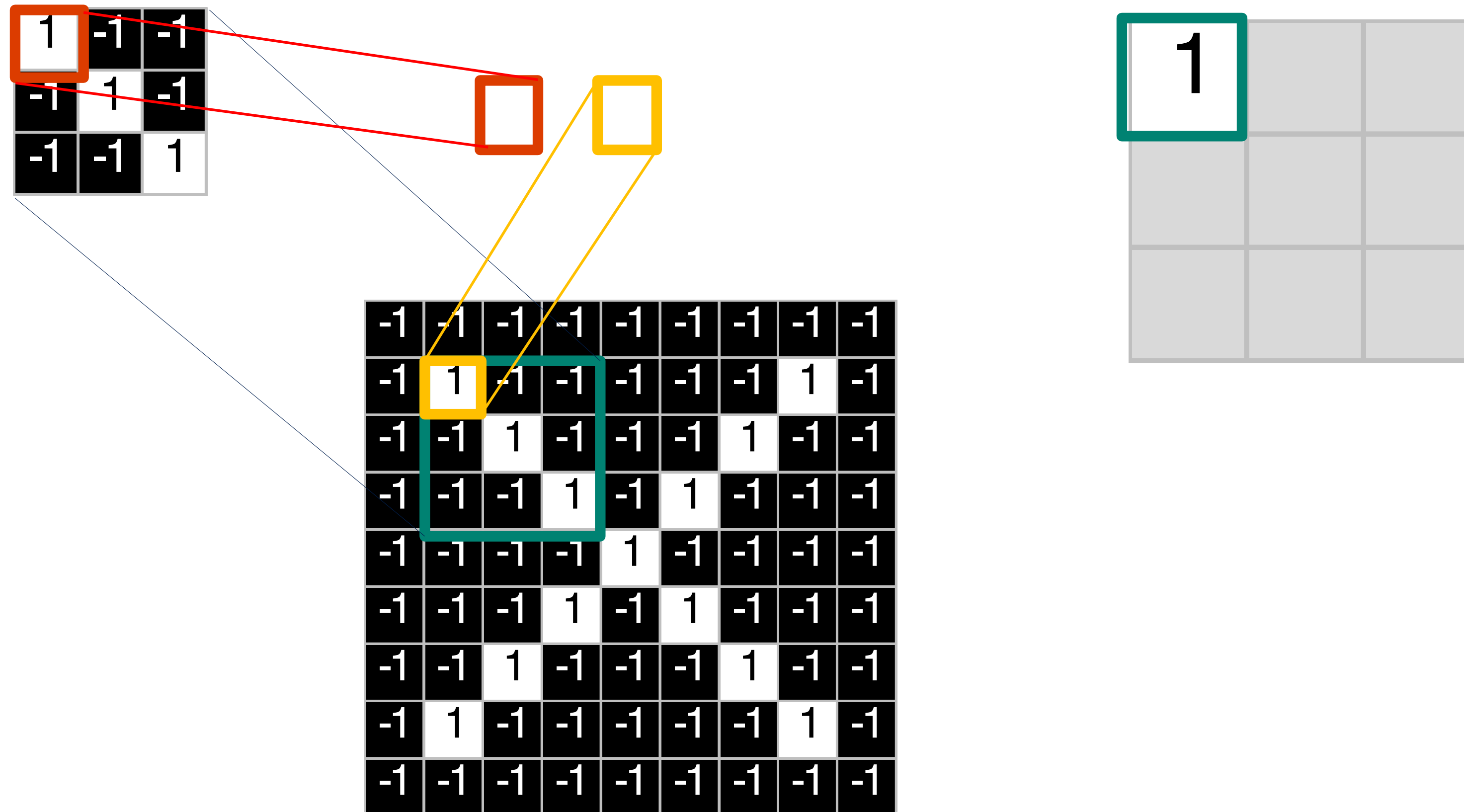
Filtering: The math behind the match

1. Line up the feature and the image patch.
2. Multiply each image pixel by the corresponding feature pixel.
3. Add them up.
4. Divide by the total number of pixels in the feature.

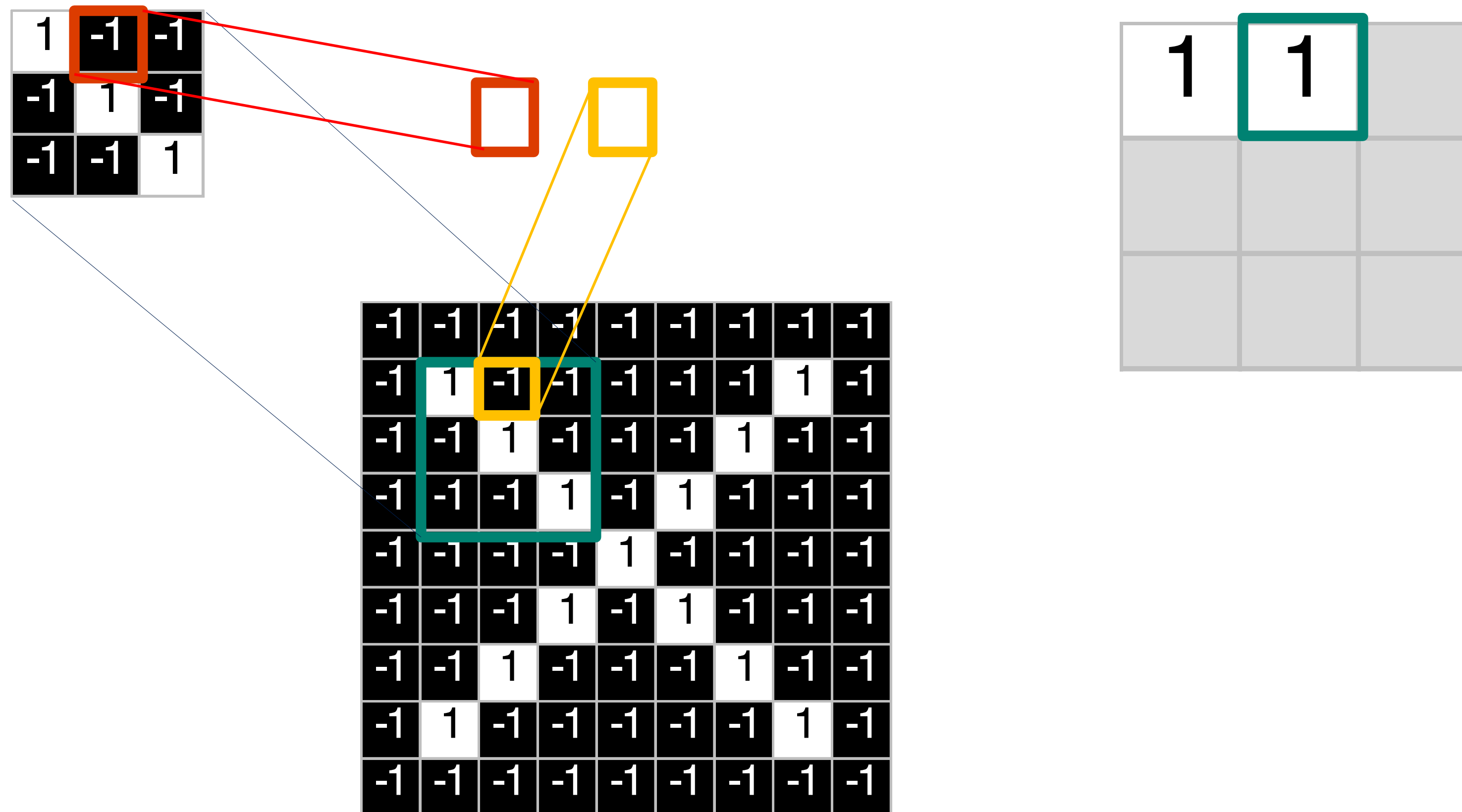
Filtering: The math behind the match



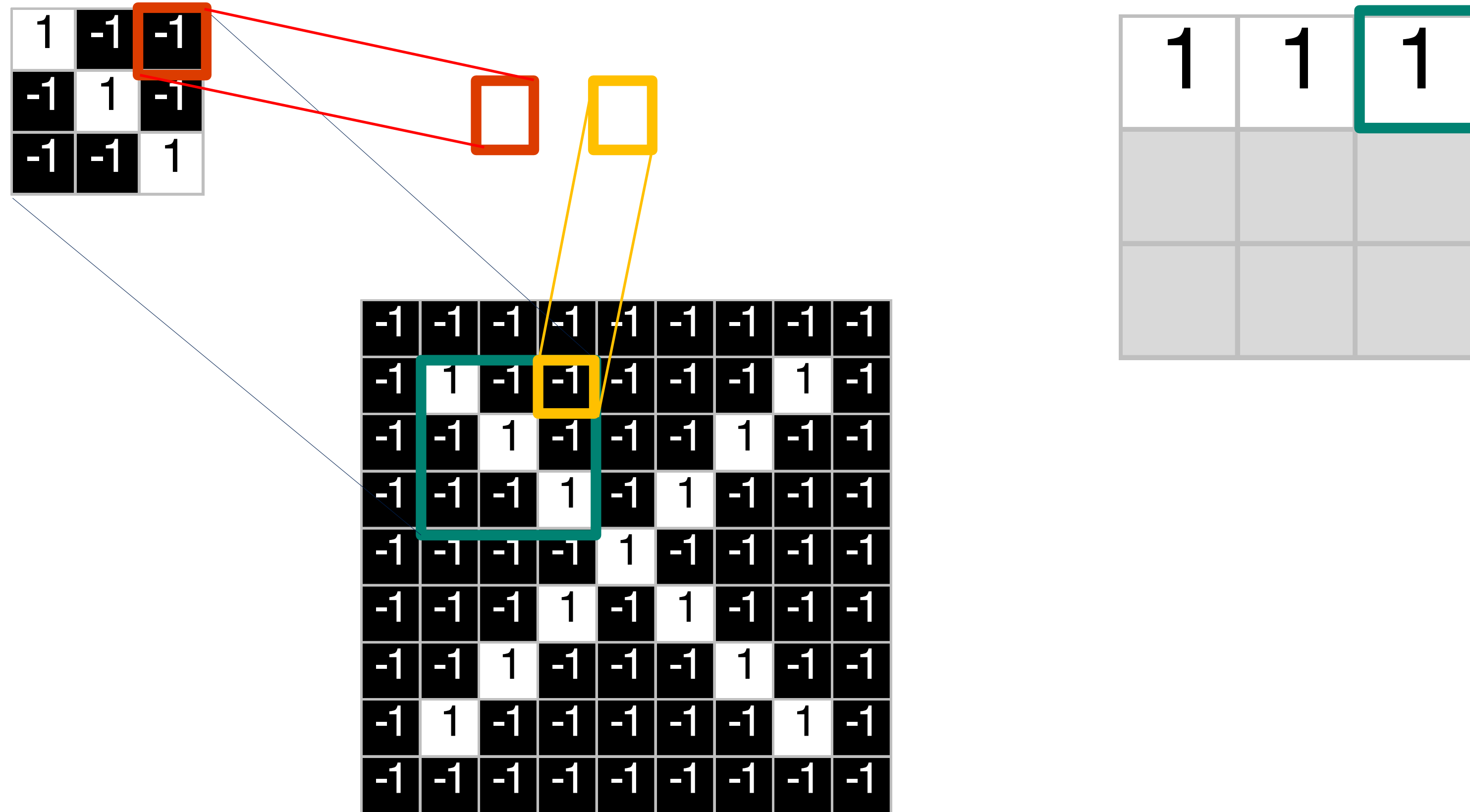
Filtering: The math behind the match



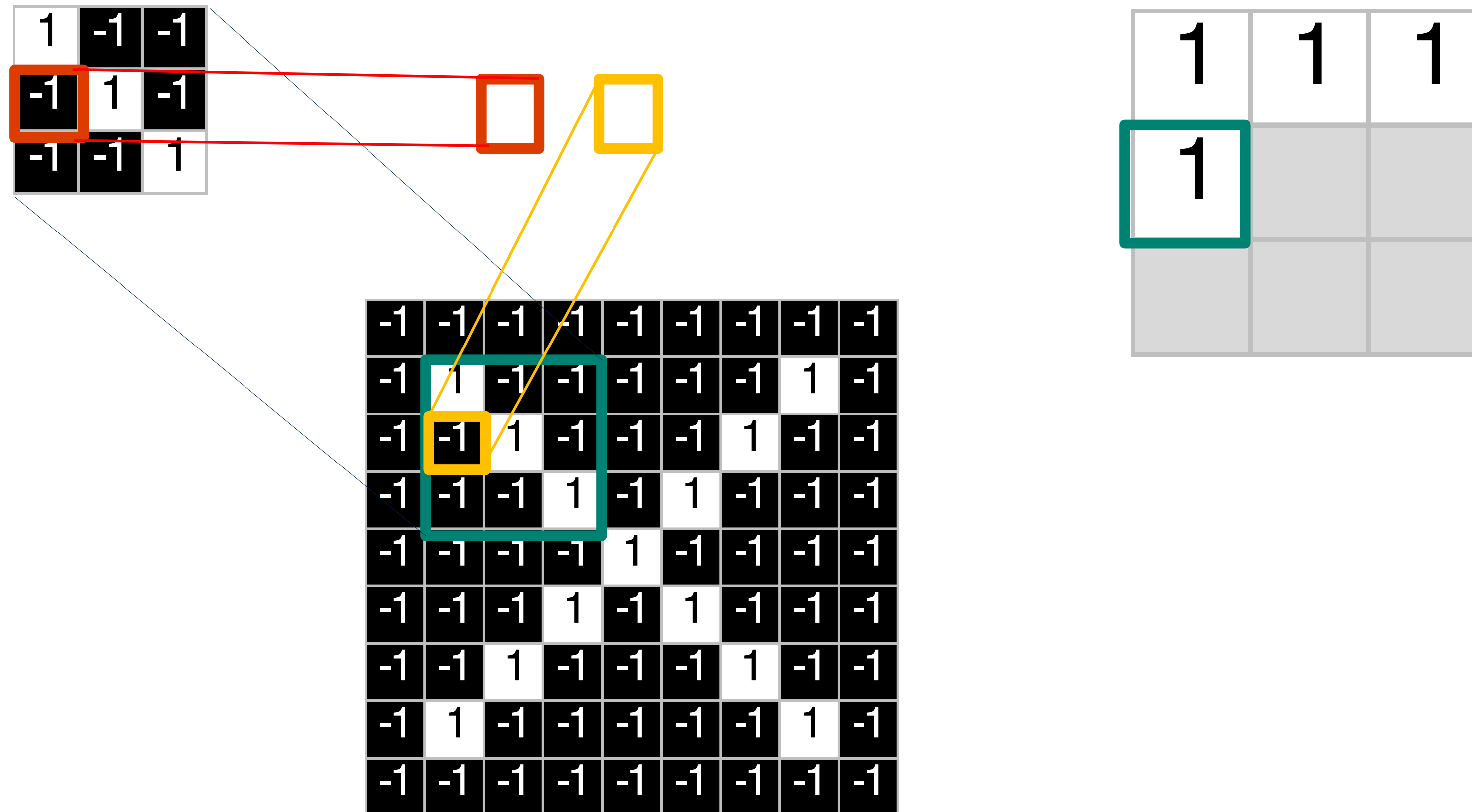
Filtering: The math behind the match



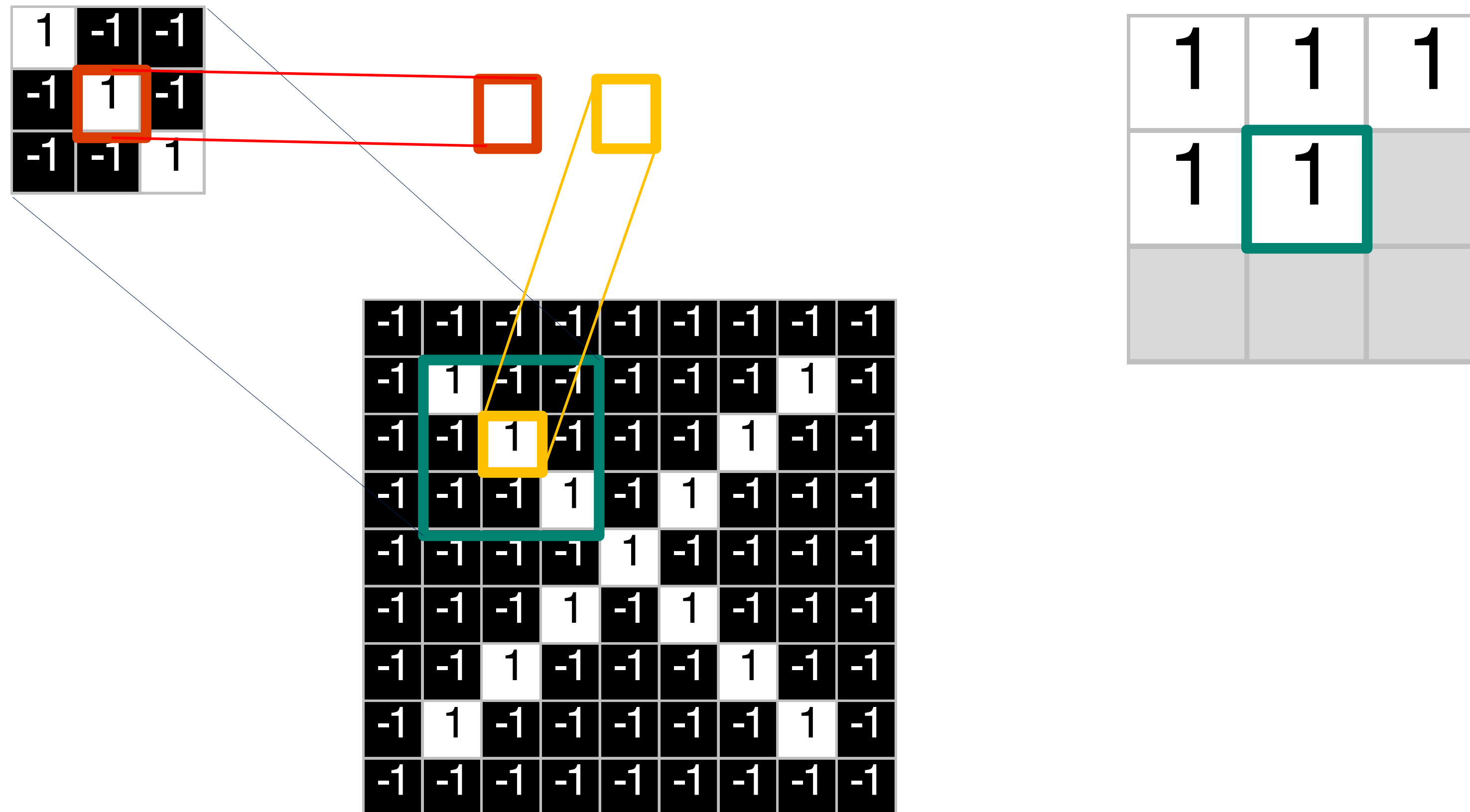
Filtering: The math behind the match



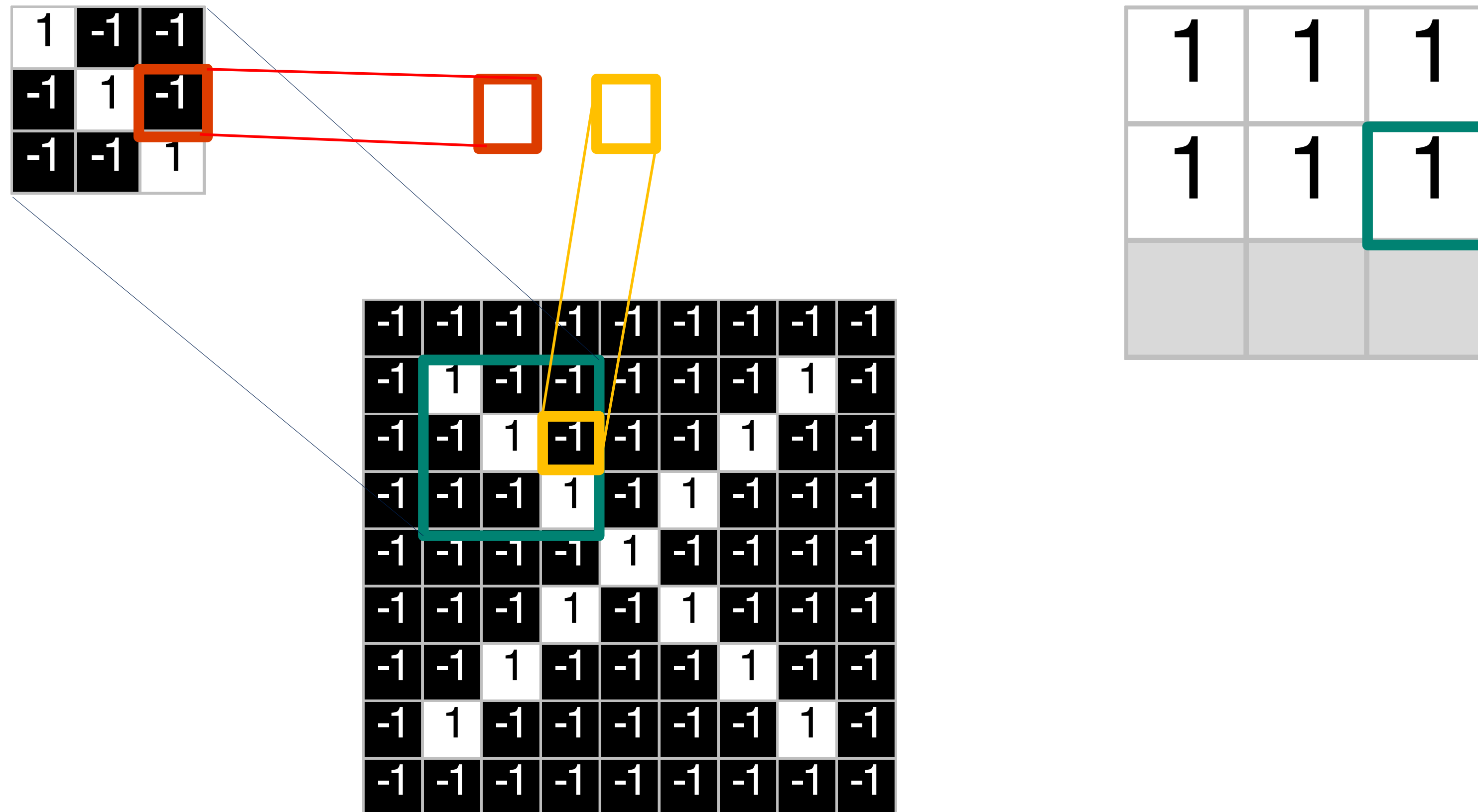
Filtering: The math behind the match



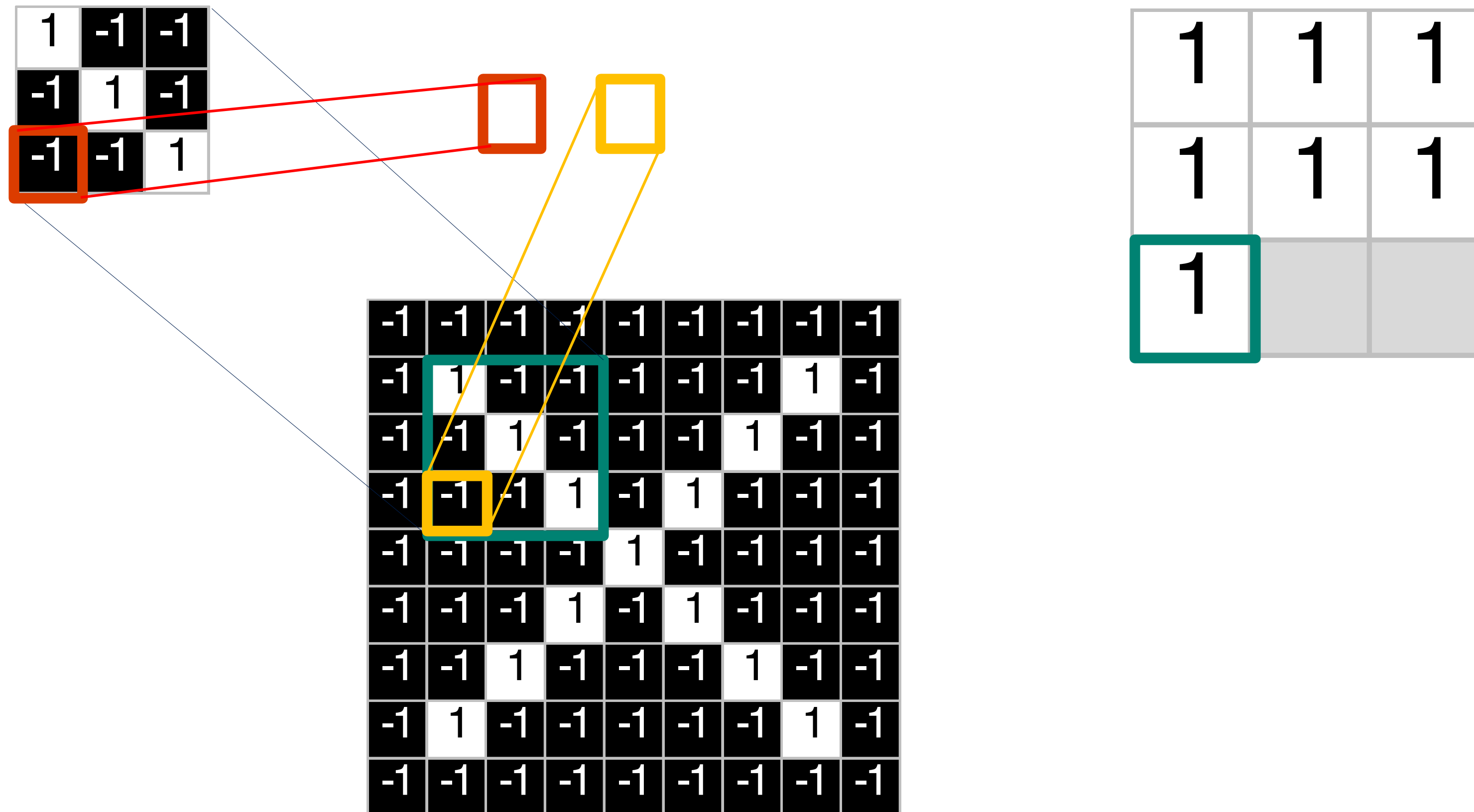
Filtering: The math behind the match



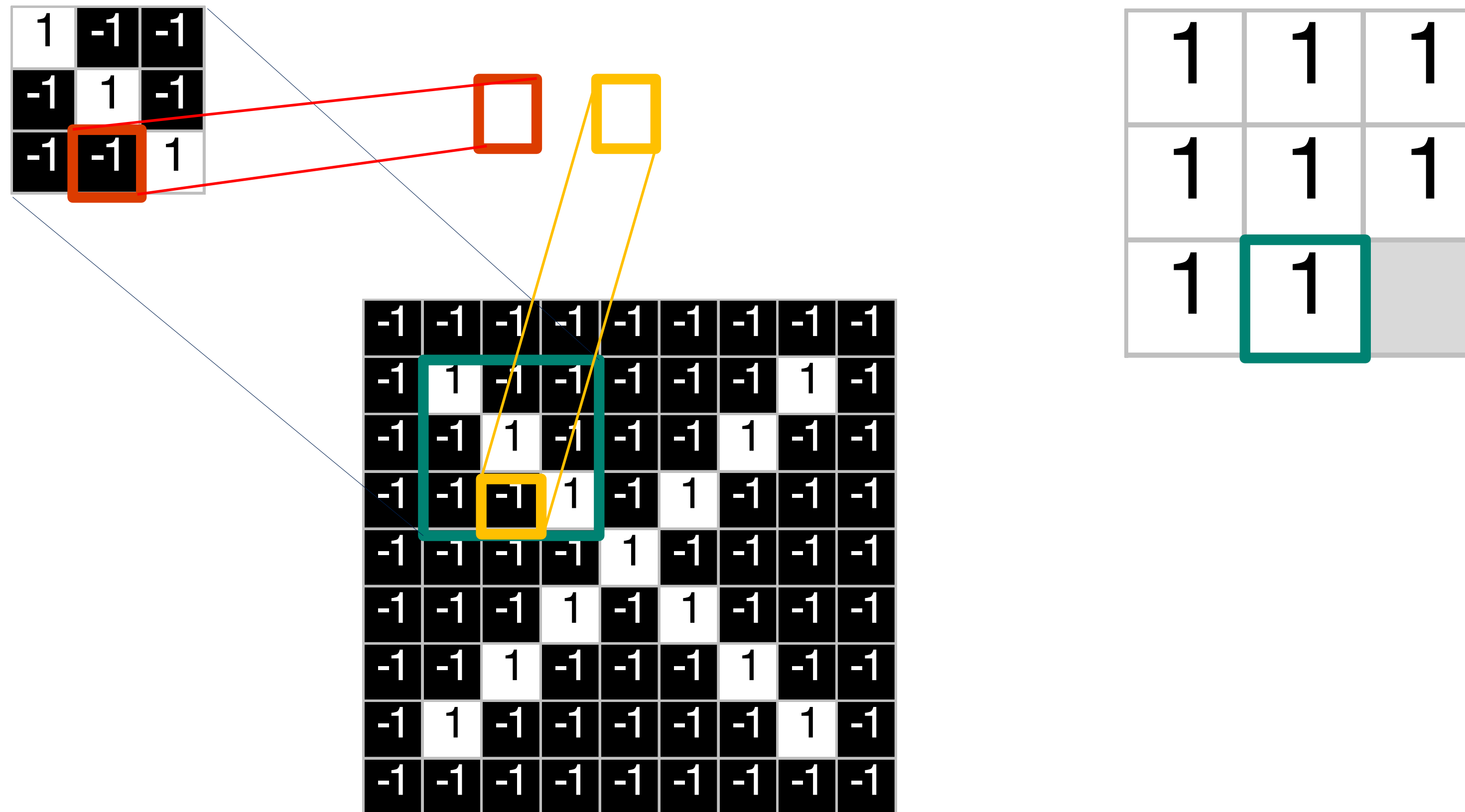
Filtering: The math behind the match



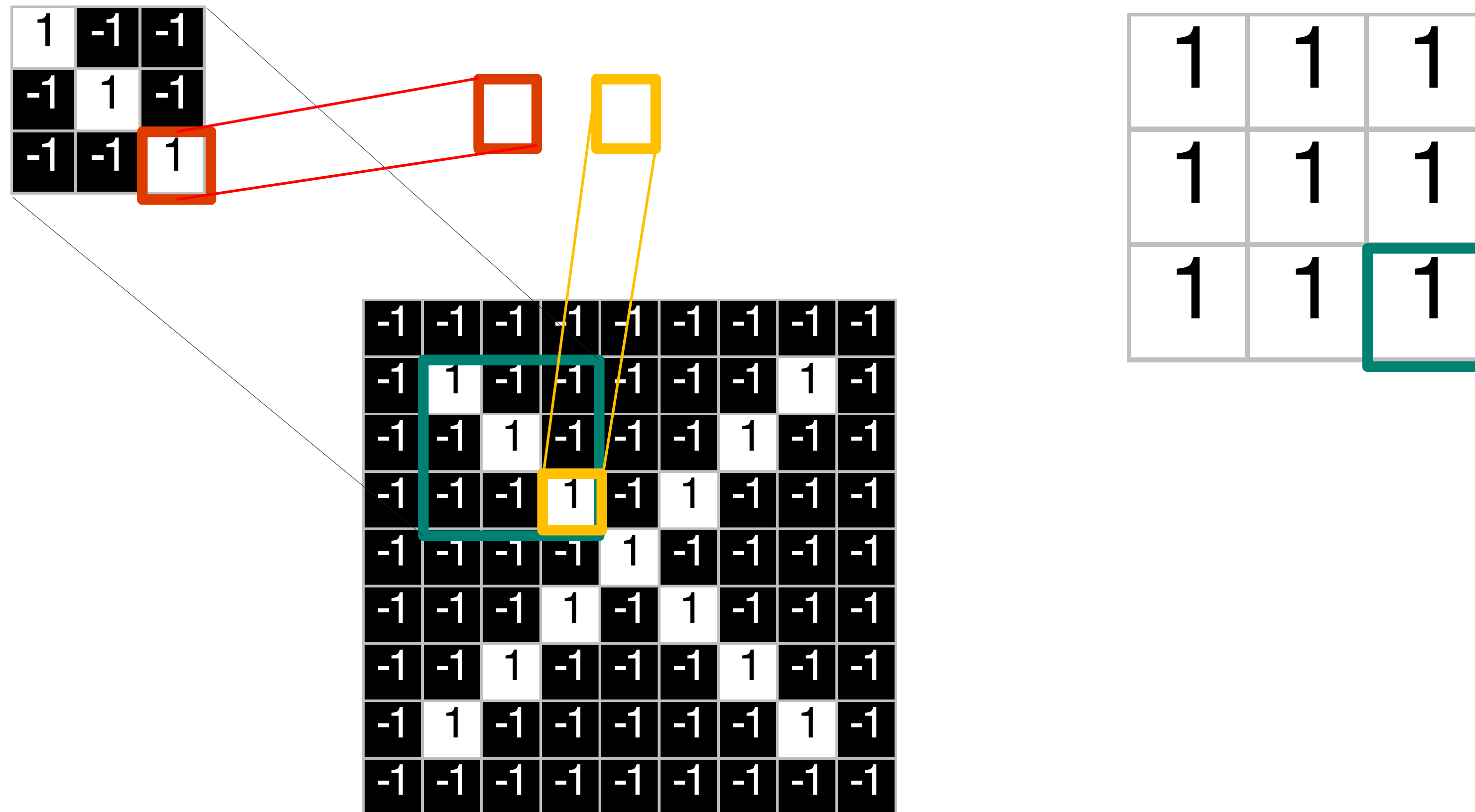
Filtering: The math behind the match



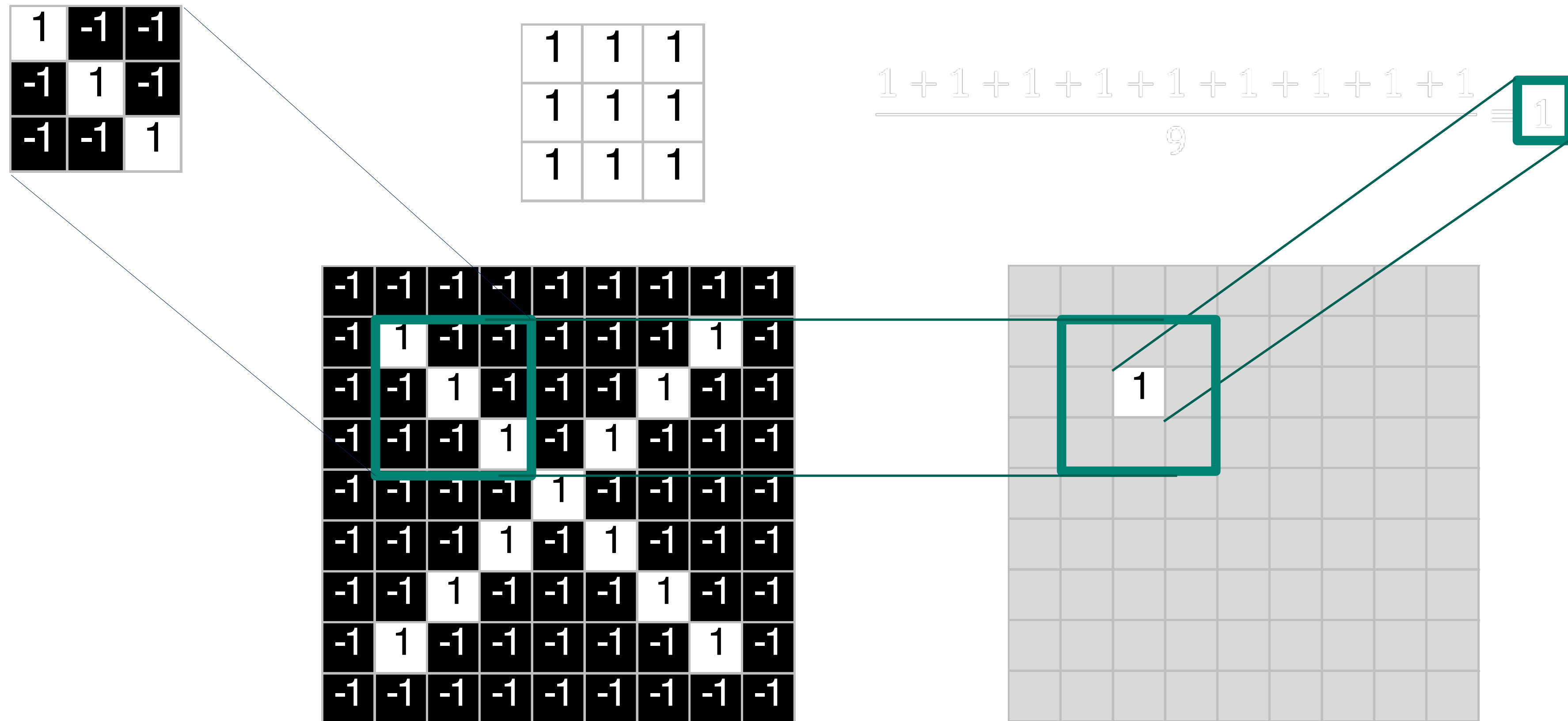
Filtering: The math behind the match



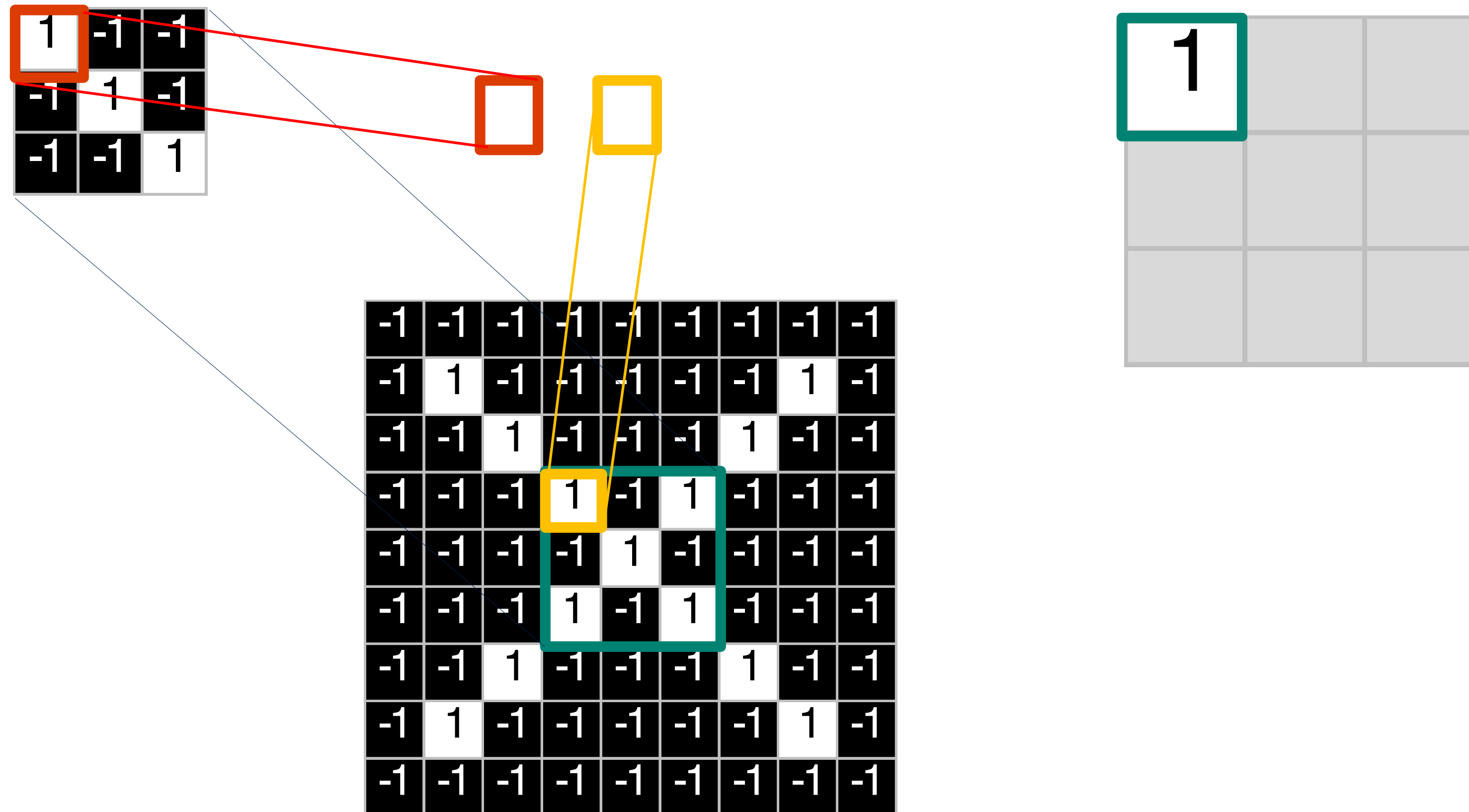
Filtering: The math behind the match



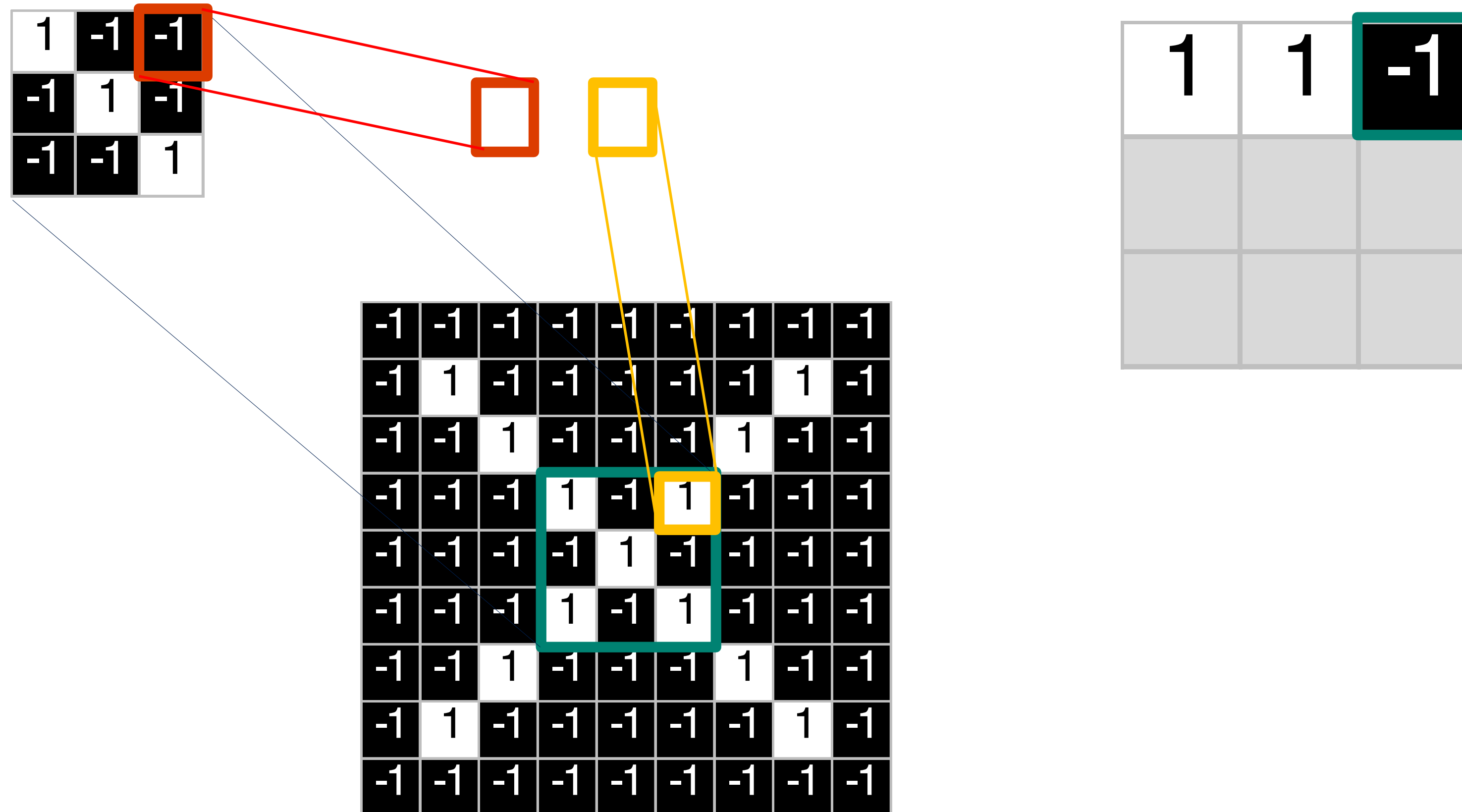
Filtering: The math behind the match



Filtering: The math behind the match



Filtering: The math behind the match



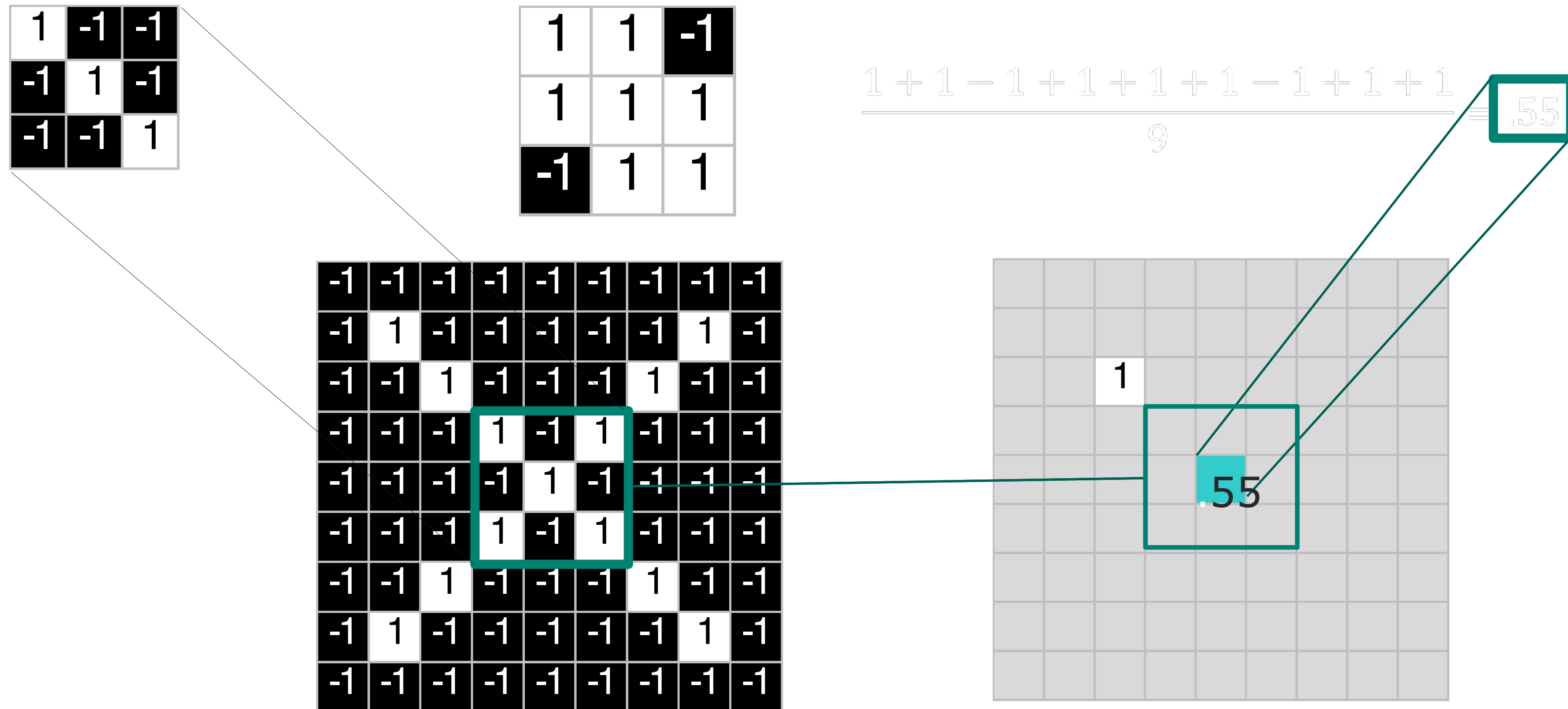
Filtering: The math behind the match

1	-1	-1
-1	1	-1
-1	-1	1

-1	-1	-1	-1	-1	-1	-1	-1	-1
-1	1	-1	-1	-1	-1	-1	1	-1
-1	-1	1	-1	-1	-1	1	-1	-1
-1	-1	-1	1	-1	1	-1	-1	-1
-1	-1	-1	-1	1	-1	-1	-1	-1
-1	-1	-1	1	-1	1	-1	-1	-1
-1	-1	1	-1	-1	-1	1	-1	-1
-1	1	-1	-1	-1	-1	-1	1	-1
-1	-1	-1	-1	-1	-1	-1	-1	-1

1	1	-1
1	1	1
-1	1	1

Filtering: The math behind the match



Convolution: Trying every possible match

1	-1	-1
-1	1	-1
-1	-1	1

-1	-1	-1	-1	-1	-1	-1	-1	-1
-1	1	-1	-1	-1	-1	-1	1	-1
-1	-1	1	-1	-1	-1	1	-1	-1
-1	-1	-1	1	-1	1	-1	-1	-1
-1	-1	-1	-1	1	-1	-1	-1	-1
-1	-1	-1	1	-1	1	-1	-1	-1
-1	-1	1	-1	-1	-1	1	-1	-1
-1	1	-1	-1	-1	-1	-1	1	-1
-1	-1	-1	-1	-1	-1	-1	-1	-1



0.77	-0.11	0.11	0.33	0.55	-0.11	0.33
-0.11	1.00	-0.11	0.33	-0.11	0.11	-0.11
0.11	-0.11	1.00	-0.33	0.11	-0.11	0.55
0.33	0.33	-0.33	0.55	-0.33	0.33	0.33
0.55	-0.11	0.11	-0.33	1.00	-0.11	0.11
-0.11	0.11	-0.11	0.33	-0.11	1.00	-0.11
0.33	-0.11	0.55	0.33	0.11	-0.11	0.77

Convolution: Trying every possible match

-1	-1	-1	-1	-1	-1	-1	-1	-1
-1	1	-1	-1	-1	-1	-1	1	-1
-1	-1	1	-1	-1	-1	1	-1	-1
-1	-1	-1	1	-1	1	-1	-1	-1
-1	-1	-1	-1	1	-1	-1	-1	-1
-1	-1	-1	1	-1	1	-1	-1	-1
-1	-1	1	-1	-1	-1	1	-1	-1
-1	1	-1	-1	-1	-1	-1	1	-1
-1	-1	-1	-1	-1	-1	-1	-1	-1



1	-1	-1
-1	1	-1
-1	-1	1

0.77	-0.11	0.11	0.33	0.55	-0.11	0.33
-0.11	1.00	-0.11	0.33	-0.11	0.11	-0.11
0.11	-0.11	1.00	-0.33	0.11	-0.11	0.55
0.33	0.33	-0.33	0.55	-0.33	0.33	0.33
0.55	-0.11	0.11	-0.33	1.00	-0.11	0.11
-0.11	0.11	-0.11	0.33	-0.11	1.00	-0.11
0.33	-0.11	0.55	0.33	0.11	-0.11	0.77

-1	-1	-1	-1	-1	-1	-1	-1	-1
-1	1	-1	-1	-1	-1	-1	1	-1
-1	-1	1	-1	-1	-1	1	-1	-1
-1	-1	-1	1	-1	1	-1	-1	-1
-1	-1	-1	-1	1	-1	-1	-1	-1
-1	-1	-1	1	-1	1	-1	-1	-1
-1	-1	1	-1	-1	-1	1	-1	-1
-1	1	-1	-1	-1	-1	-1	1	-1
-1	-1	-1	-1	-1	-1	-1	-1	-1



1	-1	-1
-1	1	-1
-1	-1	1



0.77	-0.11	0.11	0.33	0.55	-0.11	0.33
-0.11	1.00	-0.11	0.33	-0.11	0.11	-0.11
0.11	-0.11	1.00	-0.33	0.11	-0.11	0.55
0.33	0.33	-0.33	0.55	-0.33	0.33	0.33
0.55	-0.11	0.11	-0.33	1.00	-0.11	0.11
-0.11	0.11	-0.11	0.33	-0.11	1.00	-0.11
0.33	-0.11	0.55	0.33	0.11	-0.11	0.77

-1	-1	-1	-1	-1	-1	-1	-1	-1
-1	1	-1	-1	-1	-1	-1	1	-1
-1	-1	1	-1	-1	-1	1	-1	-1
-1	-1	-1	1	-1	1	-1	-1	-1
-1	-1	-1	-1	1	-1	-1	-1	-1
-1	-1	-1	1	-1	1	-1	-1	-1
-1	-1	1	-1	-1	-1	1	-1	-1
-1	1	-1	-1	-1	-1	-1	1	-1
-1	-1	-1	-1	-1	-1	-1	-1	-1



1	-1	1
-1	1	-1
1	-1	1



0.33	-0.55	0.11	-0.11	0.11	-0.55	0.33
-0.55	0.55	-0.55	0.33	-0.55	0.55	-0.55
0.11	-0.55	0.55	-0.77	0.55	-0.55	0.11
-0.11	0.33	-0.77	1.00	-0.77	0.33	-0.11
0.11	-0.55	0.55	-0.77	0.55	-0.55	0.11
-0.55	0.55	-0.55	0.33	-0.55	0.55	-0.55
0.33	-0.55	0.11	-0.11	0.11	-0.55	0.33

-1	-1	-1	-1	-1	-1	-1	-1	-1
-1	1	-1	-1	-1	-1	-1	1	-1
-1	-1	1	-1	-1	-1	1	-1	-1
-1	-1	-1	1	-1	1	-1	-1	-1
-1	-1	-1	-1	1	-1	-1	-1	-1
-1	-1	-1	1	-1	1	-1	-1	-1
-1	-1	1	-1	-1	-1	1	-1	-1
-1	1	-1	-1	-1	-1	-1	1	-1
-1	-1	-1	-1	-1	-1	-1	-1	-1



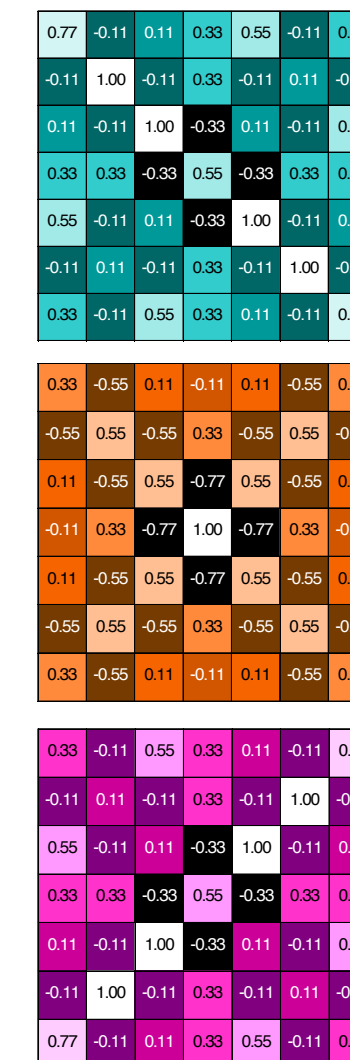
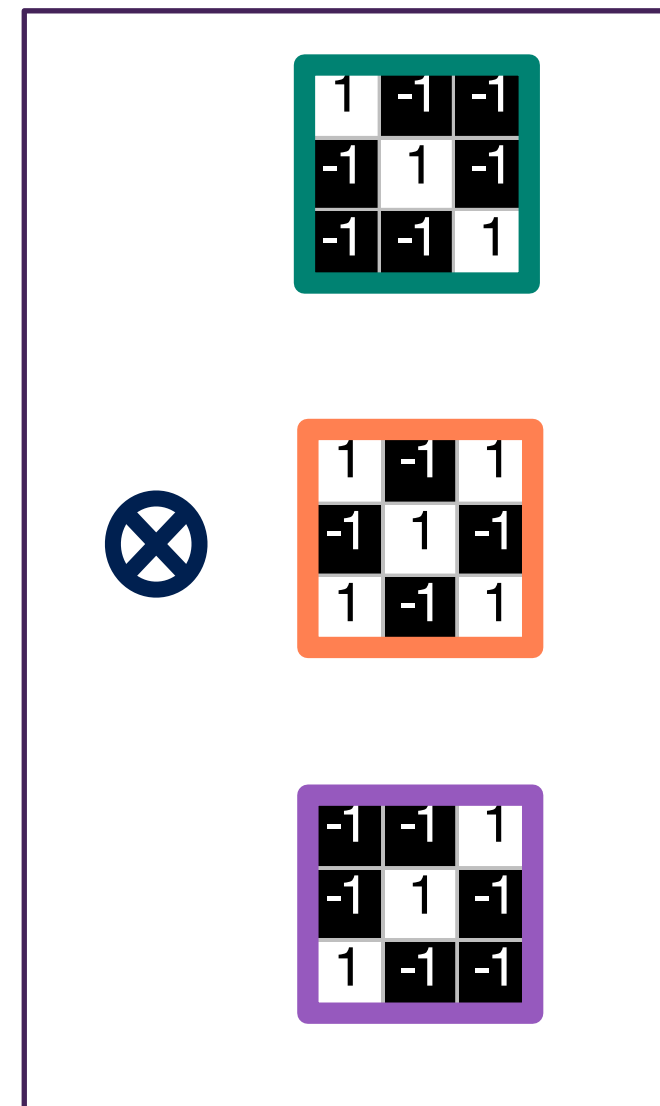
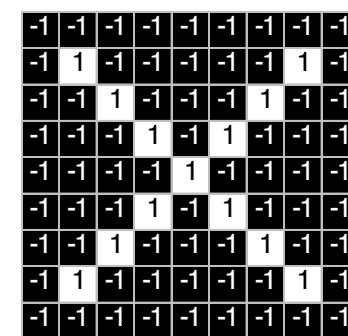
-1	-1	1
-1	1	-1
1	-1	-1



0.33	-0.11	0.55	0.33	0.11	-0.11	0.77
-0.11	0.11	-0.11	0.33	-0.11	1.00	-0.11
0.55	-0.11	0.11	-0.33	1.00	-0.11	0.11
0.33	0.33	-0.33	0.55	-0.33	0.33	0.33
0.11	-0.11	1.00	-0.33	0.11	-0.11	0.55
-0.11	1.00	-0.11	0.33	-0.11	0.11	-0.11
0.77	-0.11	0.11	0.33	0.55	-0.11	0.33

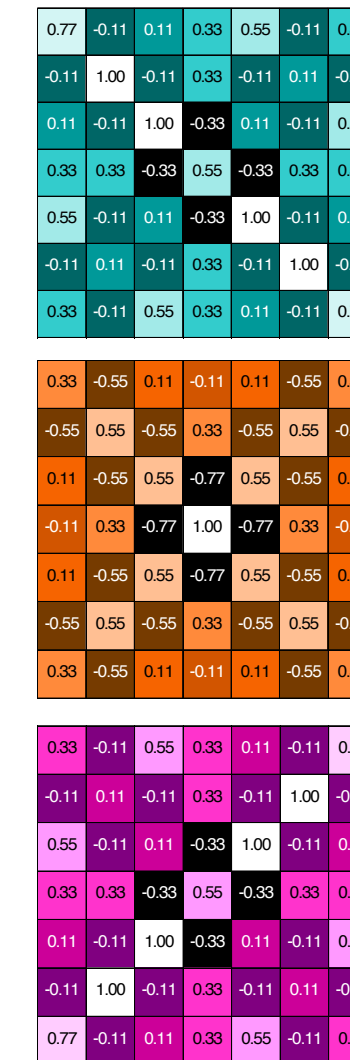
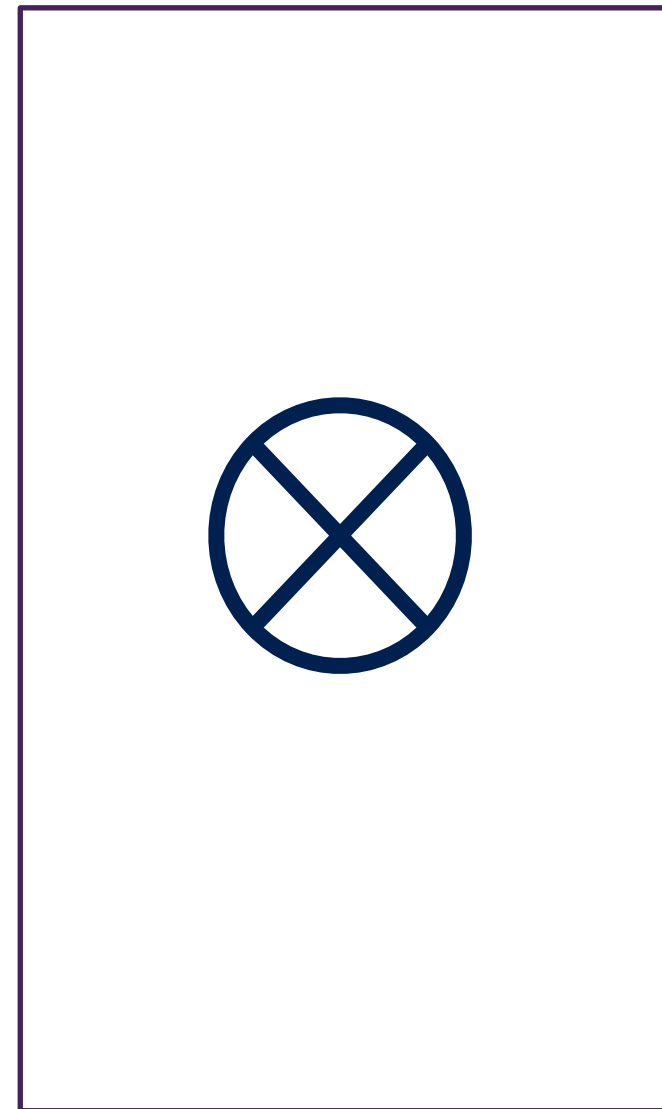
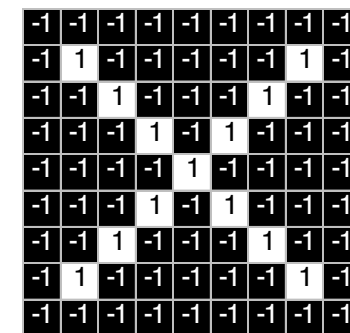
Convolution layer

One image becomes a stack of filtered images



Convolution layer

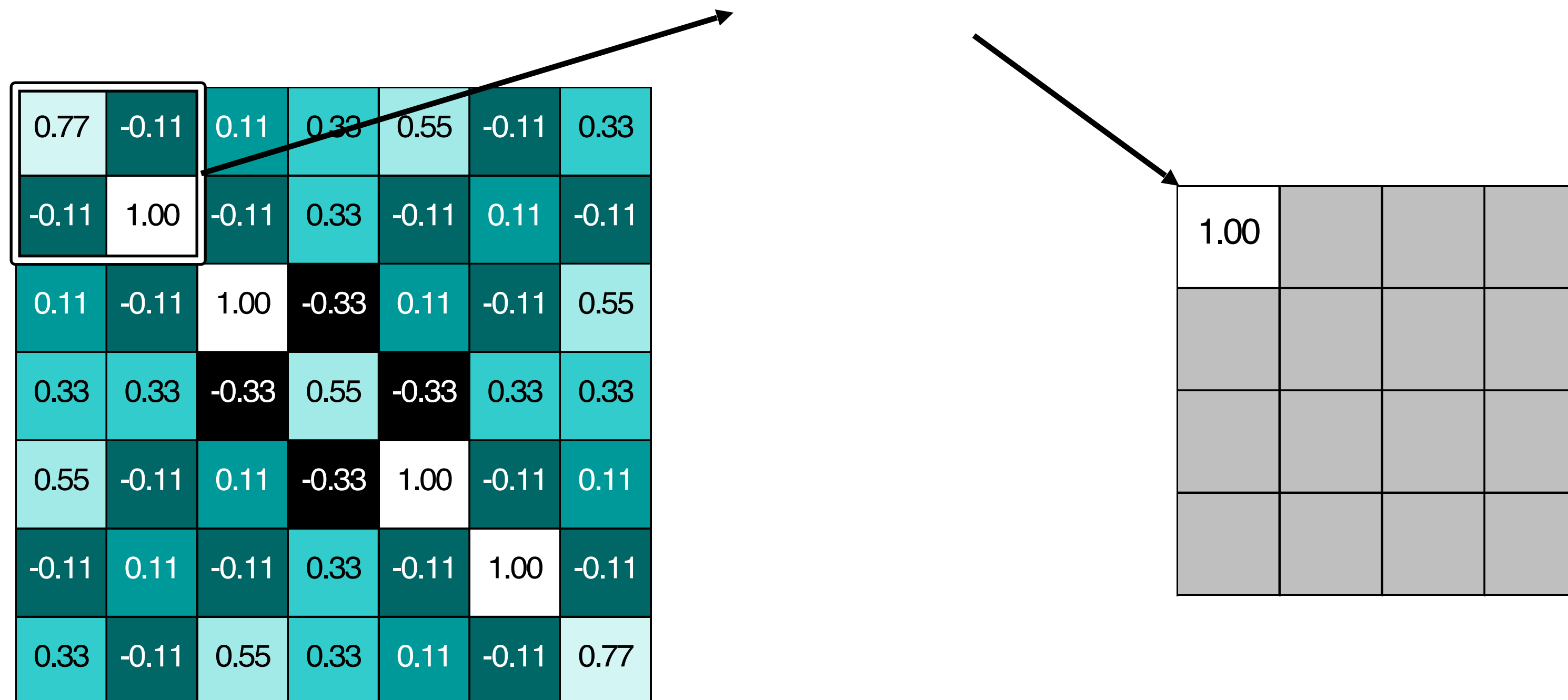
One image becomes a stack of filtered images



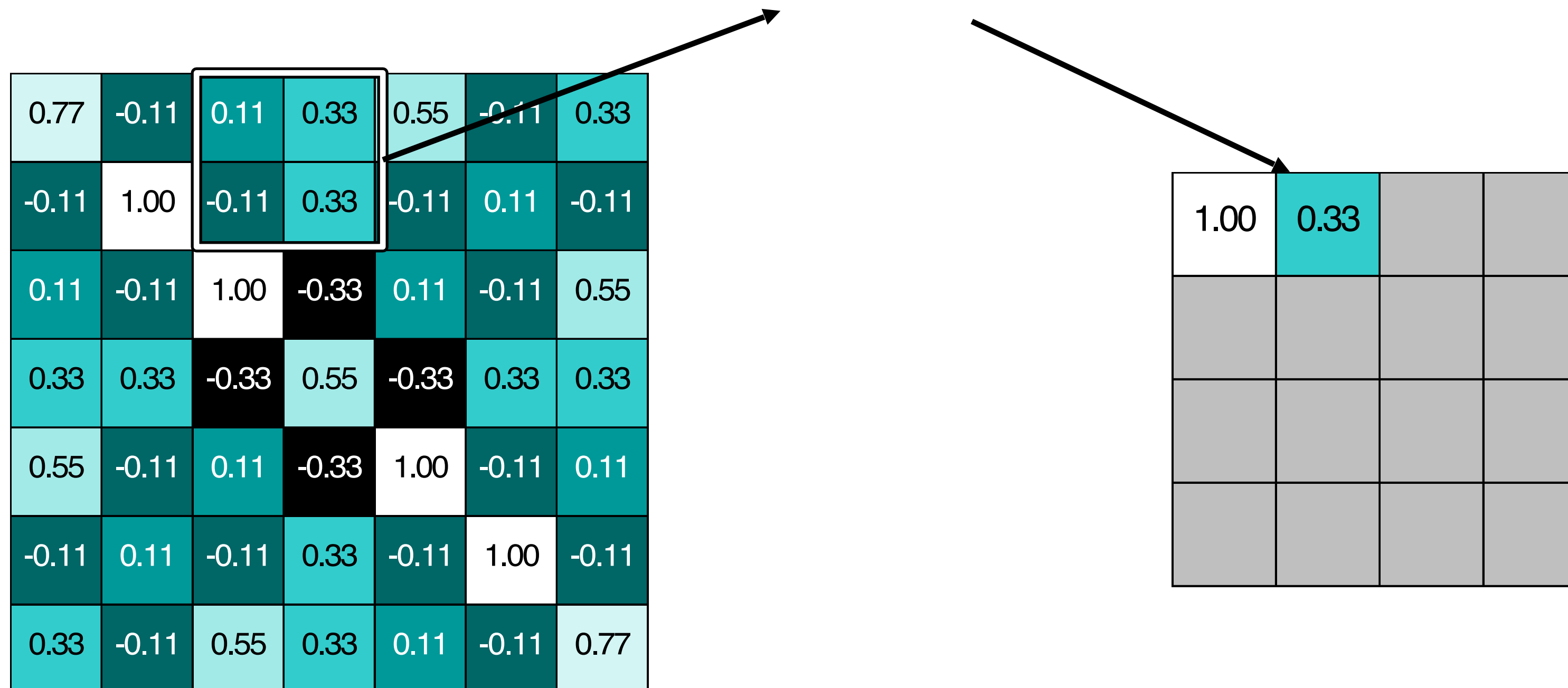
Pooling: Shrinking the image stack

1. Pick a window size (usually 2 or 3).
2. Pick a stride (usually 2).
3. Walk your window across your filtered images.
4. From each window, take the maximum value.

Pooling



Pooling



Pooling

0.77	-0.11	0.11	0.33	0.55	-0.11	0.33
-0.11	1.00	-0.11	0.33	-0.11	0.11	-0.11
0.11	-0.11	1.00	-0.33	0.11	-0.11	0.55
0.33	0.33	-0.33	0.55	-0.33	0.33	0.33
0.55	-0.11	0.11	-0.33	1.00	-0.11	0.11
-0.11	0.11	-0.11	0.33	-0.11	1.00	-0.11
0.33	-0.11	0.55	0.33	0.11	-0.11	0.77

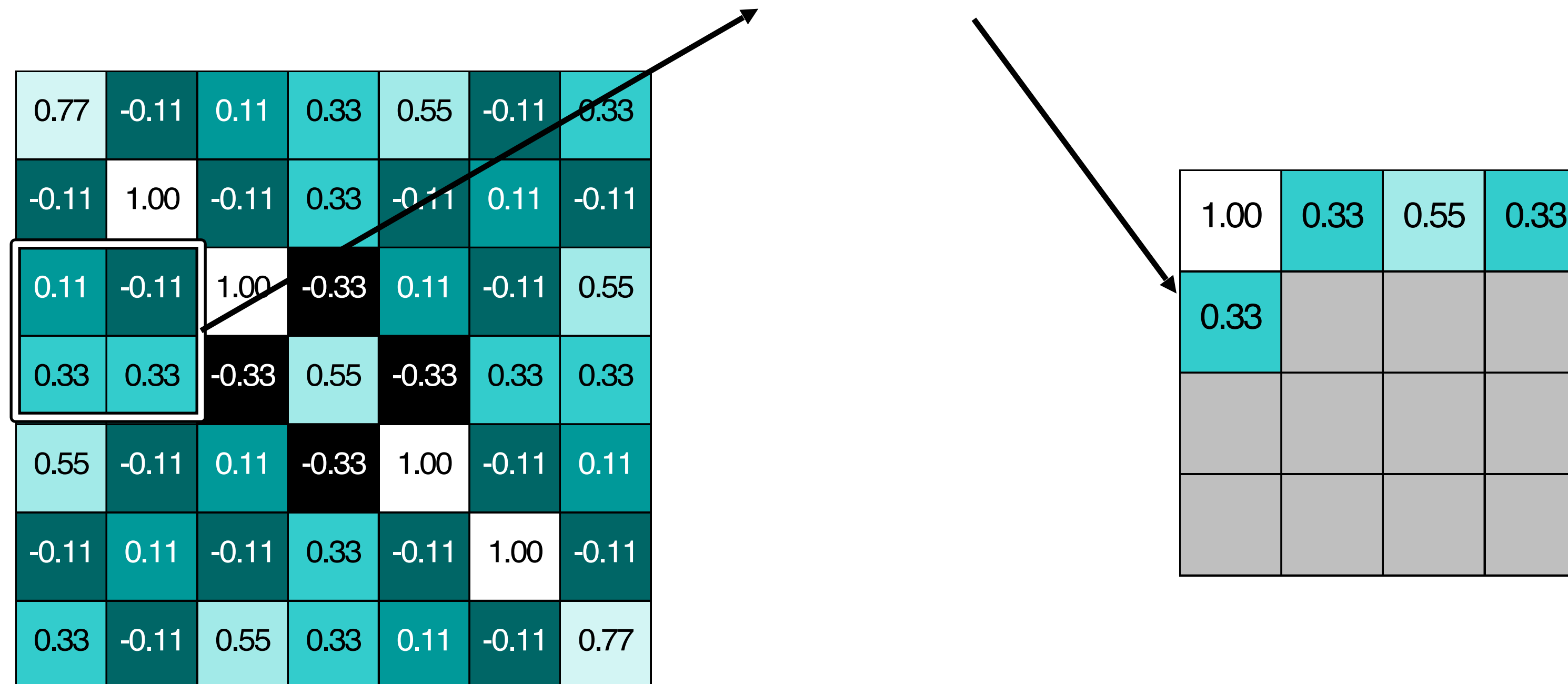
1.00	0.33	0.55	

Pooling

0.77	-0.11	0.11	0.33	0.55	-0.11	0.33	
-0.11	1.00	-0.11	0.33	-0.11	0.11	-0.11	
0.11	-0.11	1.00	-0.33	0.11	-0.11	0.55	
0.33	0.33	-0.33	0.55	-0.33	0.33	0.33	
0.55	-0.11	0.11	-0.33	1.00	-0.11	0.11	
-0.11	0.11	-0.11	0.33	-0.11	1.00	-0.11	
0.33	-0.11	0.55	0.33	0.11	-0.11	0.77	

1.00	0.33	0.55	0.33

Pooling



Pooling

0.77	-0.11	0.11	0.33	0.55	-0.11	0.33
-0.11	1.00	-0.11	0.33	-0.11	0.11	-0.11
0.11	-0.11	1.00	-0.33	0.11	-0.11	0.55
0.33	0.33	-0.33	0.55	-0.33	0.33	0.33
0.55	-0.11	0.11	-0.33	1.00	-0.11	0.11
-0.11	0.11	-0.11	0.33	-0.11	1.00	-0.11
0.33	-0.11	0.55	0.33	0.11	-0.11	0.77



1.00	0.33	0.55	0.33
0.33	1.00	0.33	0.55
0.55	0.33	1.00	0.11
0.33	0.55	0.11	0.77

0.77	-0.11	0.11	0.33	0.55	-0.11	0.33
-0.11	1.00	-0.11	0.33	-0.11	0.11	-0.11
0.11	-0.11	1.00	-0.33	0.11	-0.11	0.55
0.33	0.33	-0.33	0.55	-0.33	0.33	0.33
0.55	-0.11	0.11	-0.33	1.00	-0.11	0.11
-0.11	0.11	-0.11	0.33	-0.11	1.00	-0.11
0.33	-0.11	0.55	0.33	0.11	-0.11	0.77



1.00	0.33	0.55	0.33
0.33	1.00	0.33	0.55
0.55	0.33	1.00	0.11
0.33	0.55	0.11	0.77

0.33	-0.55	0.11	-0.11	0.11	-0.55	0.33
-0.55	0.55	-0.55	0.33	-0.55	0.55	-0.55
0.11	-0.55	0.55	-0.77	0.55	-0.55	0.11
-0.11	0.33	-0.77	1.00	-0.77	0.33	-0.11
0.11	-0.55	0.55	-0.77	0.55	-0.55	0.11
-0.55	0.55	-0.55	0.33	-0.55	0.55	-0.55
0.33	-0.55	0.11	-0.11	0.11	-0.55	0.33



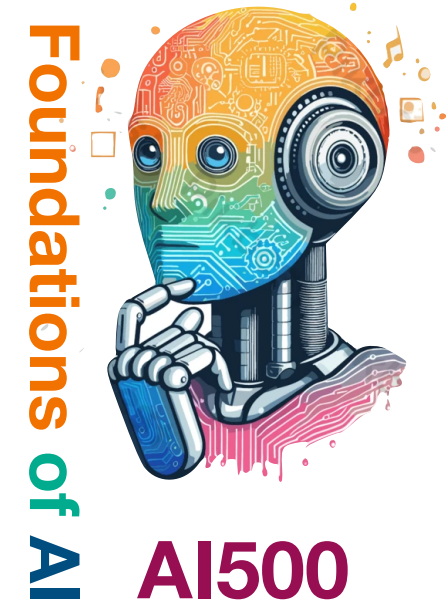
0.55	0.33	0.55	0.33
0.33	1.00	0.55	0.11
0.55	0.55	0.55	0.11
0.33	0.11	0.11	0.33

0.33	-0.11	0.55	0.33	0.11	-0.11	0.77
-0.11	0.11	-0.11	0.33	-0.11	1.00	-0.11
0.55	-0.11	0.11	-0.33	1.00	-0.11	0.11
0.33	0.33	-0.33	0.55	-0.33	0.33	0.33
0.11	-0.11	1.00	-0.33	0.11	-0.11	0.55
-0.11	1.00	-0.11	0.33	-0.11	0.11	-0.11
0.77	-0.11	0.11	0.33	0.55	-0.11	0.33



0.33	0.55	1.00	0.77
0.55	0.55	1.00	0.33
1.00	1.00	0.11	0.55
0.77	0.33	0.55	0.33

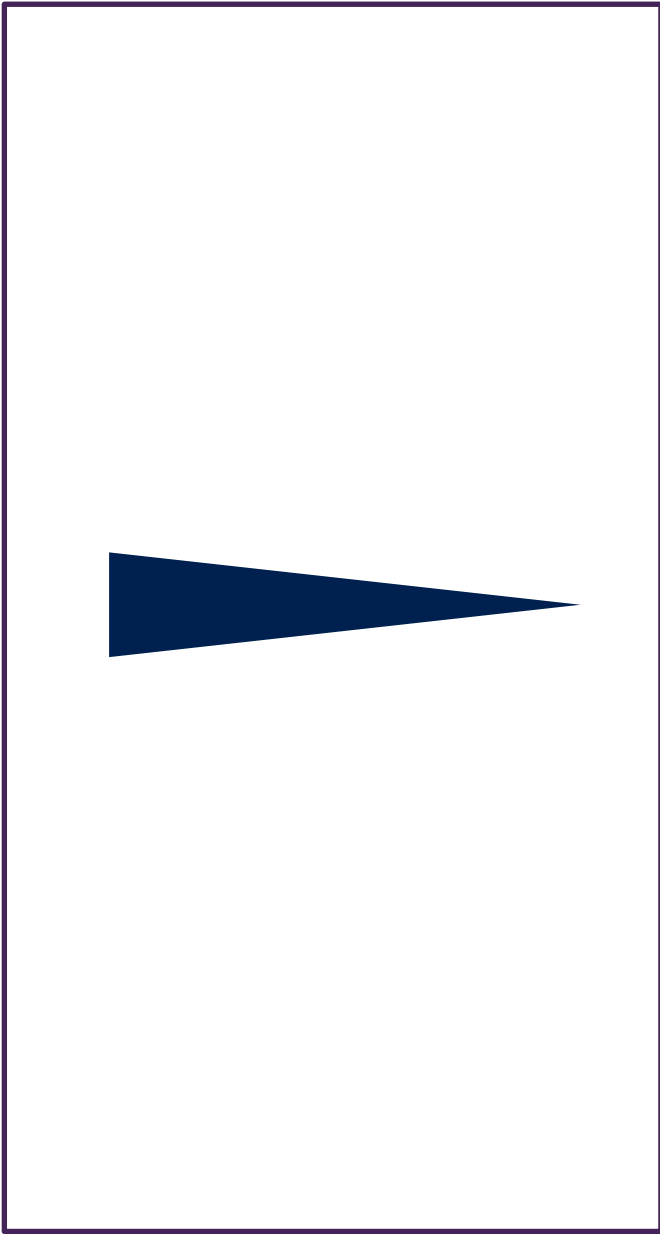
Pooling layer



0.77	-0.11	0.11	0.33	0.55	-0.11	0.33
-0.11	1.00	-0.11	0.33	-0.11	0.11	-0.11
0.11	-0.11	1.00	-0.33	0.11	-0.11	0.55
0.33	0.33	-0.33	0.55	-0.33	0.33	0.33
0.55	-0.11	0.11	-0.33	1.00	-0.11	0.11
-0.11	0.11	-0.11	0.33	-0.11	1.00	-0.11
0.33	-0.11	0.55	0.33	0.11	-0.11	0.77

0.33	-0.55	0.11	-0.11	0.11	-0.55	0.33
-0.55	0.55	-0.55	0.33	-0.55	0.55	-0.55
0.11	-0.55	0.55	-0.77	0.55	-0.55	0.11
-0.11	0.33	-0.77	1.00	-0.77	0.33	-0.11
0.11	-0.55	0.55	-0.77	0.55	-0.55	0.11
-0.55	0.55	-0.55	0.33	-0.55	0.55	-0.55
0.33	-0.55	0.11	-0.11	0.11	-0.55	0.33

0.33	-0.11	0.55	0.33	0.11	-0.11	0.77
-0.11	0.11	-0.11	0.33	-0.11	1.00	-0.11
0.55	-0.11	0.11	-0.33	1.00	-0.11	0.11
0.33	0.33	-0.33	0.55	-0.33	0.33	0.33
0.11	-0.11	1.00	-0.33	0.11	-0.11	0.55
-0.11	1.00	-0.11	0.33	-0.11	0.11	-0.11
0.77	-0.11	0.11	0.33	0.55	-0.11	0.33



1.00	0.33	0.55	0.33
0.33	1.00	0.33	0.55
0.55	0.33	1.00	0.11
0.33	0.55	0.11	0.77

0.55	0.33	0.55	0.33
0.33	1.00	0.55	0.11
0.55	0.55	0.55	0.11
0.33	0.11	0.11	0.33

0.33	0.55	1.00	0.77
0.55	0.55	1.00	0.33
1.00	1.00	0.11	0.55
0.77	0.33	0.55	0.33

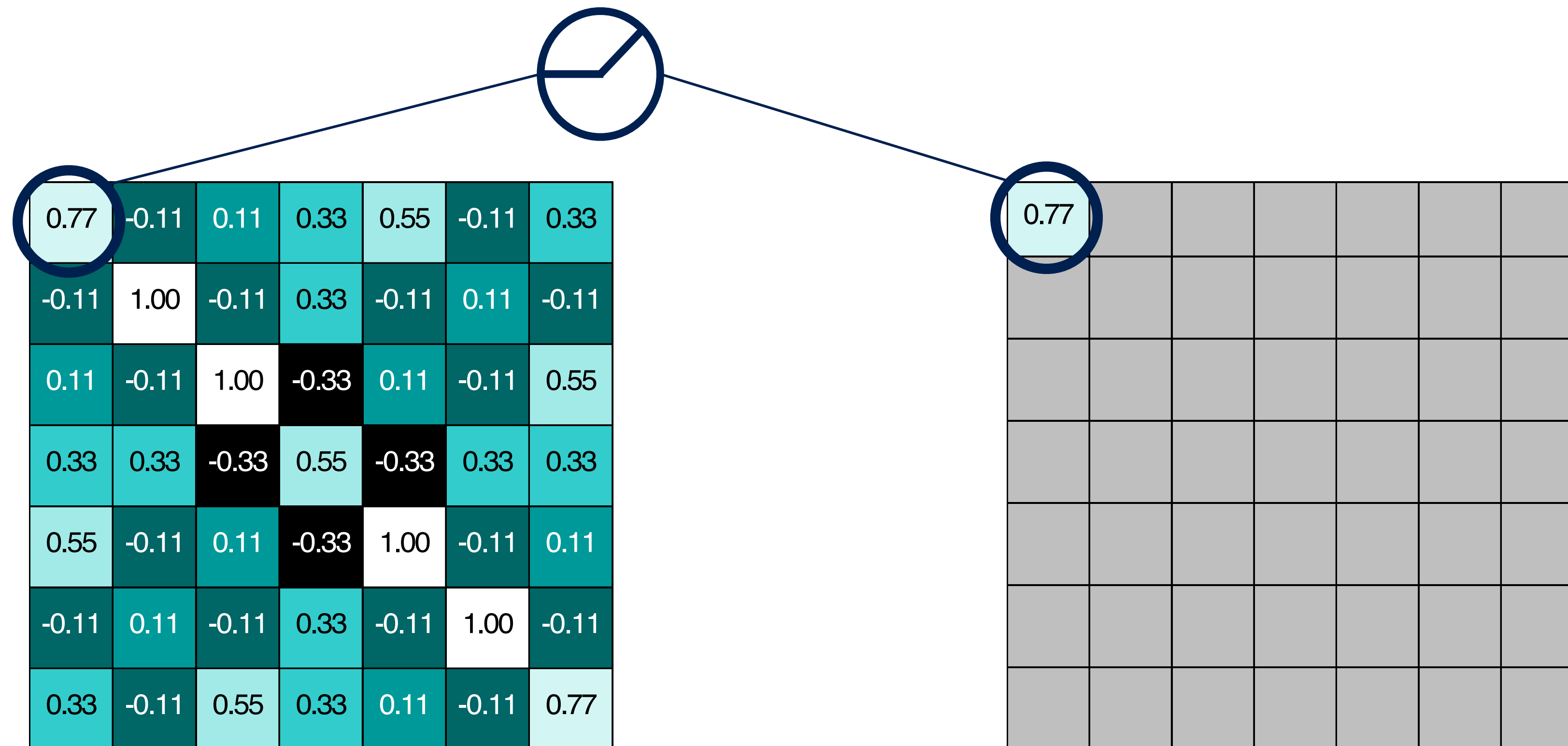
Normalization



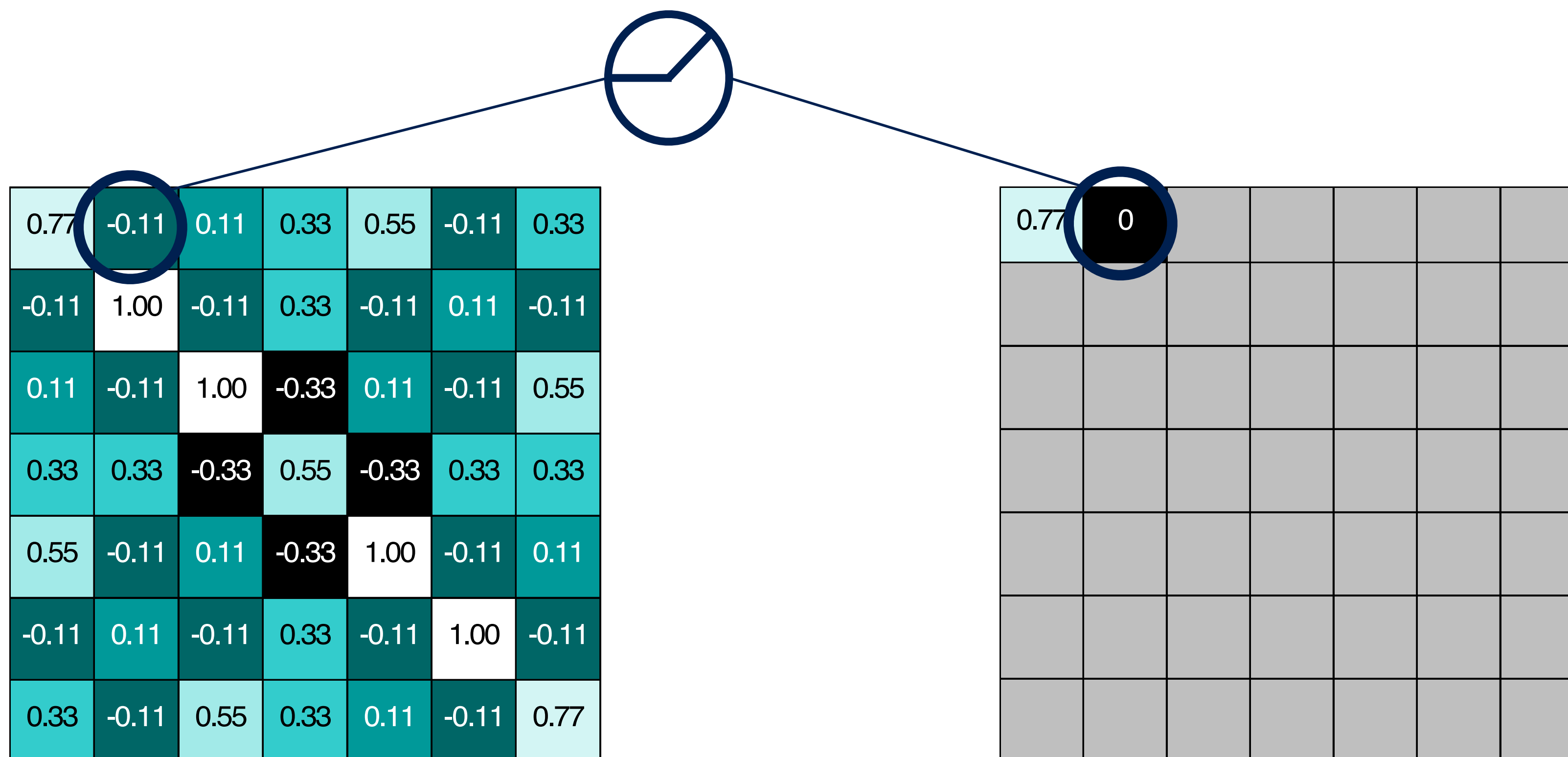
Keep the math from breaking by tweaking each of the values just a bit.

Change everything negative to zero.

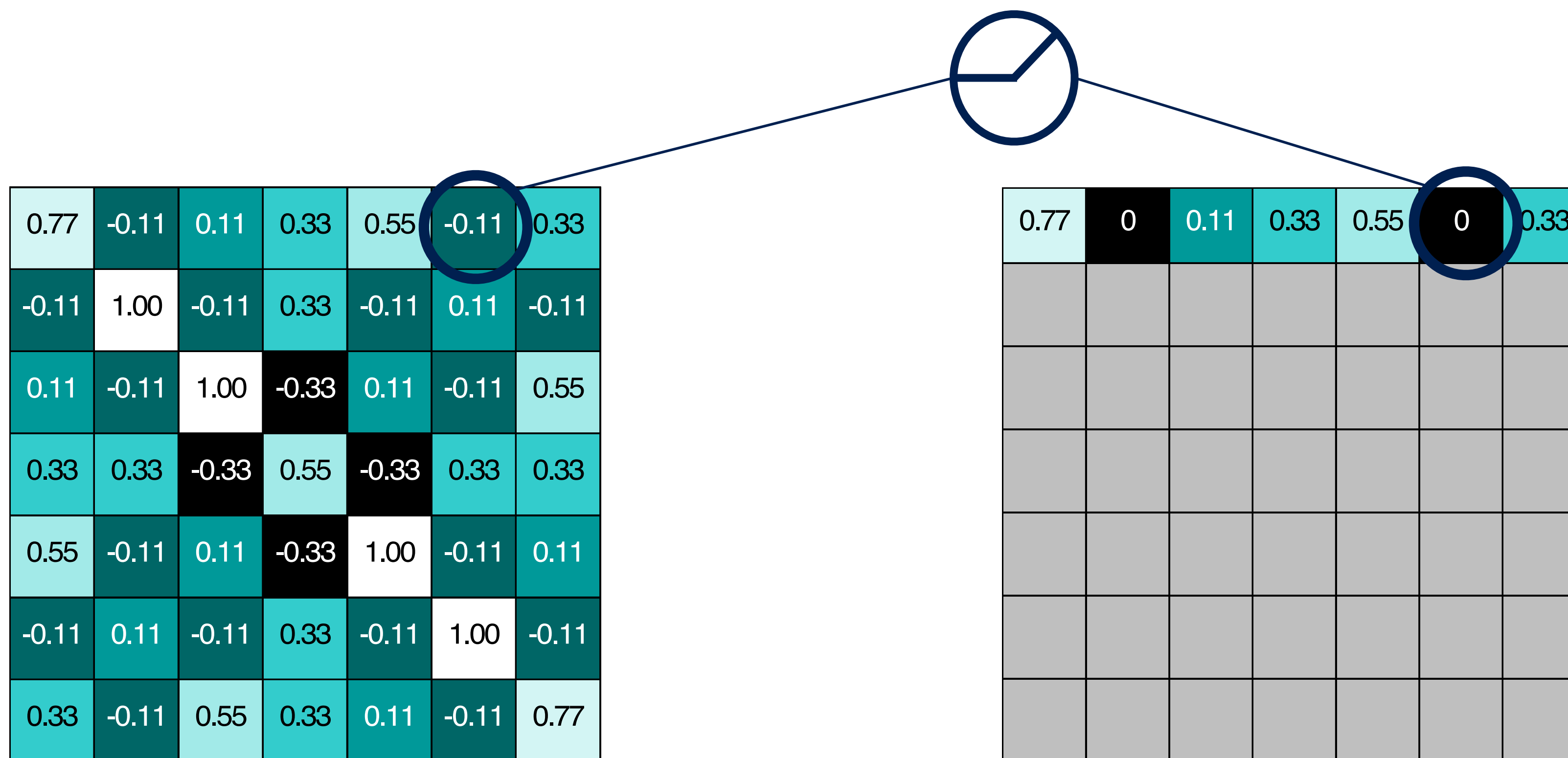
Rectified Linear Units (ReLUs)



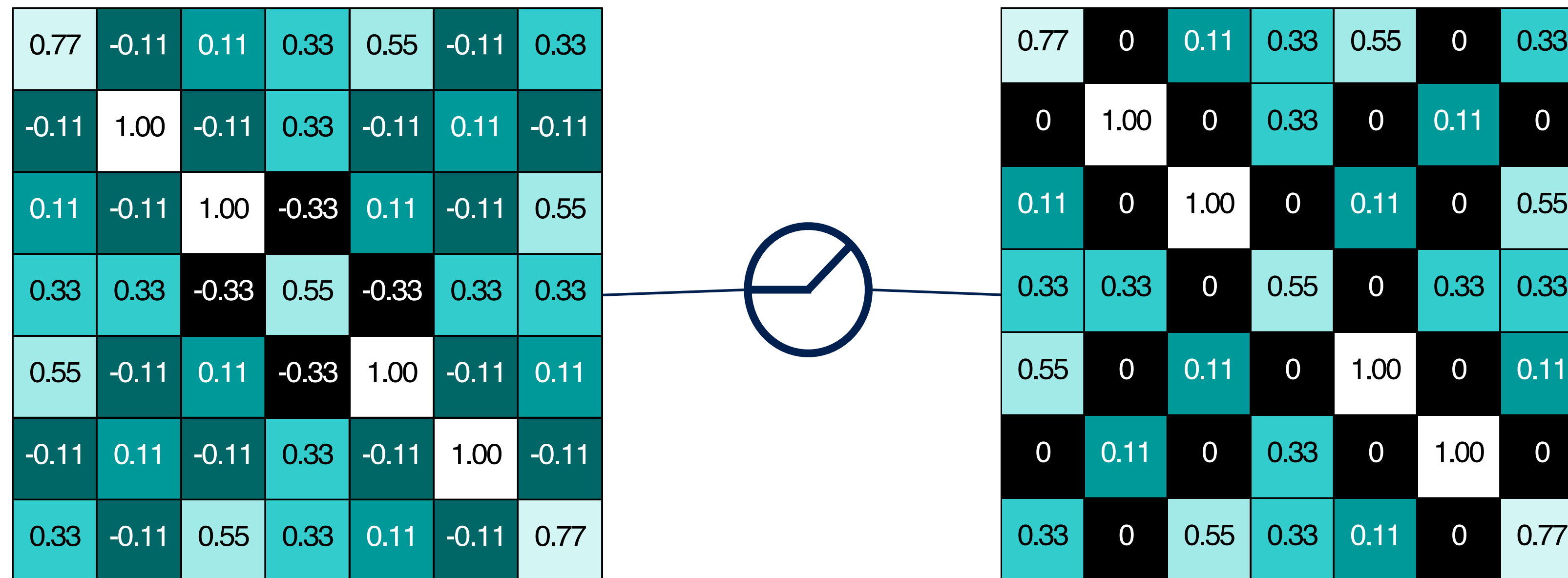
Rectified Linear Units (ReLU)



Rectified Linear Units (ReLUs)

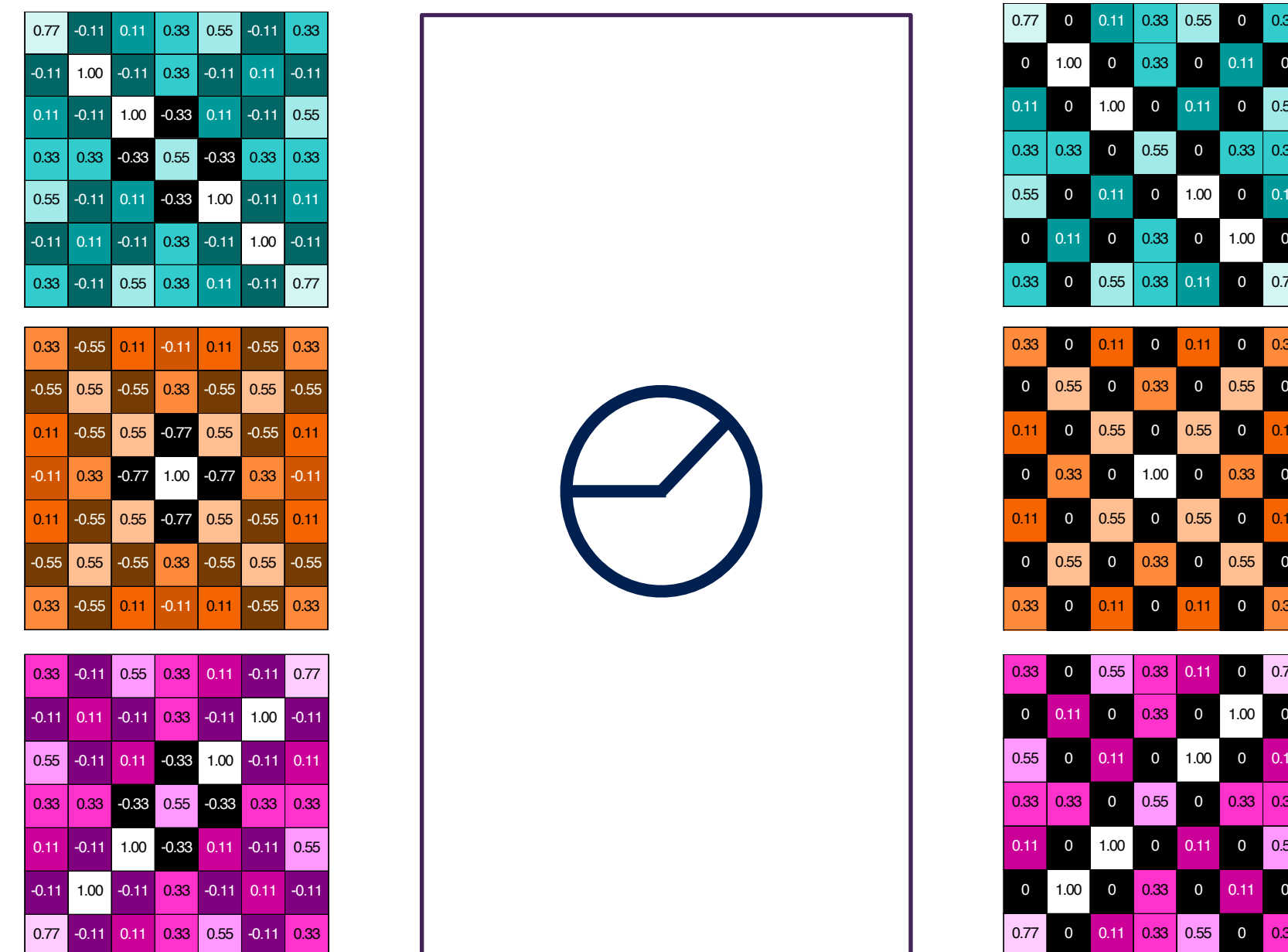


Rectified Linear Units (ReLUs)



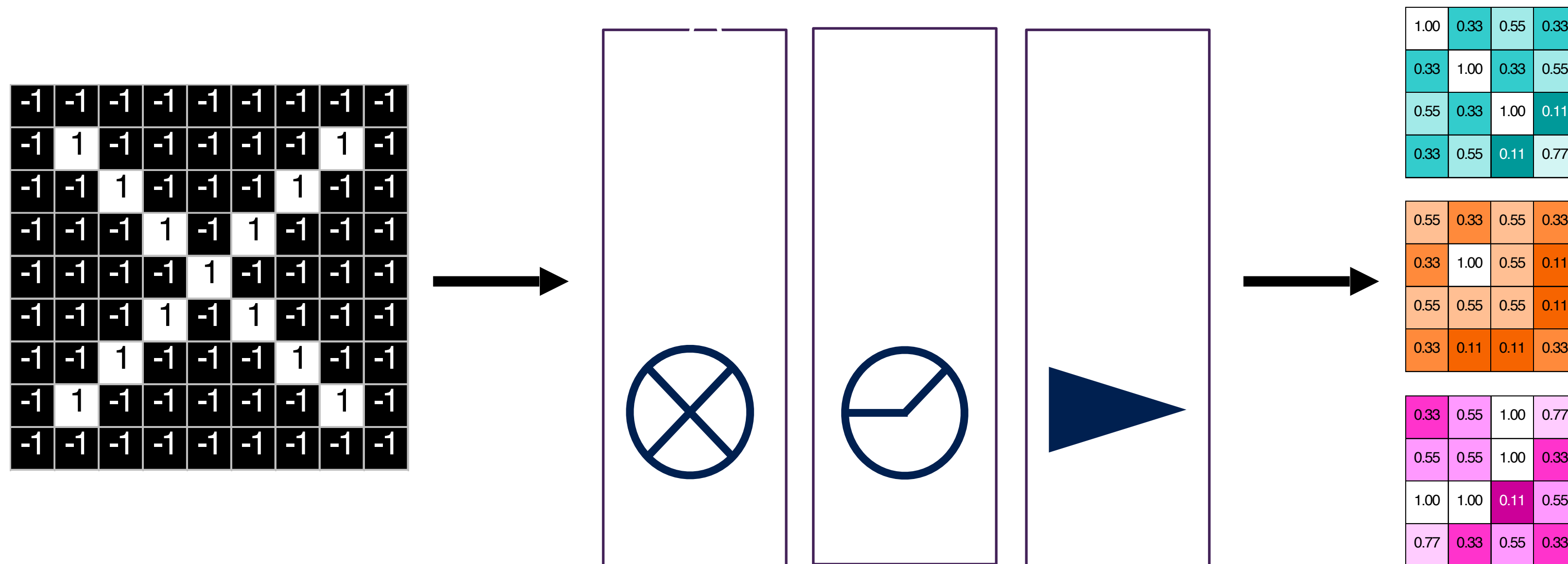
ReLU layer

A stack of images becomes a stack of images with no negative values.



Layers get stacked

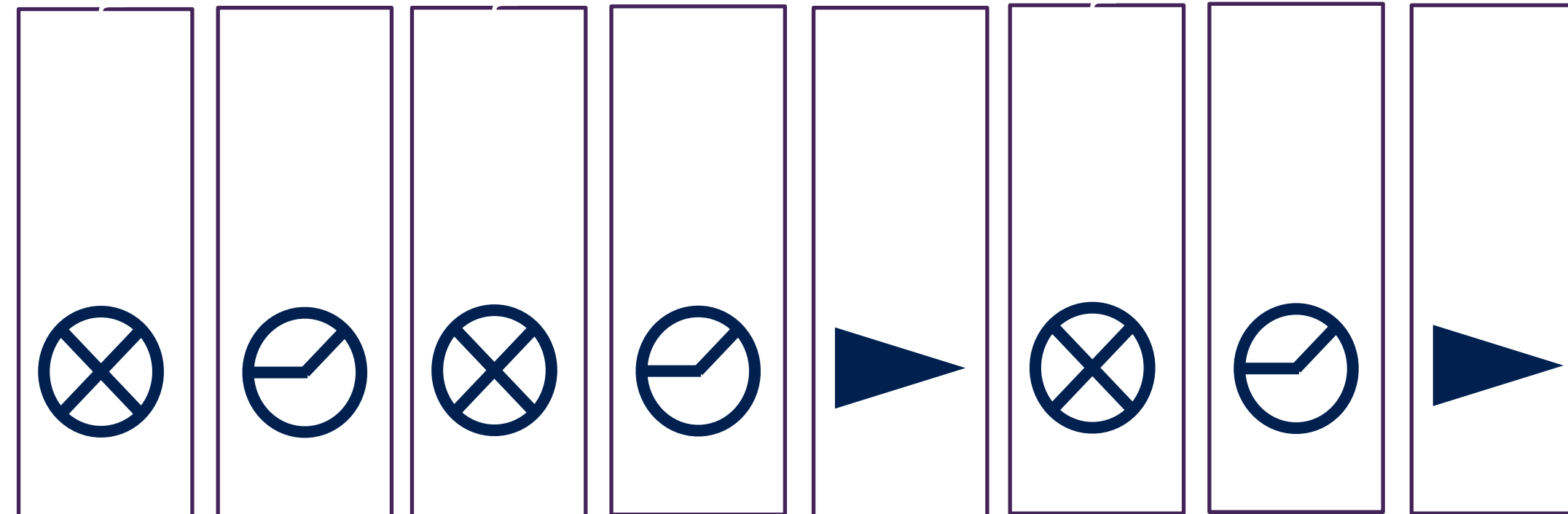
The output of one becomes the input of the next.



Deep stacking

Layers can be repeated several (or many) times.

-1	-1	-1	-1	-1	-1	-1	-1	-1	-1
-1	1	-1	-1	-1	-1	-1	1	-1	-1
-1	-1	1	-1	-1	-1	1	-1	-1	-1
-1	-1	-1	1	-1	1	-1	-1	-1	-1
-1	-1	-1	-1	1	-1	-1	-1	-1	-1
-1	-1	-1	1	-1	1	-1	-1	-1	-1
-1	-1	1	-1	-1	-1	1	-1	-1	-1
-1	1	-1	-1	-1	-1	-1	1	-1	-1
-1	-1	-1	-1	-1	-1	-1	-1	-1	-1



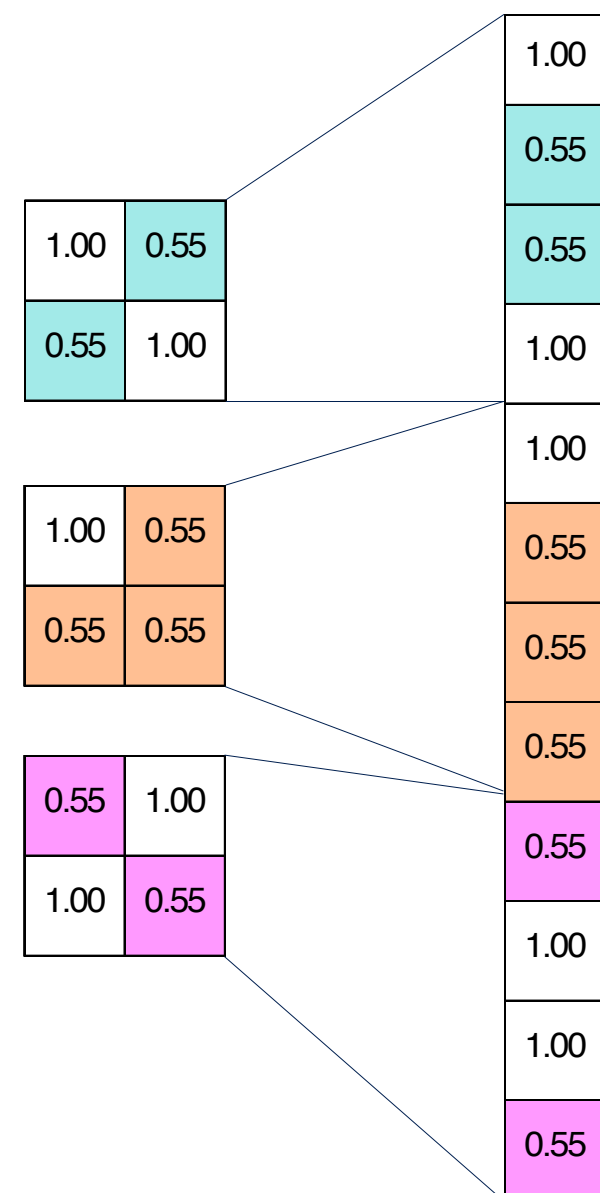
1.00	0.55
0.55	1.00

1.00	0.55
0.55	0.55

0.55	1.00
1.00	0.55

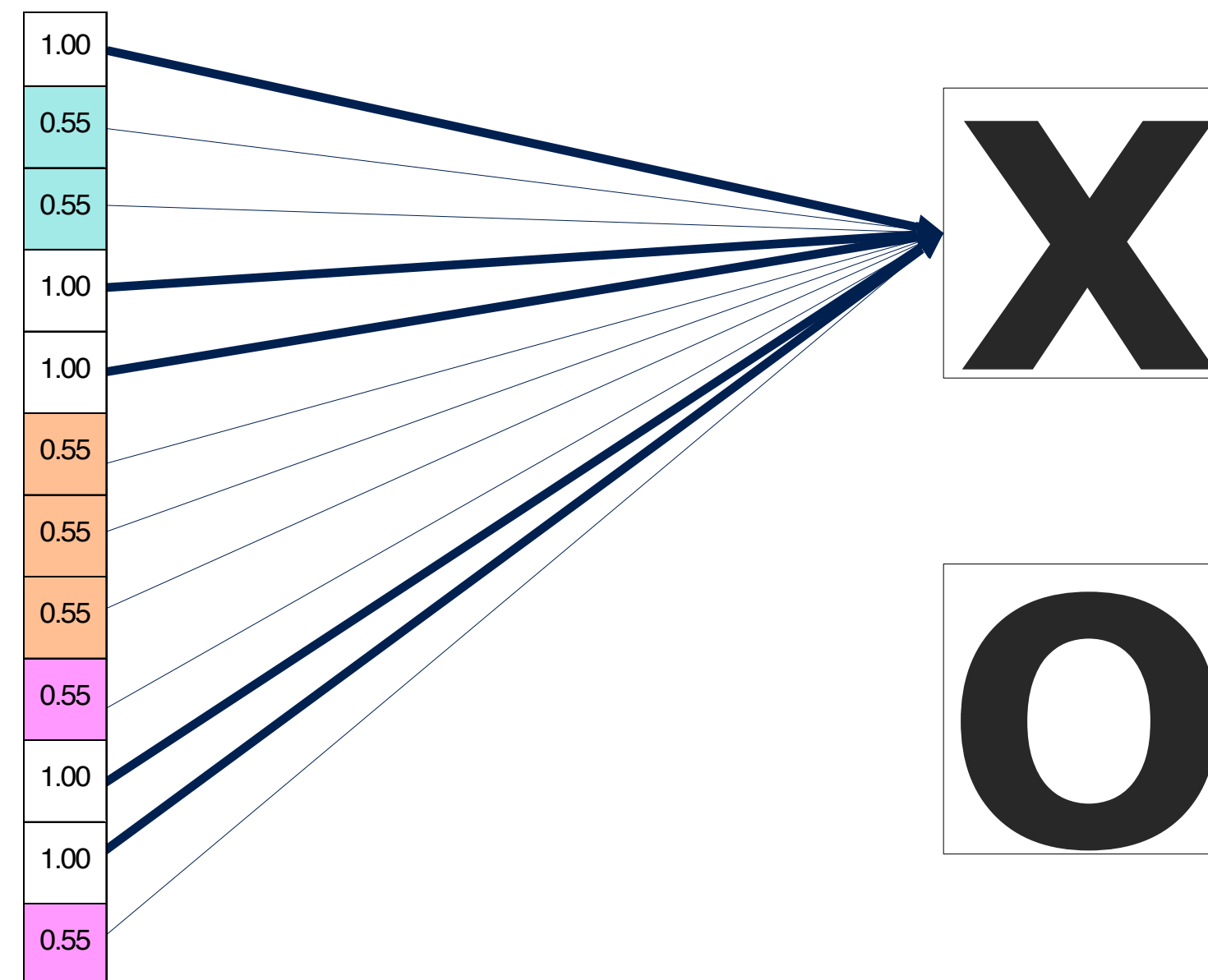
Fully connected layer

Every value gets a vote



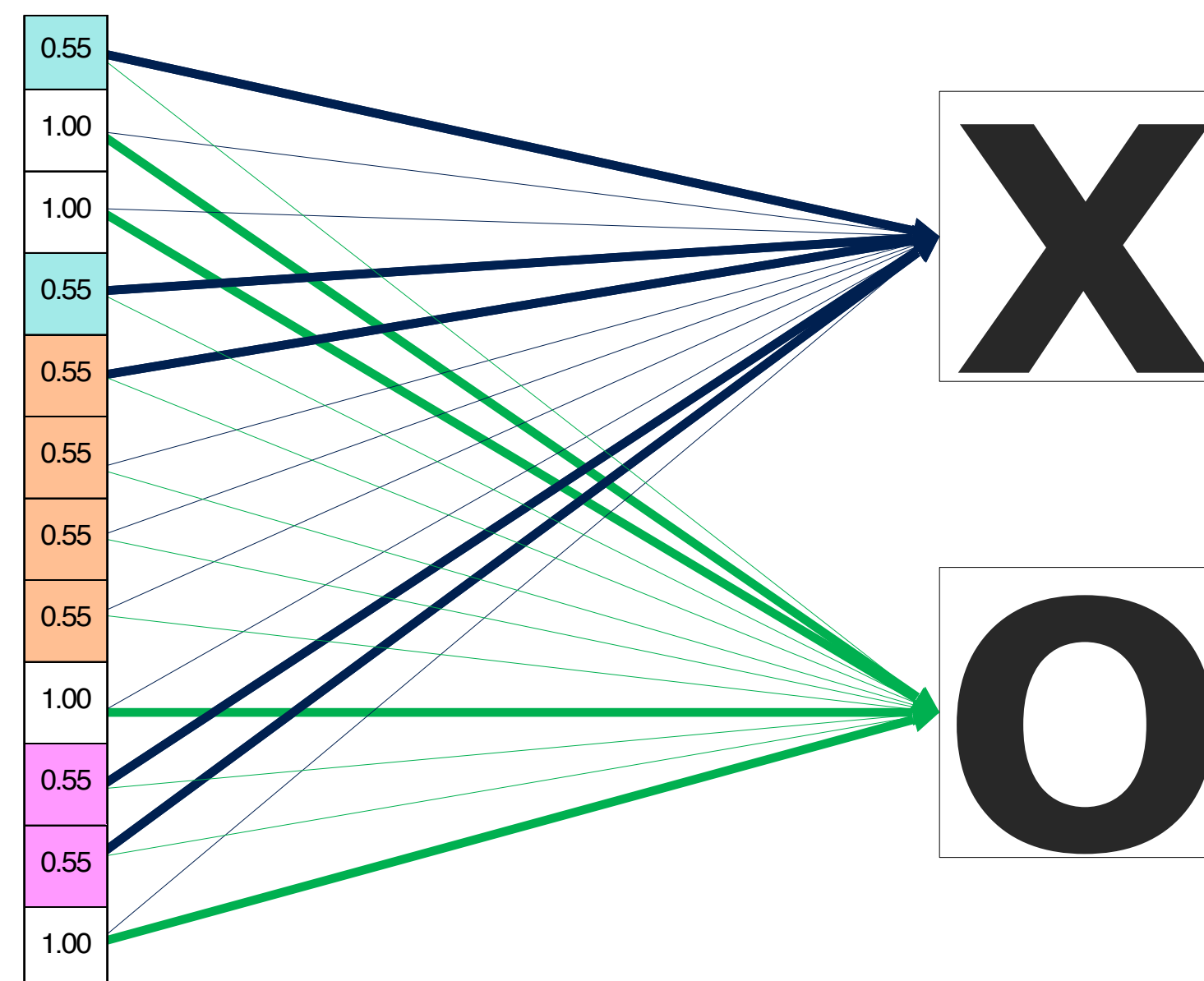
Fully connected layer

Vote depends on how strongly a value predicts X or O



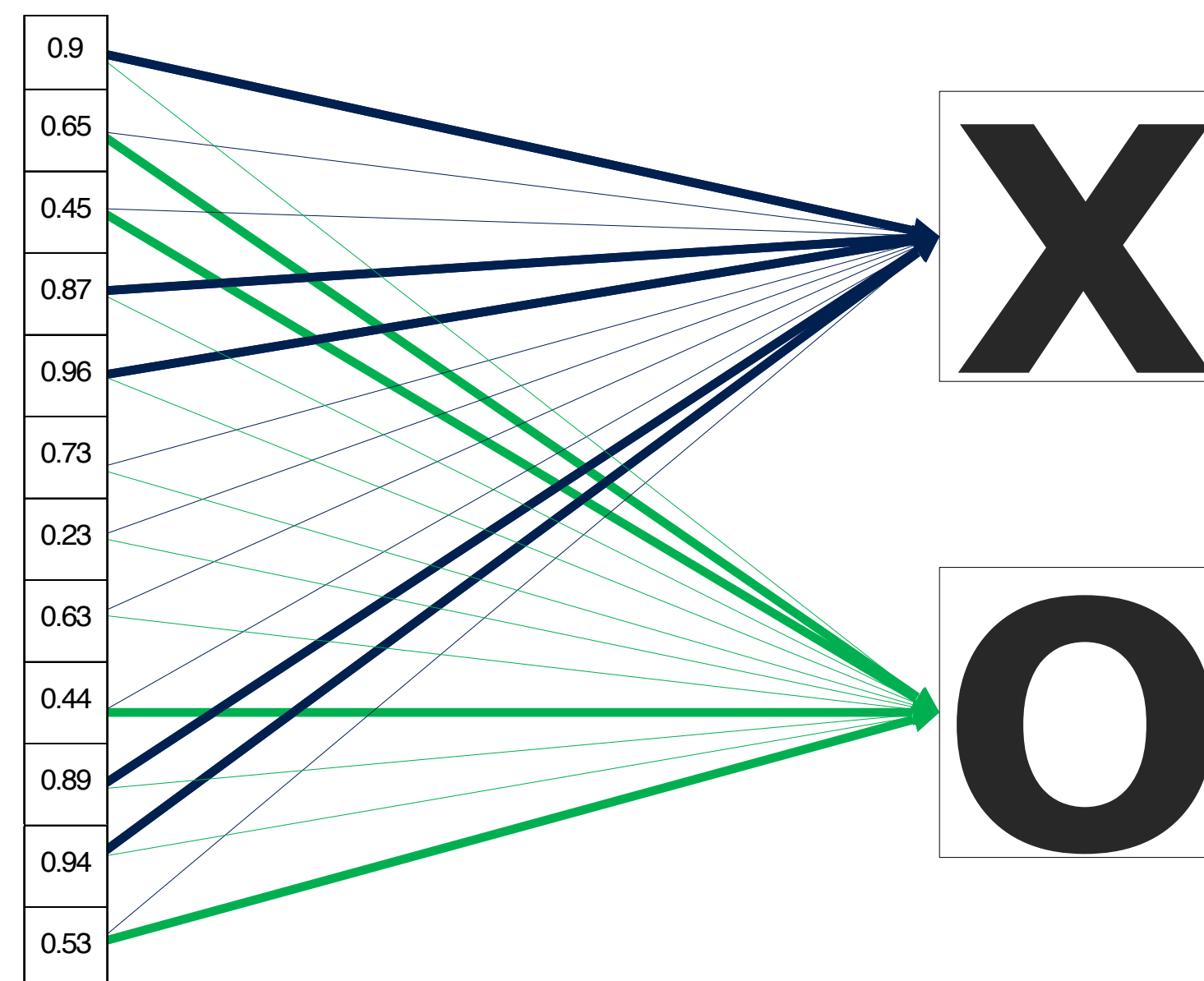
Fully connected layer

Vote depends on how strongly a value predicts X or O



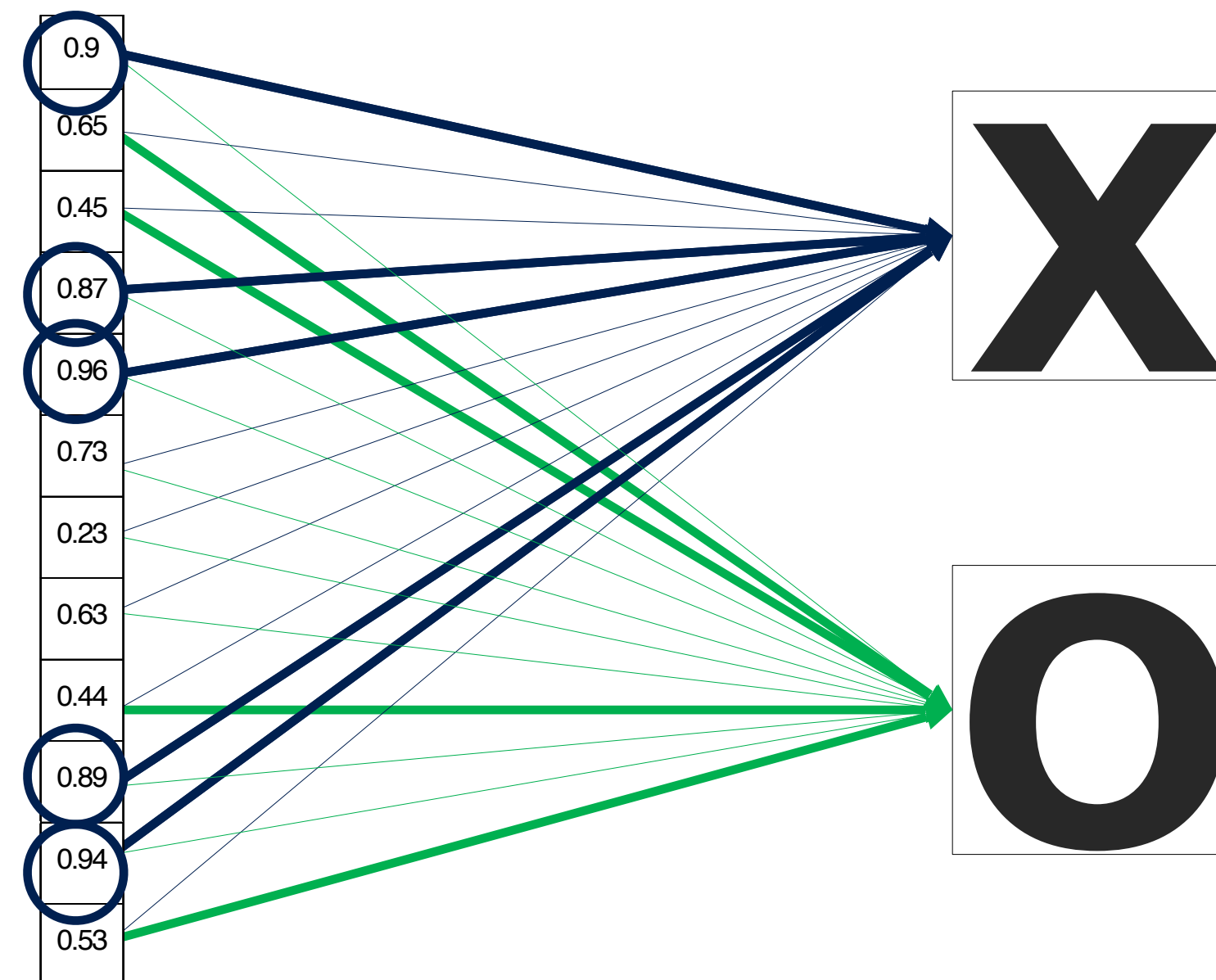
Fully connected layer

Future values vote on X or O



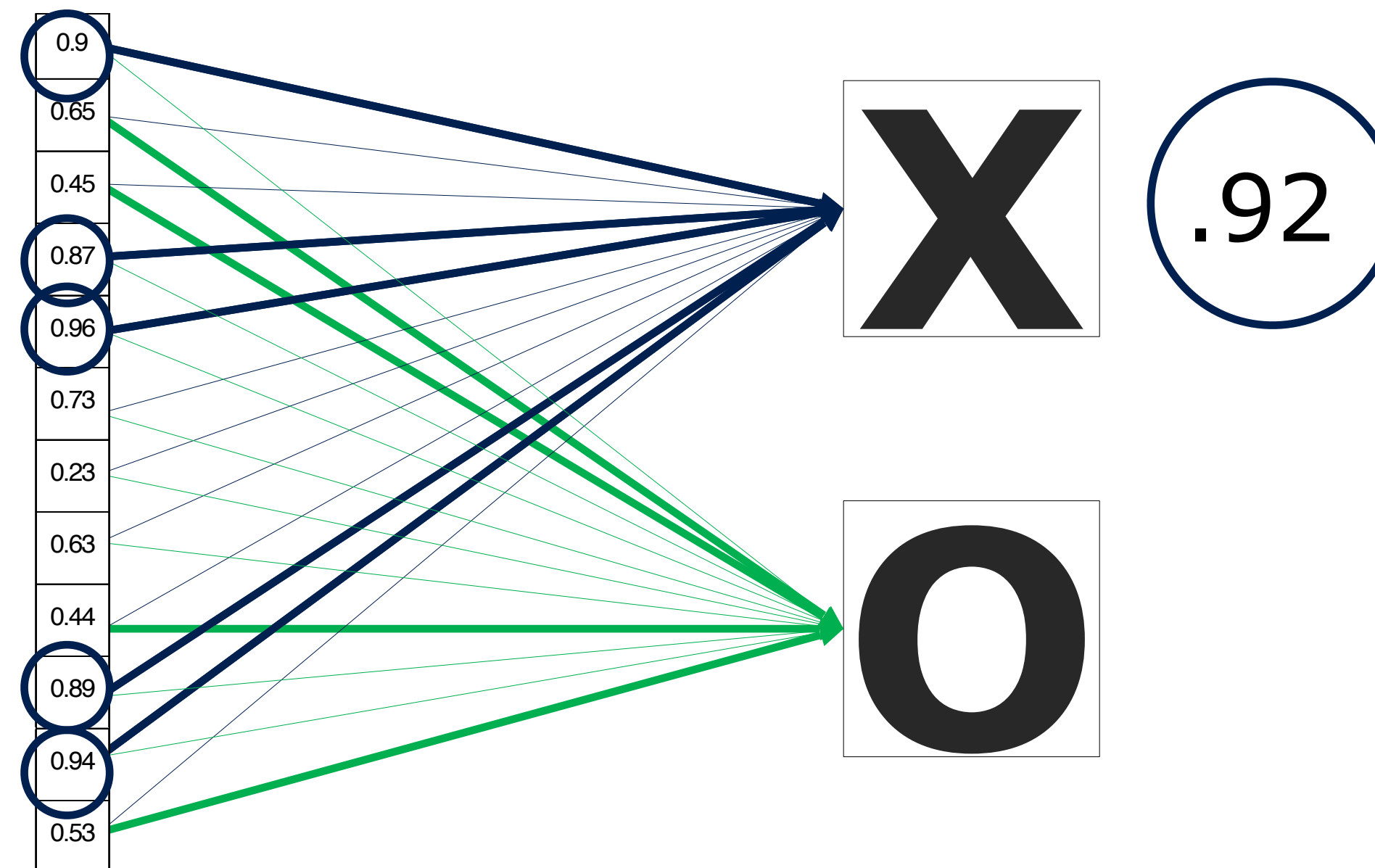
Fully connected layer

Future values vote on X or O



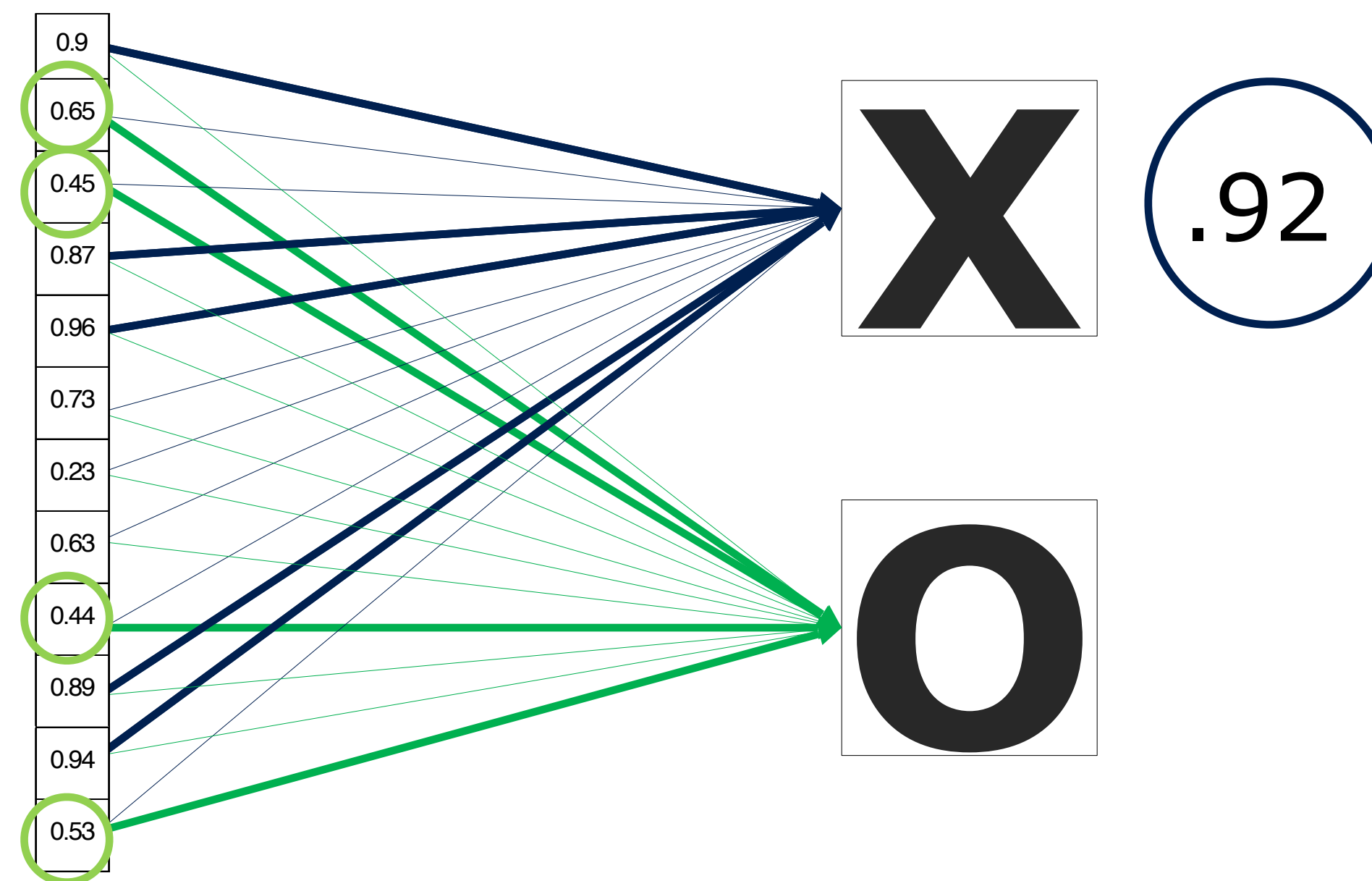
Fully connected layer

Future values vote on X or O



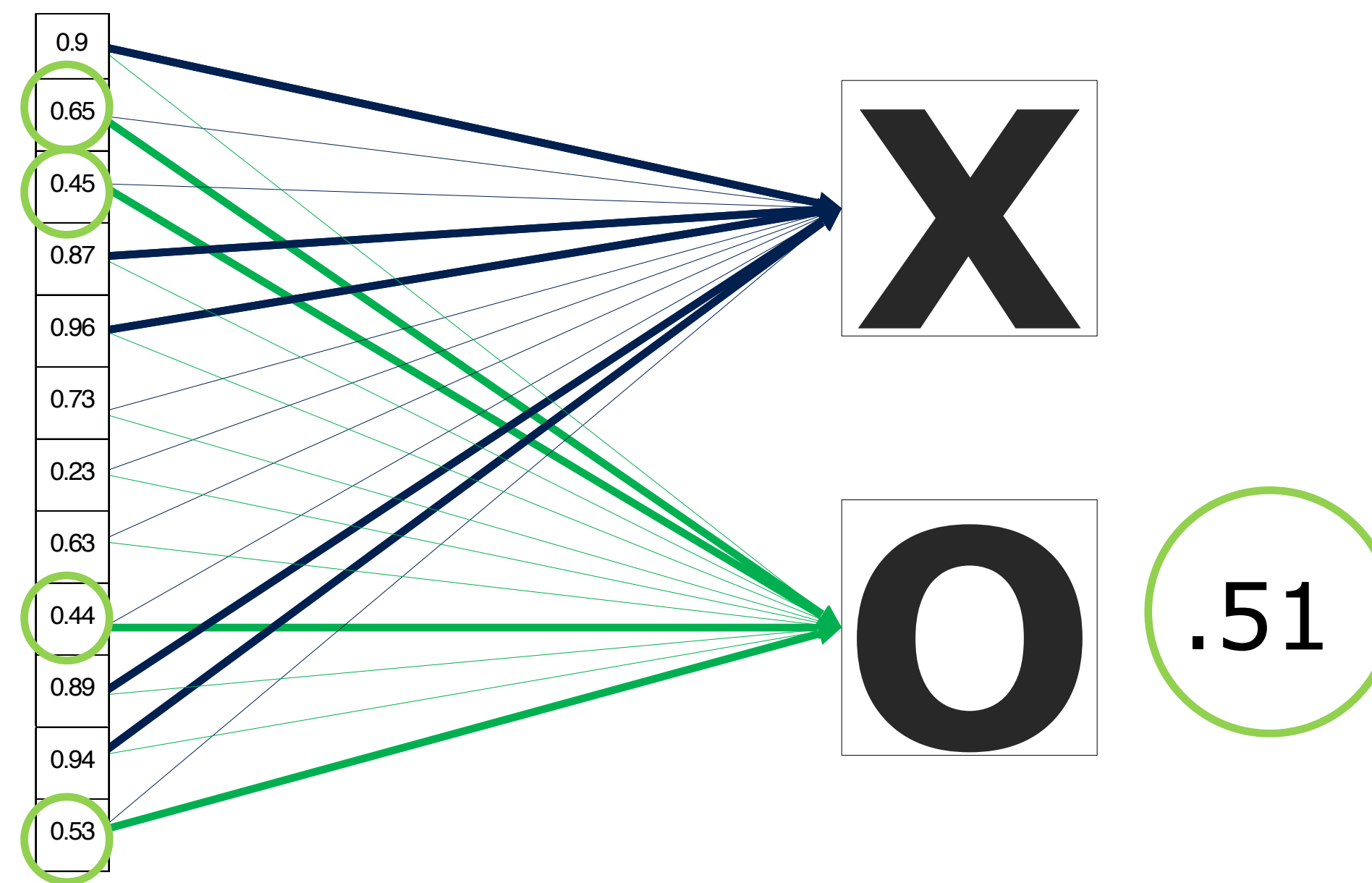
Fully connected layer

Future values vote on X or O



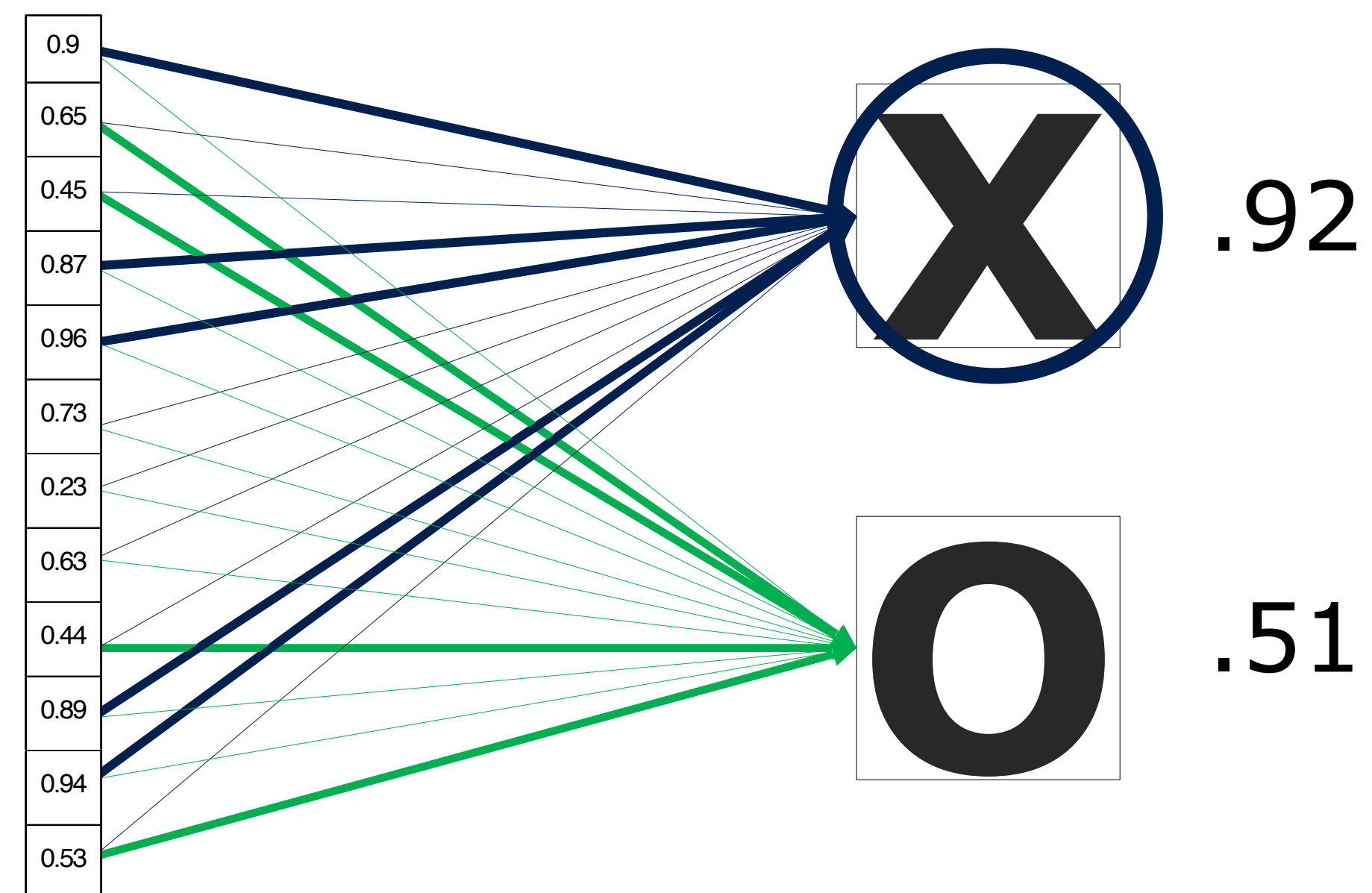
Fully connected layer

Future values vote on X or O



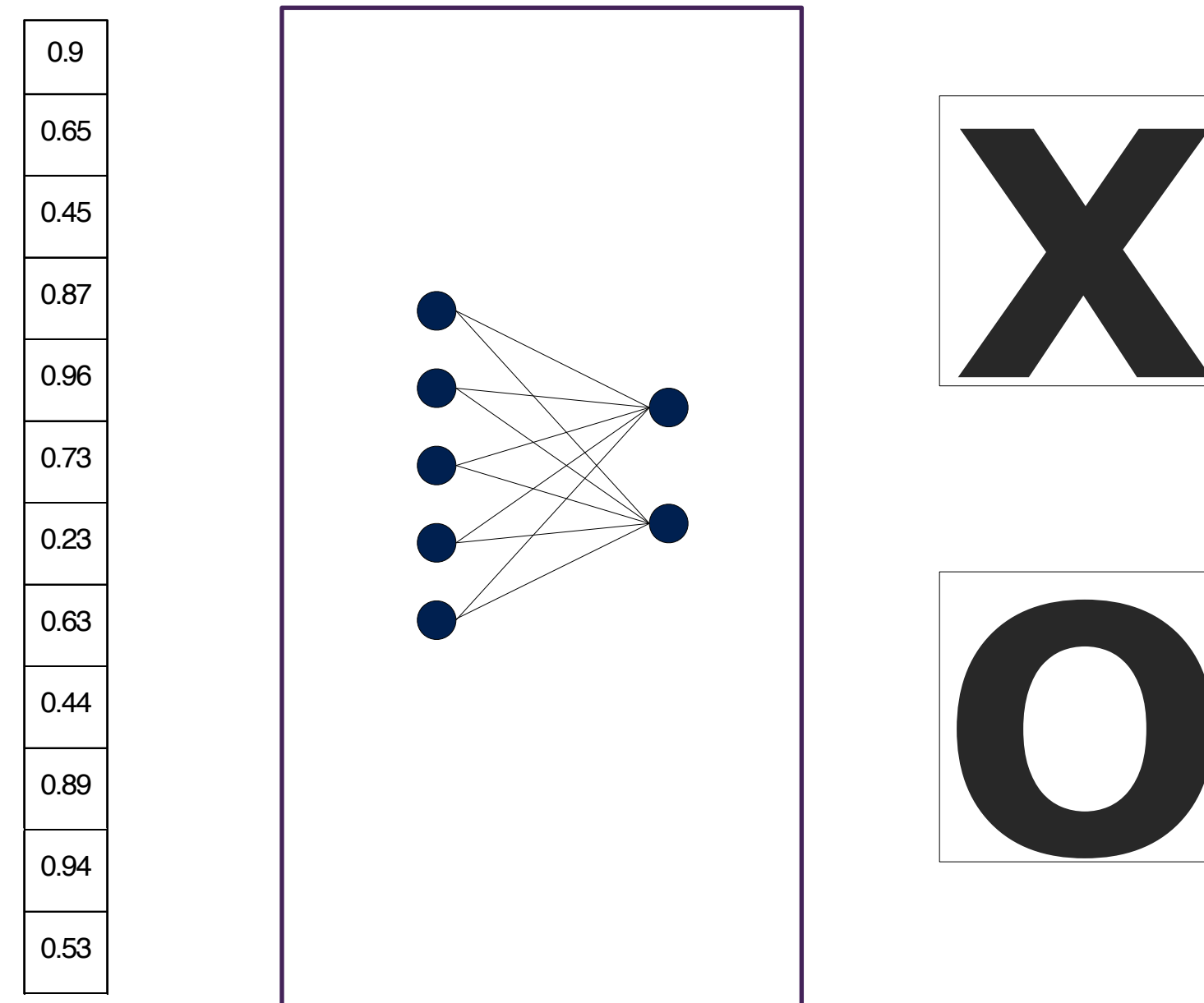
Fully connected layer

Future values vote on X or O



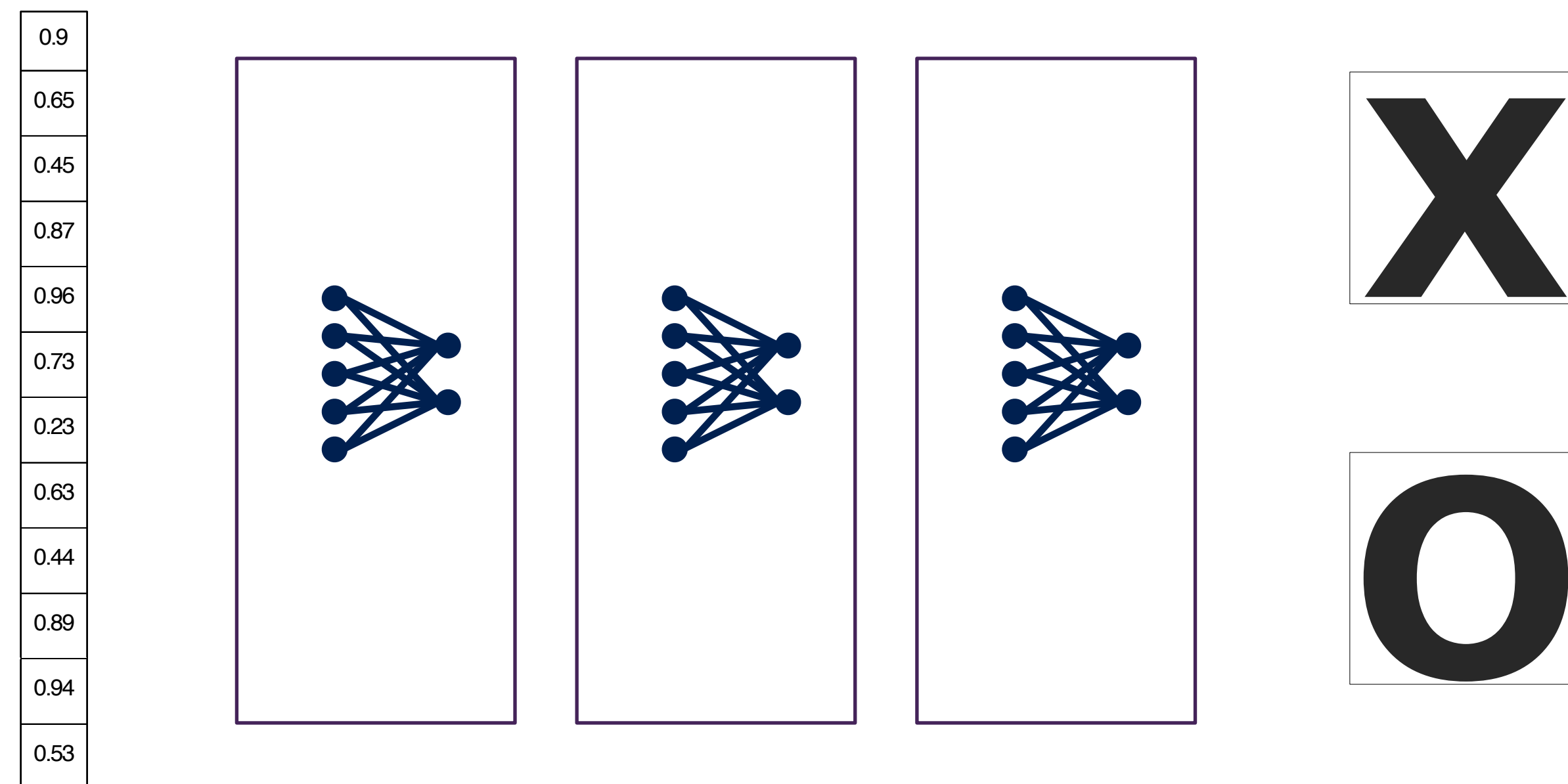
Fully connected layer

A list of feature values becomes a list of votes.



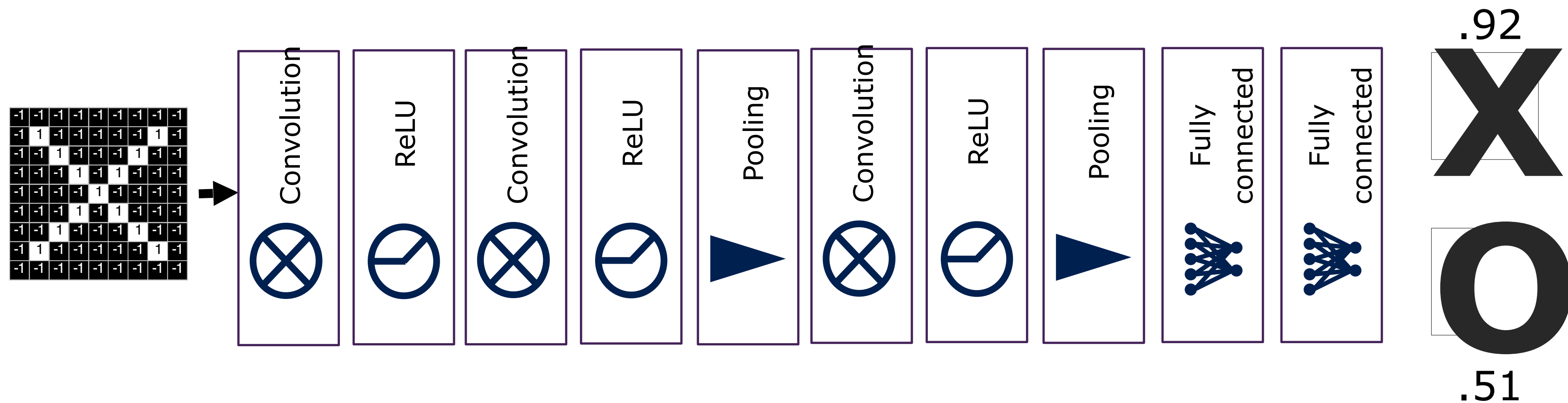
Fully connected layer

These can also be stacked.



Putting it all together

A set of pixels becomes a set of votes.



Learning



Q: Where do all the magic numbers come from?

Features in convolutional layers

Voting weights in fully connected layers

A: Backpropagation

Hyperparameters (knobs)

Convolution

- Number of features

- Size of features

Pooling

- Window size

- Window stride

Fully Connected

- Number of neurons

Architecture



How many of each type of layer?

In what order?

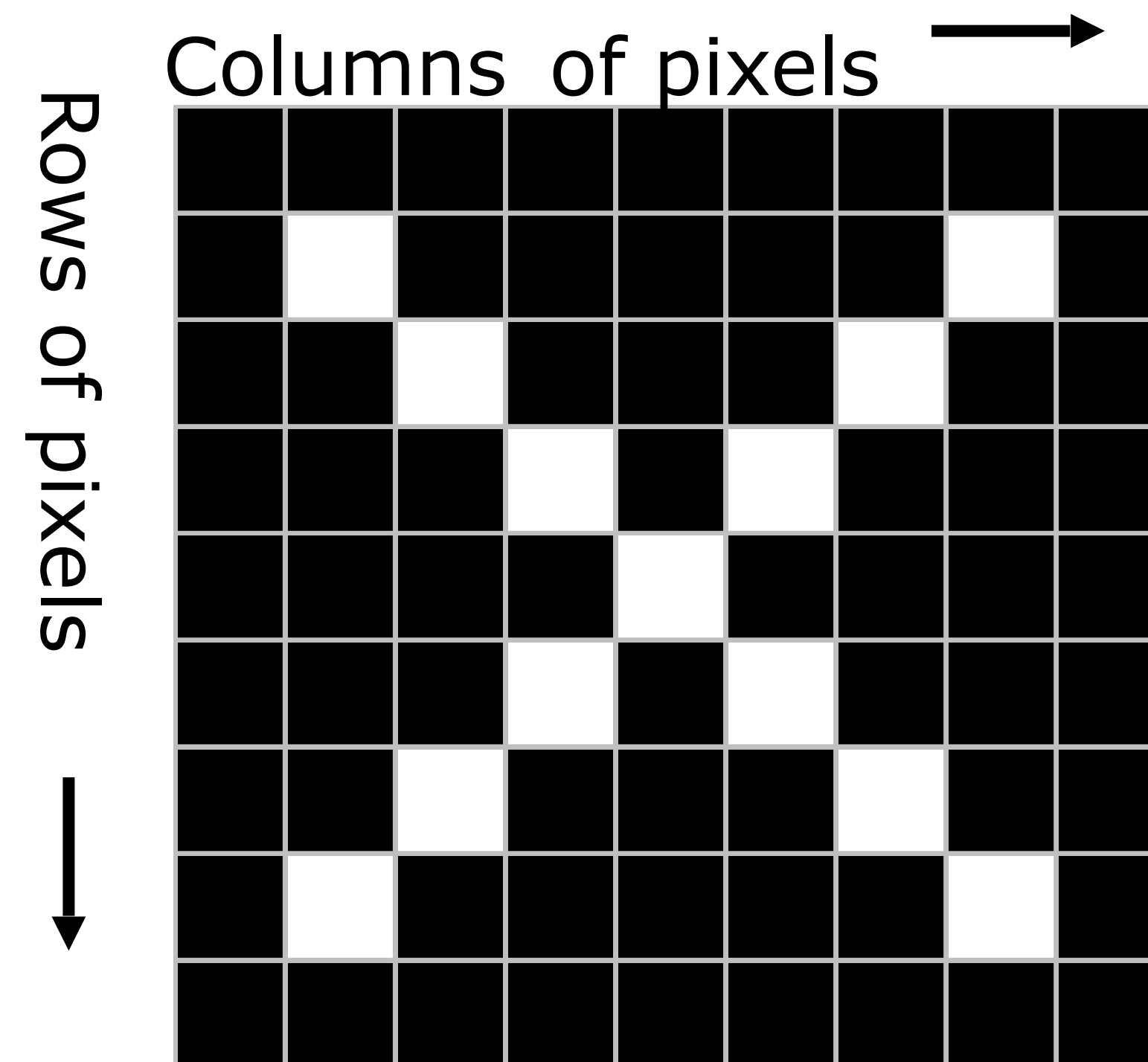
Not just images



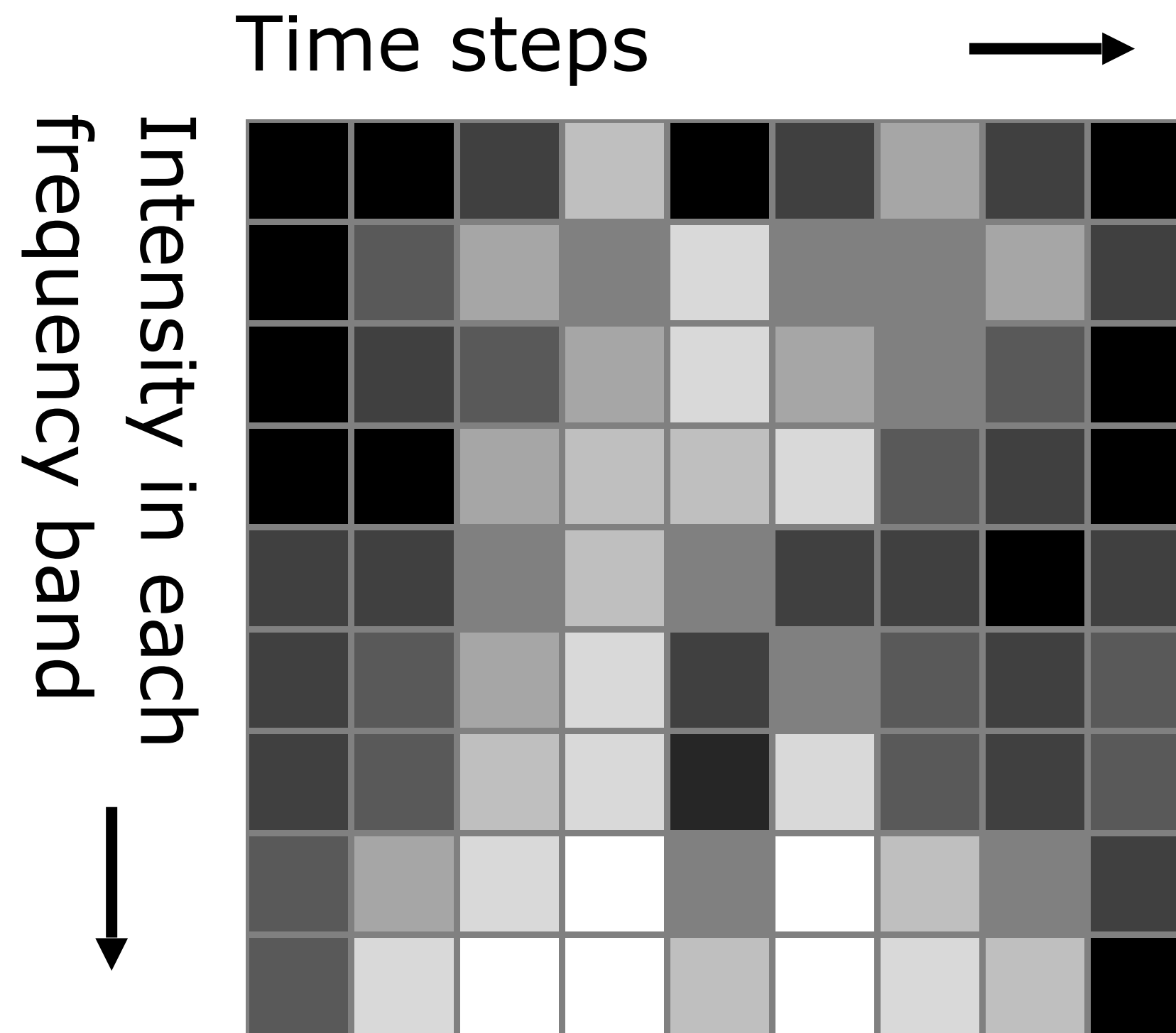
Any 2D (or 3D) data.

Things closer together are more closely related than things far away.

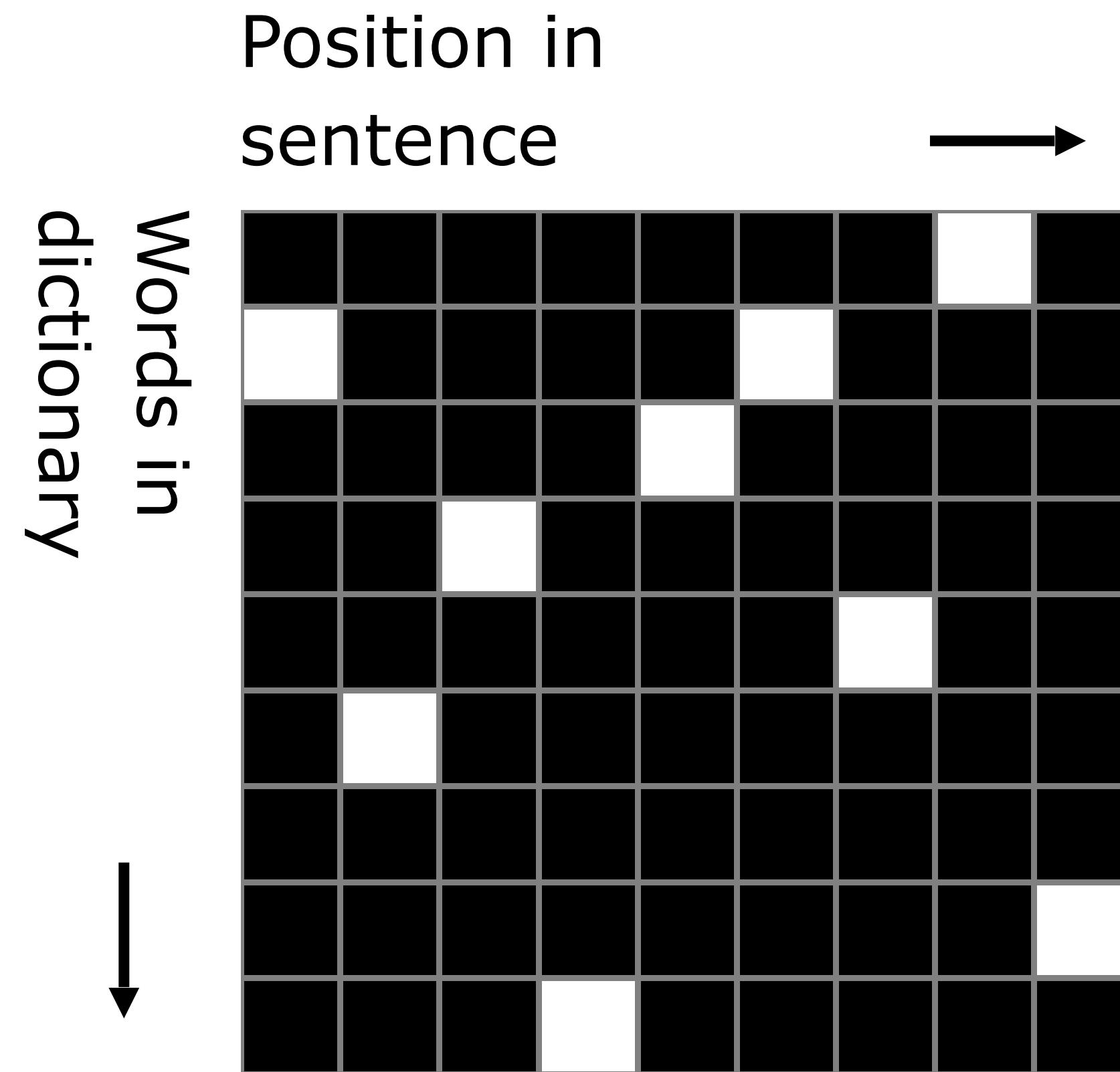
Images



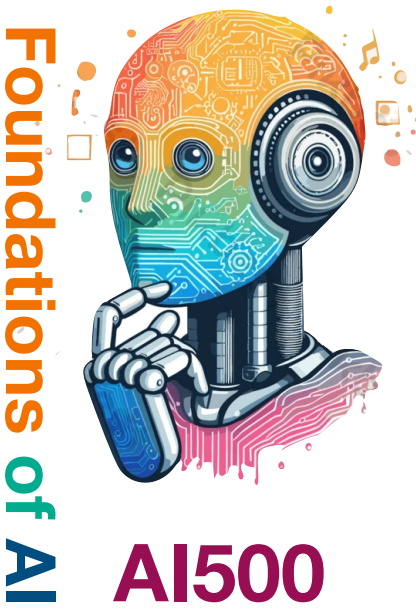
Sound



Text



Limitations



ConvNets only capture local “spatial” patterns in data.

If the data can’t be made to look like an image, ConvNets are less useful.

Customer data



Name, age,
address, email,
purchases,
browsing activity,...

→

Customers



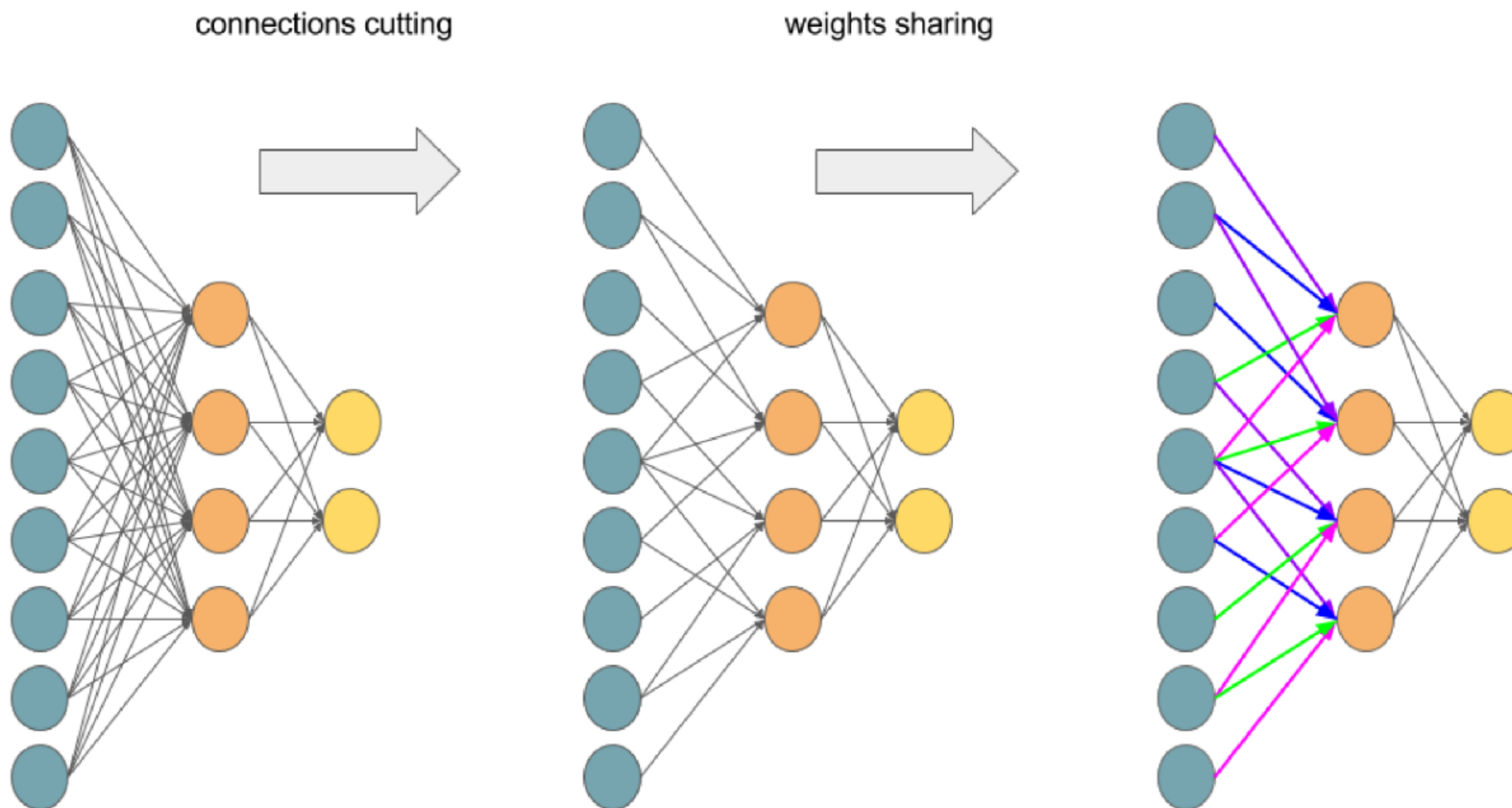
A	22	1A	a@a	1	aa	a1.a	123	aa1
B	33	2B	b@b	2	bb	b2.b	234	bb2
C	44	3C	c@c	3	cc	c3.c	345	cc3
D	55	4D	d@d	4	dd	d4.d	456	dd4
E	66	5E	e@e	5	ee	e5.e	567	ee5
F	77	6F	f@f	6	ff	f6.f	678	ff6
G	88	7G	g@g	7	gg	g7.g	789	gg7
H	99	8H	h@h	8	hh	h8.h	890	hh8
I	111	9I	i@i	9	ii	i9.i	901	ii9

Readings

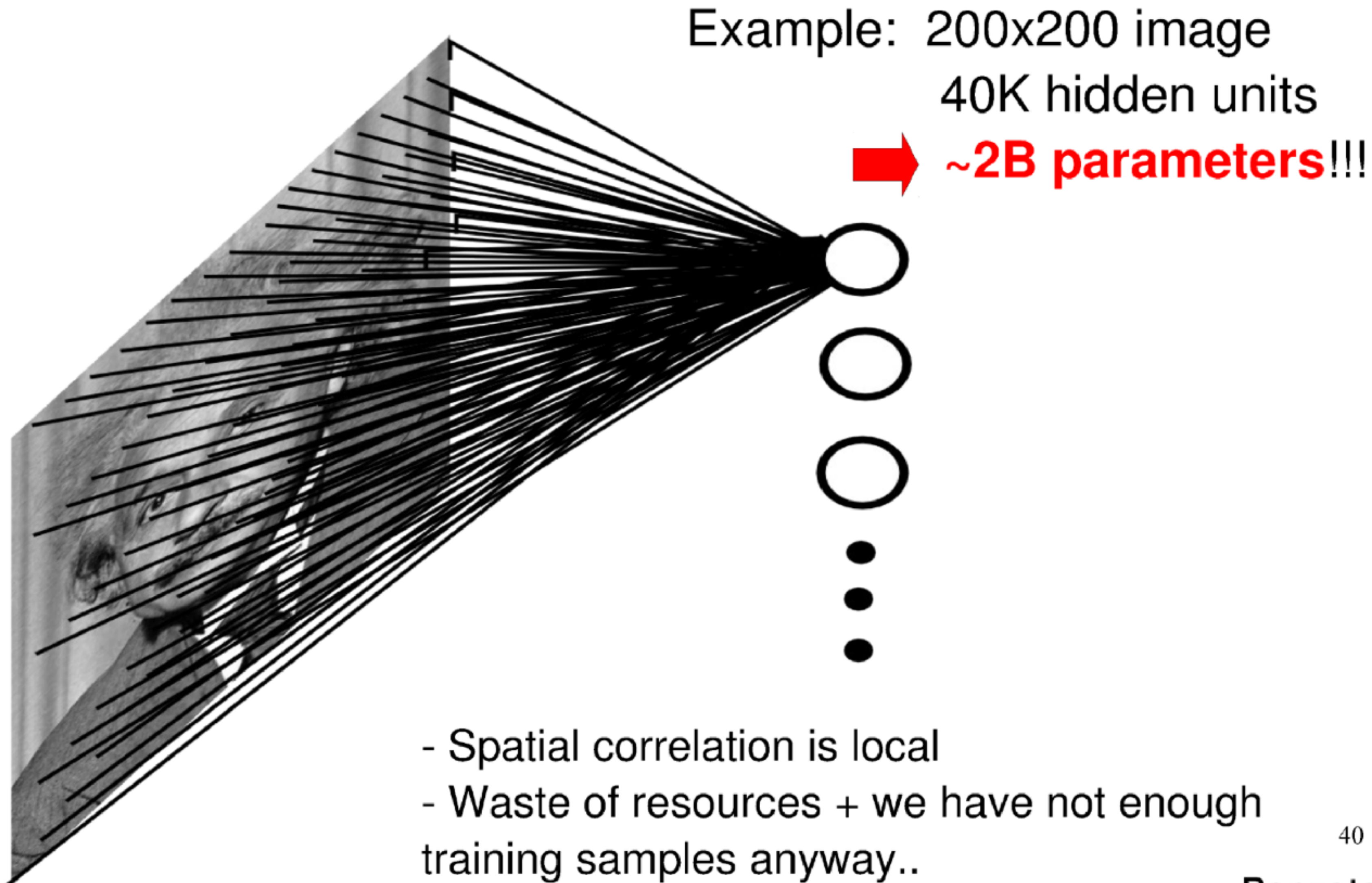
- DL Book: Convolutional Networks
 - <https://www.deeplearningbook.org/contents/convnets.html>



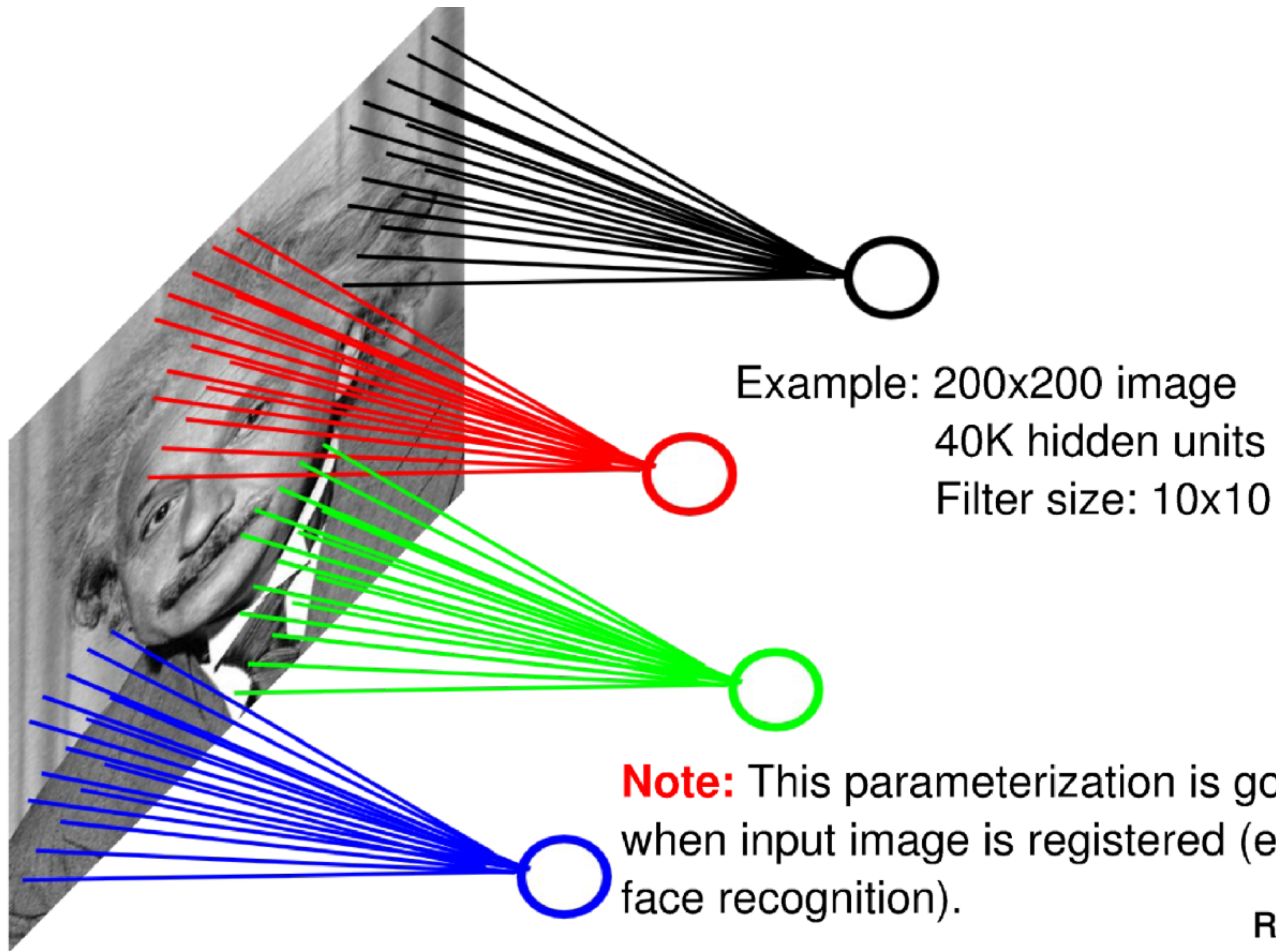
Transforming Multilayer Perceptron to CNN



Fully Connected Layer



Locally Connected Layer



Volumetric Convolution Filter

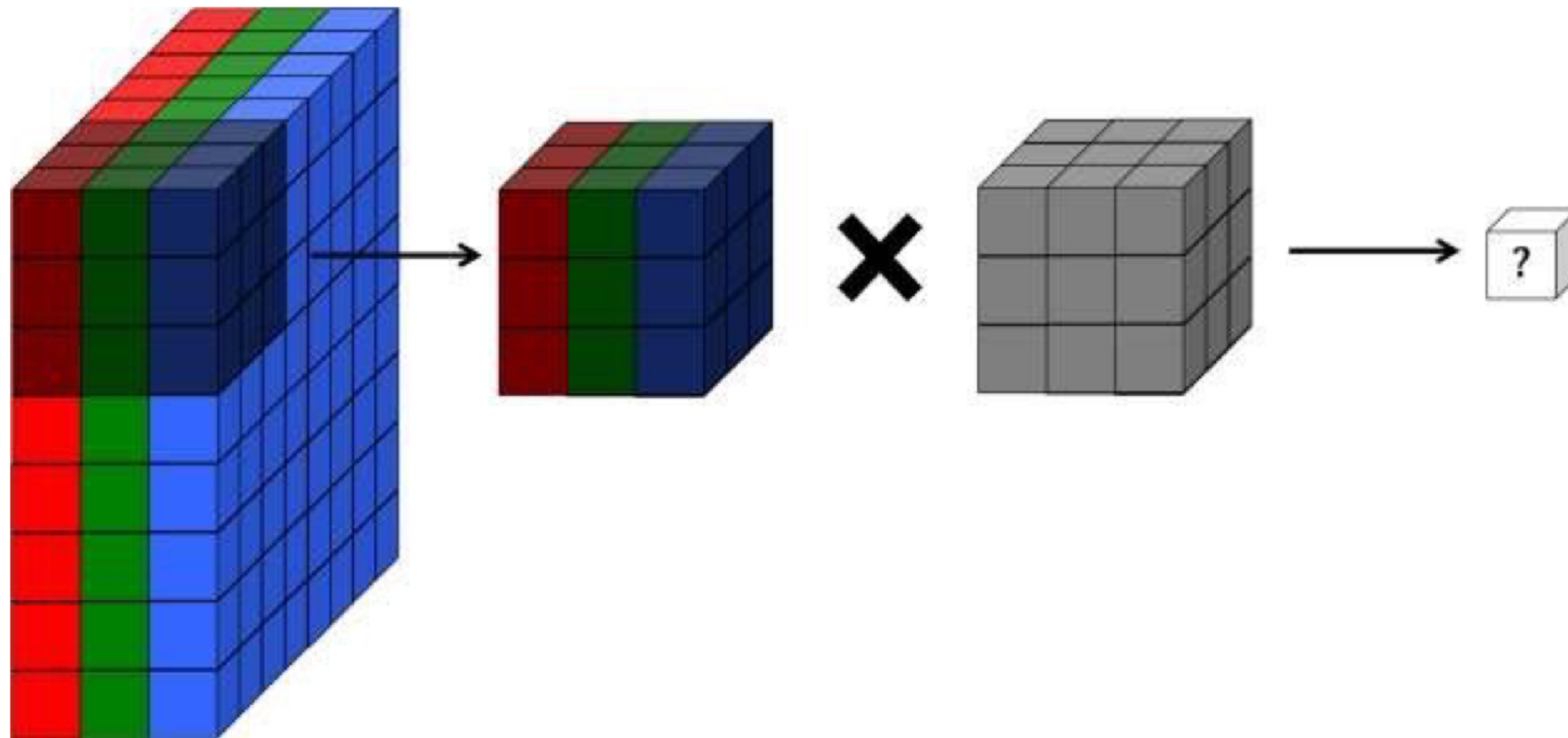


Figure 5-8. Representing a full-color RGB image as a volume and applying a volumetric convolutional filter

Volumetric Convolution Filter

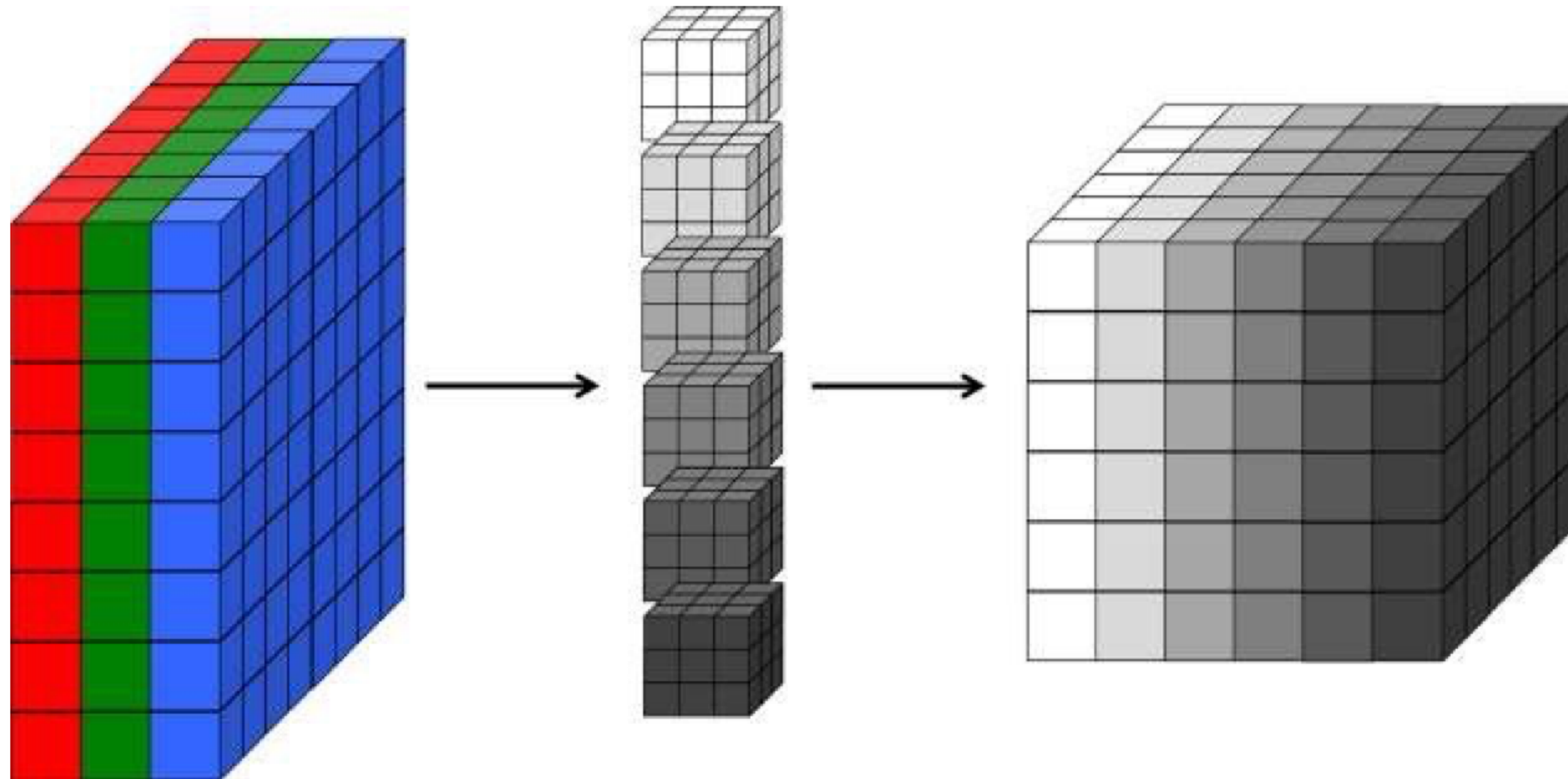
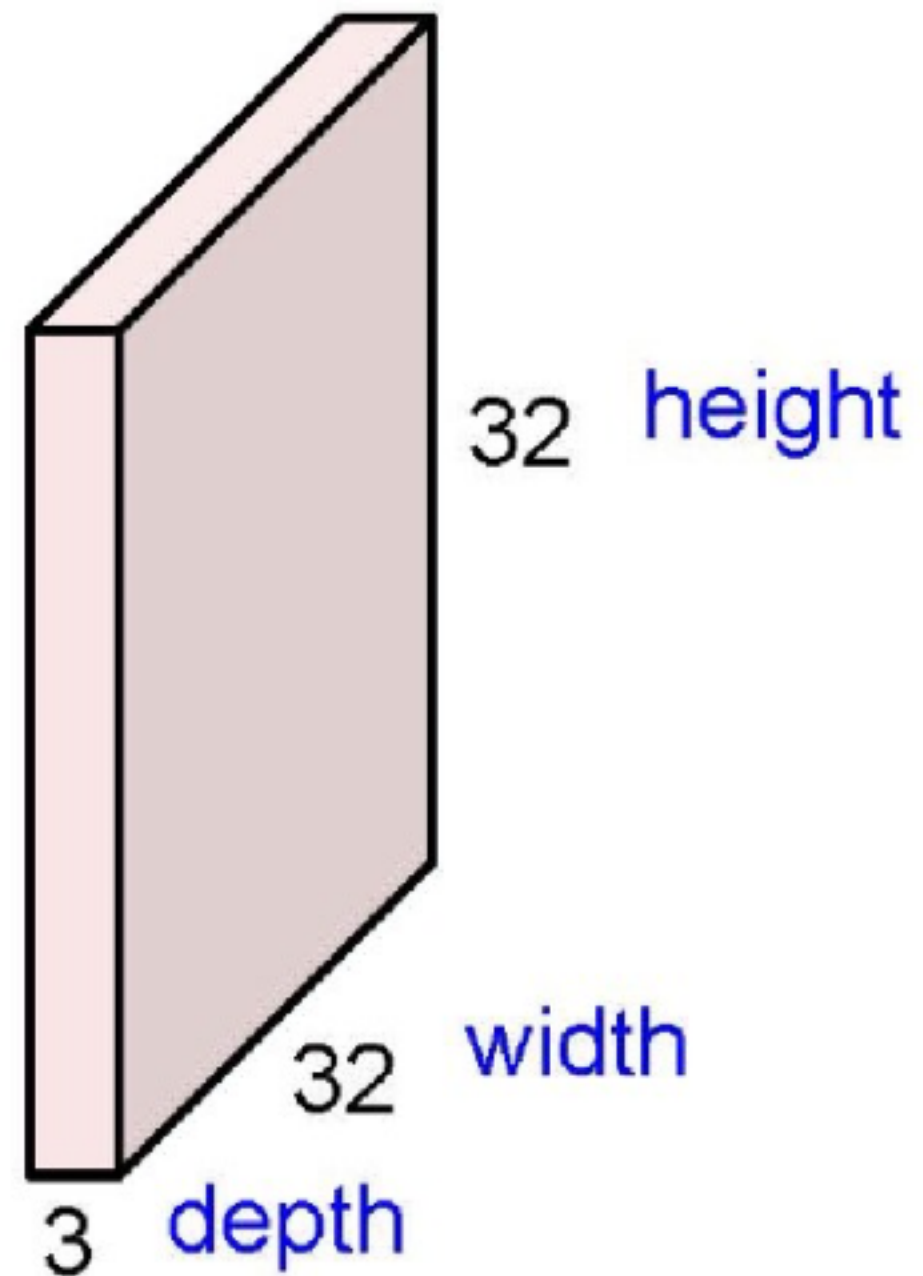


Figure 5-9. A three-dimensional visualization of a convolutional layer, where each filter corresponds to a slice in the resulting output volume

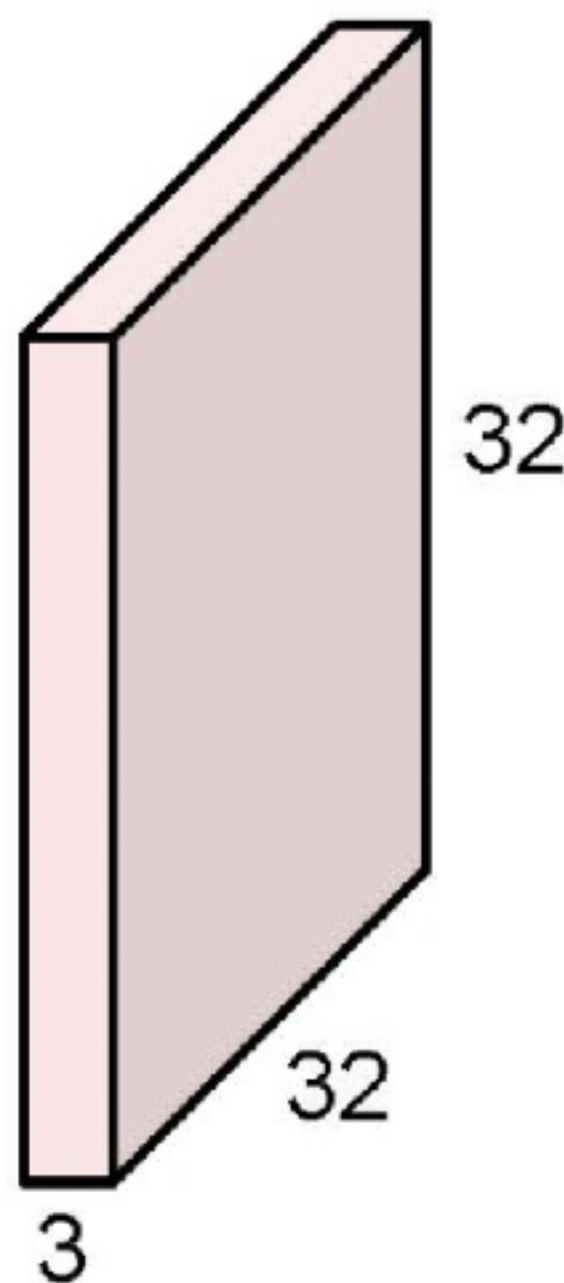
Convolution Layer

32x32x3 image



Convolution Layer

32x32x3 image

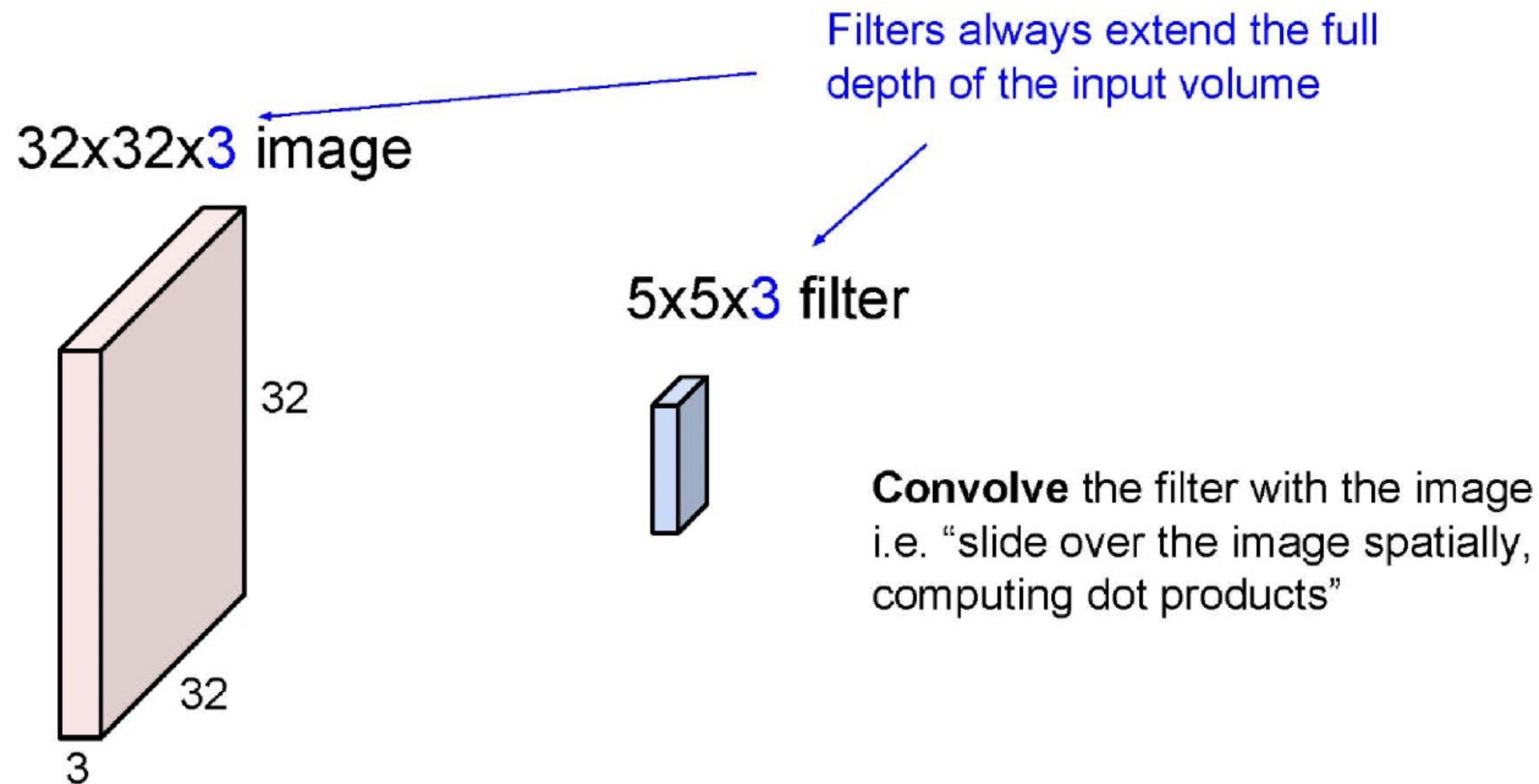


5x5x3 filter

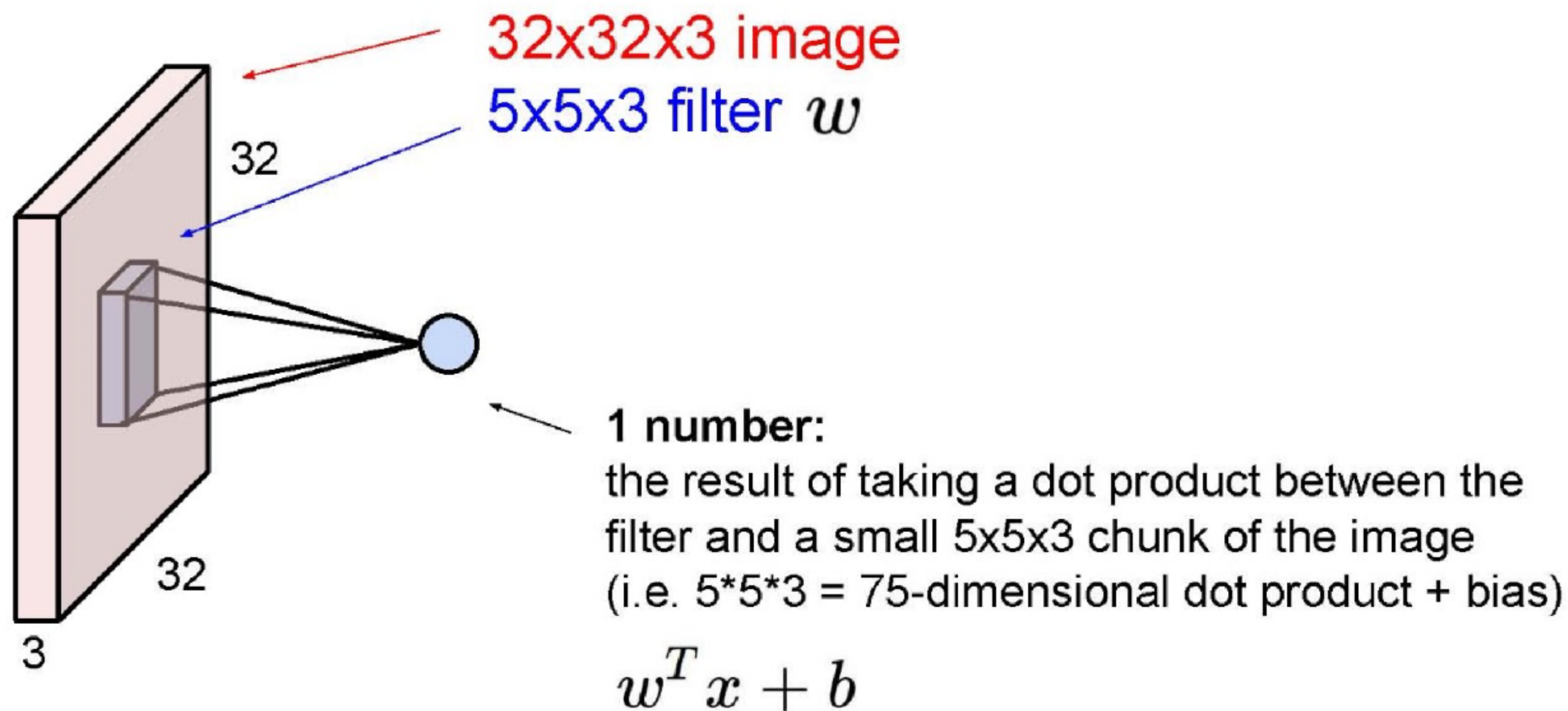


Convolve the filter with the image
i.e. “slide over the image spatially,
computing dot products”

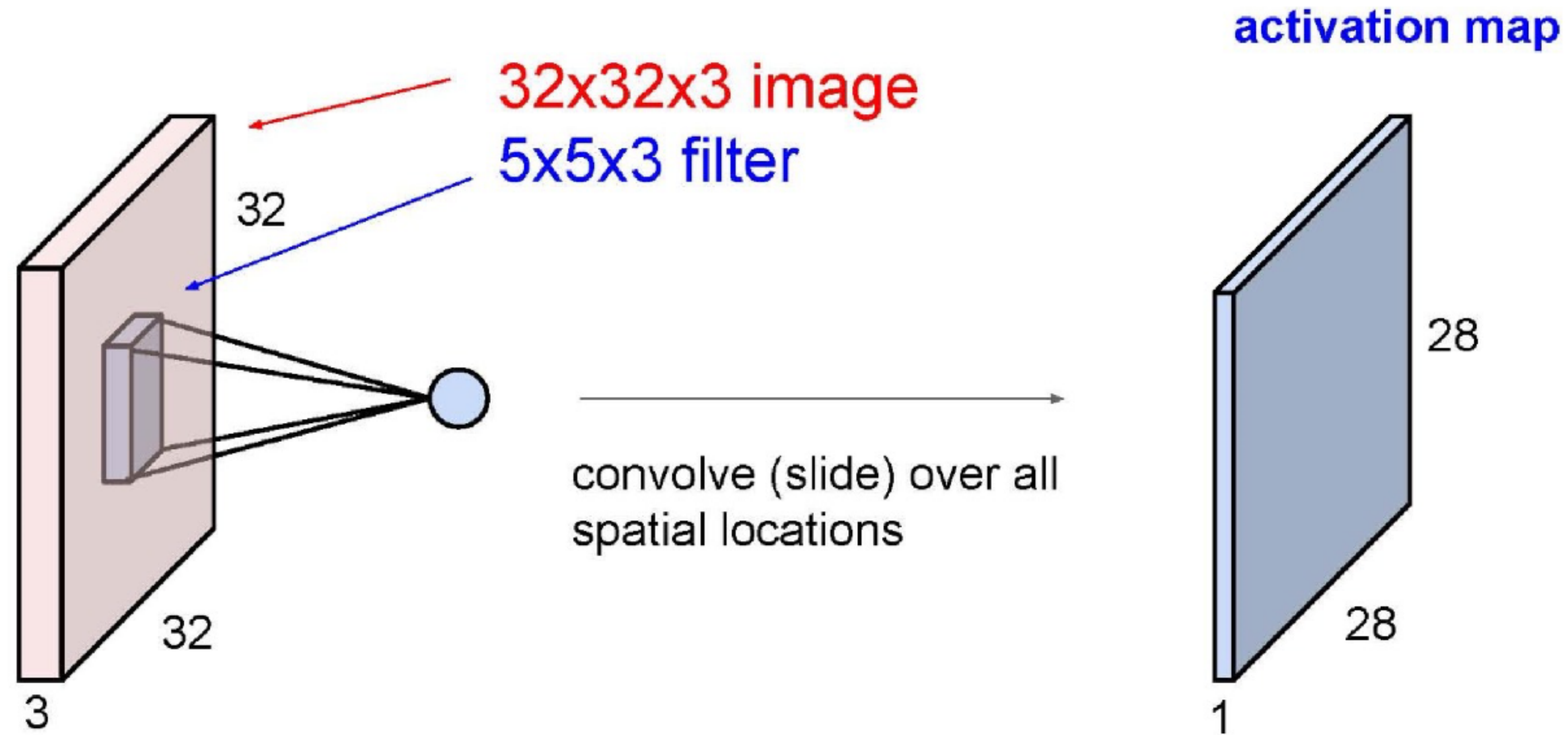
Convolution Layer



Convolution Layer

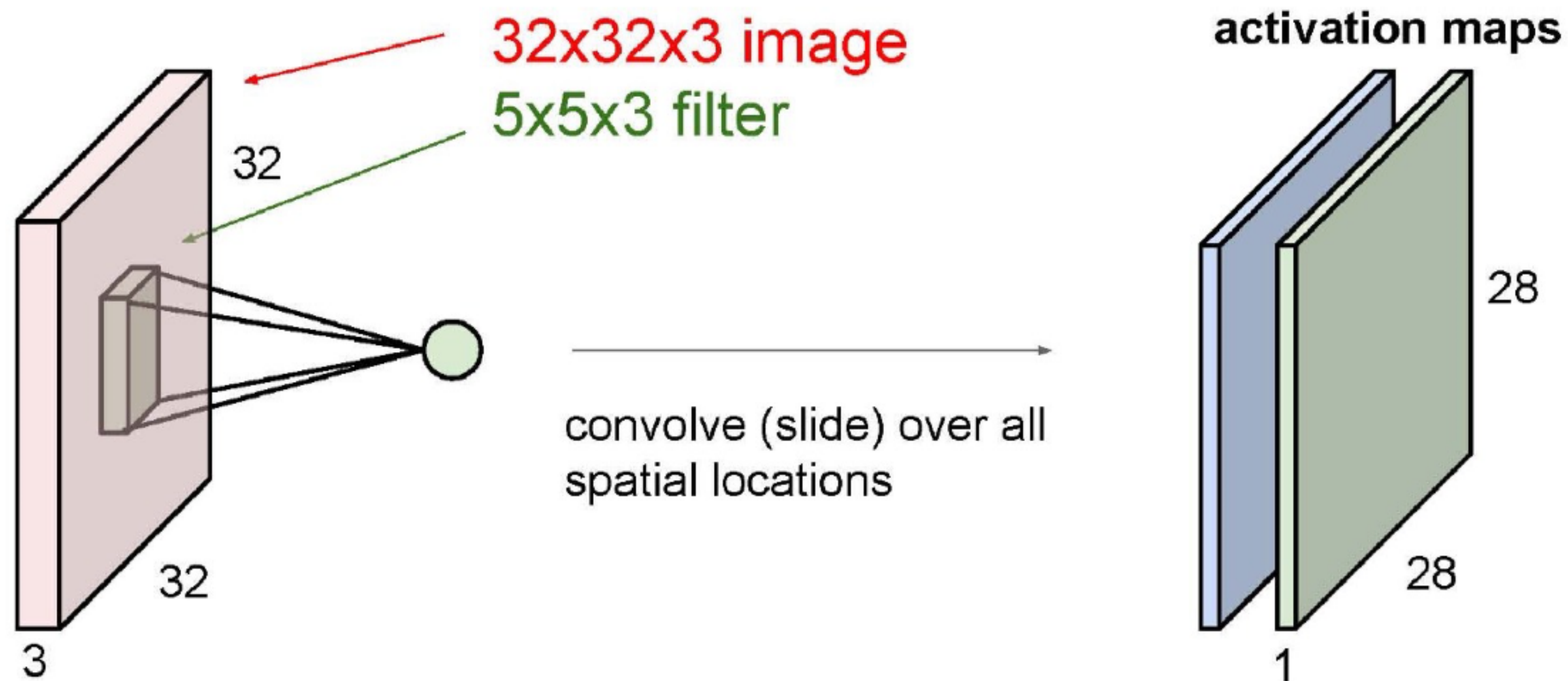


Convolution Layer



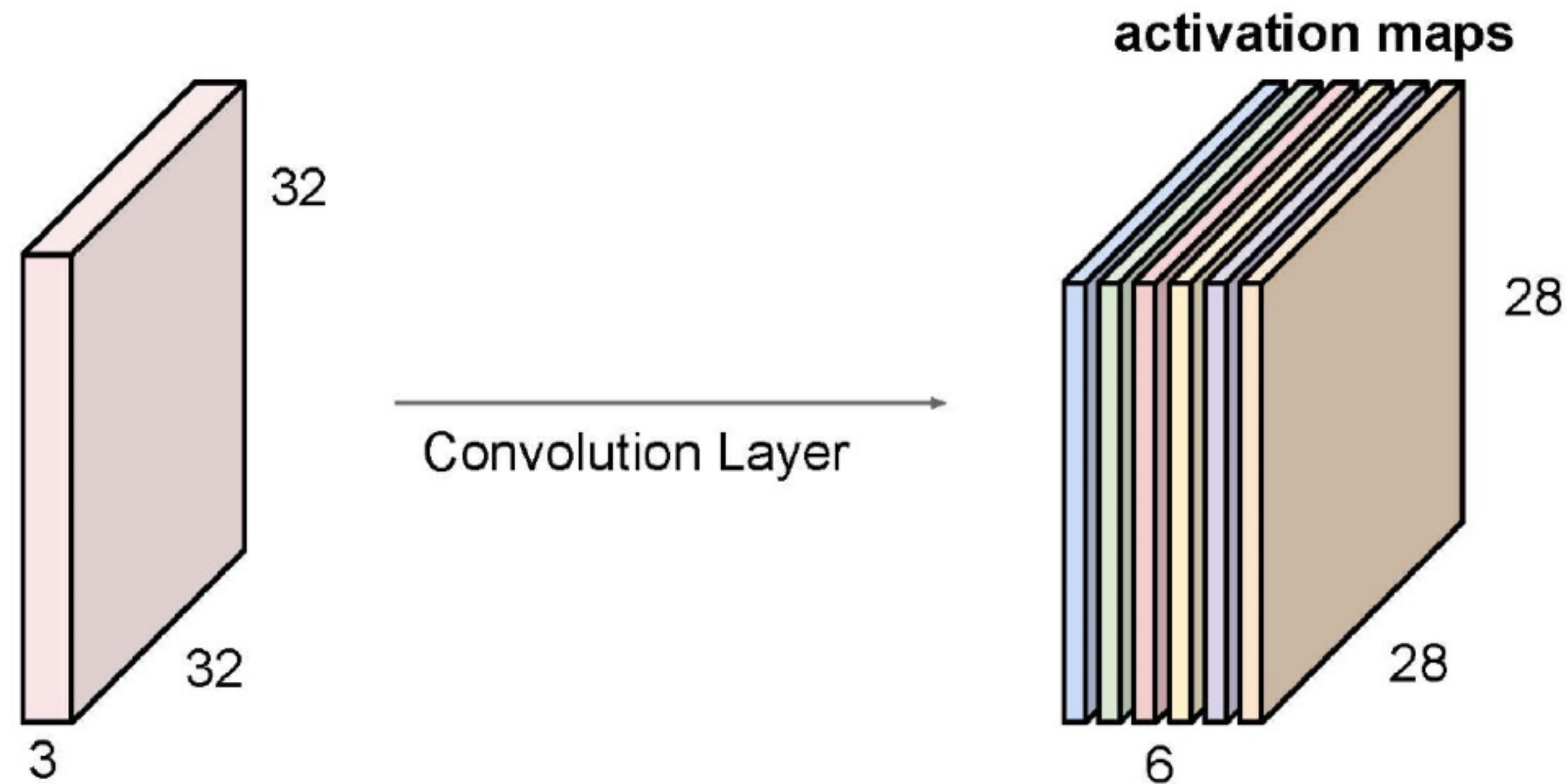
Convolution Layer

consider a second, **green** filter



Convolution Layer

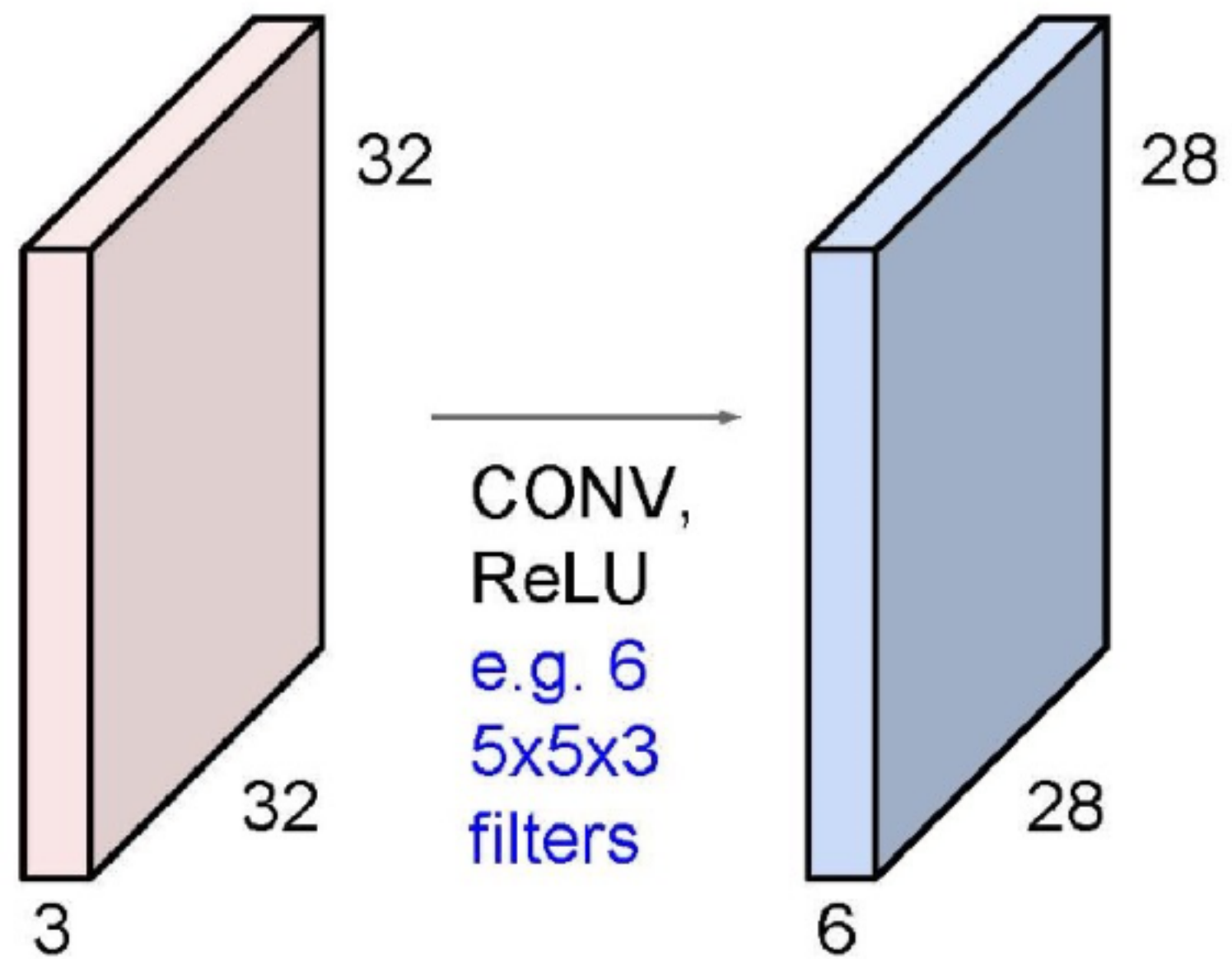
For example, if we had 6 5x5 filters, we'll get 6 separate activation maps:



We stack these up to get a “new image” of size 28x28x6!

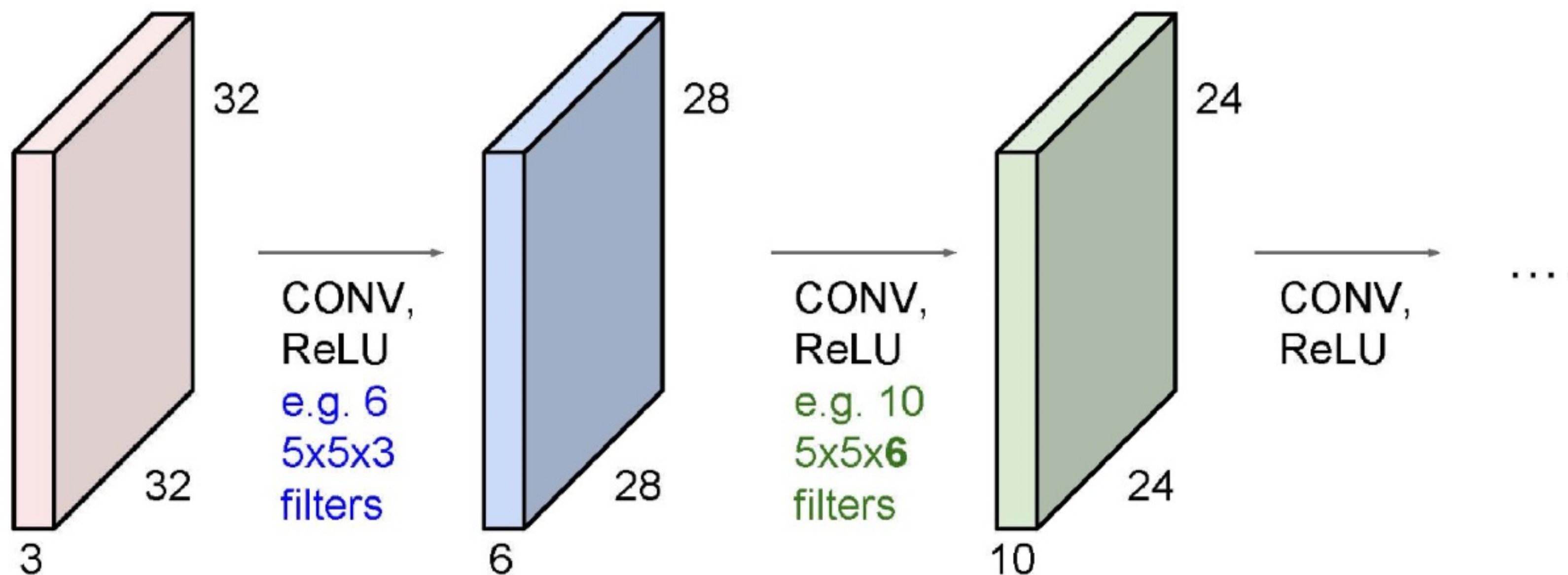
convNet

Preview: ConvNet is a sequence of Convolution Layers, interspersed with activation functions



ConvNet

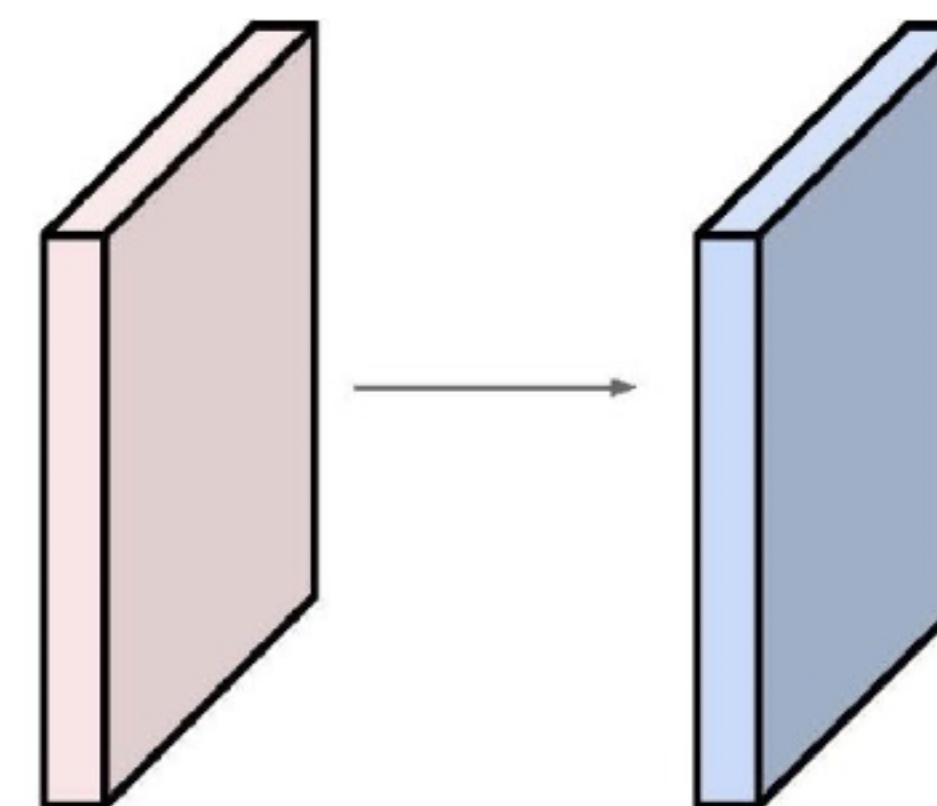
Preview: ConvNet is a sequence of Convolutional Layers, interspersed with activation functions



ConvNet

Examples time:

Input volume: **32x32x3**
10 5x5 filters with stride 1, pad 2



Output volume size: ?

32x32x10

Mirror Padding 2 for Filter Size 5x5

Image										Filter					Conv Output					
128	128	128	128	128	52	52	52	52	52											
128	128	128	128	128	52	52	52	52	52											
128	128	128	128	128	52	52	52	52	52	1	1	1	1	1	x1/25	128	128	109	90	71
128	128	128	128	128	52	52	52	52	52	1	1	1	1	1		128	128	109	90	71
128	128	128	128	128	52	52	52	52	52	1	1	1	1	1		109	109	100	90	81
52	52	52	52	52	128	128	128	128	128	1	1	1	1	1		90	90	90	90	90
52	52	52	52	52	128	128	128	128	128	1	1	1	1	1		71	71	81	90	100
52	52	52	52	52	128	128	128	128	128							52	52	71	90	109
52	52	52	52	52	128	128	128	128	128	Ignoring flip										
52	52	52	52	52	128	128	128	128	128											

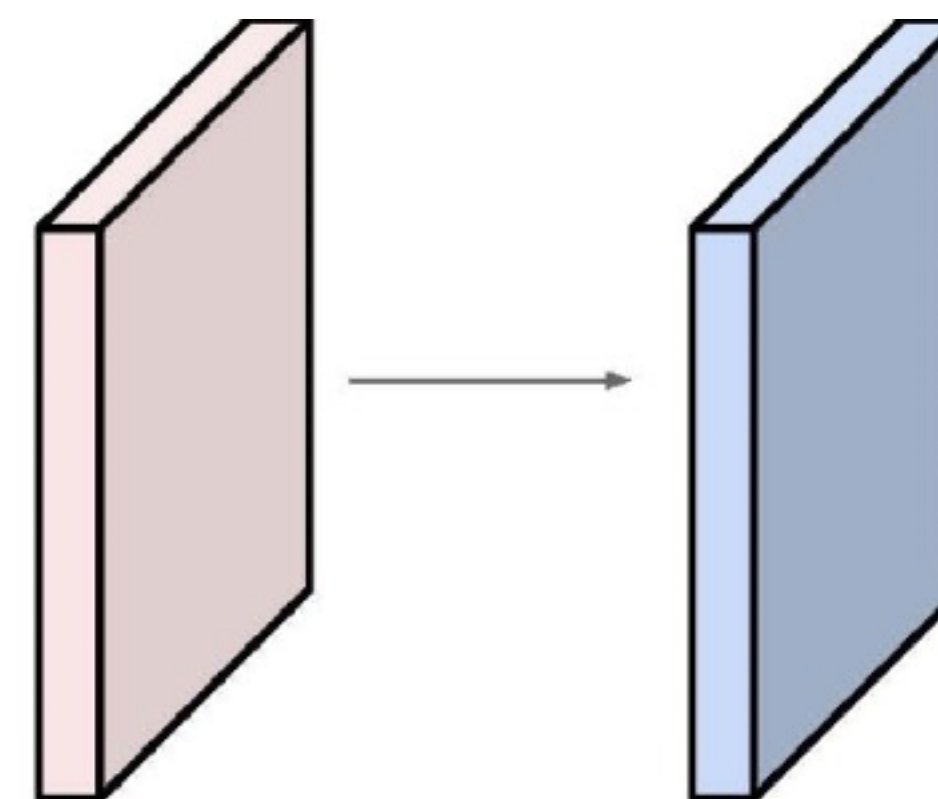
ConvNet

Examples time:

Input volume: **32x32x3**
10 5x5 filters with stride **2**, pad 2

Output volume size: ?

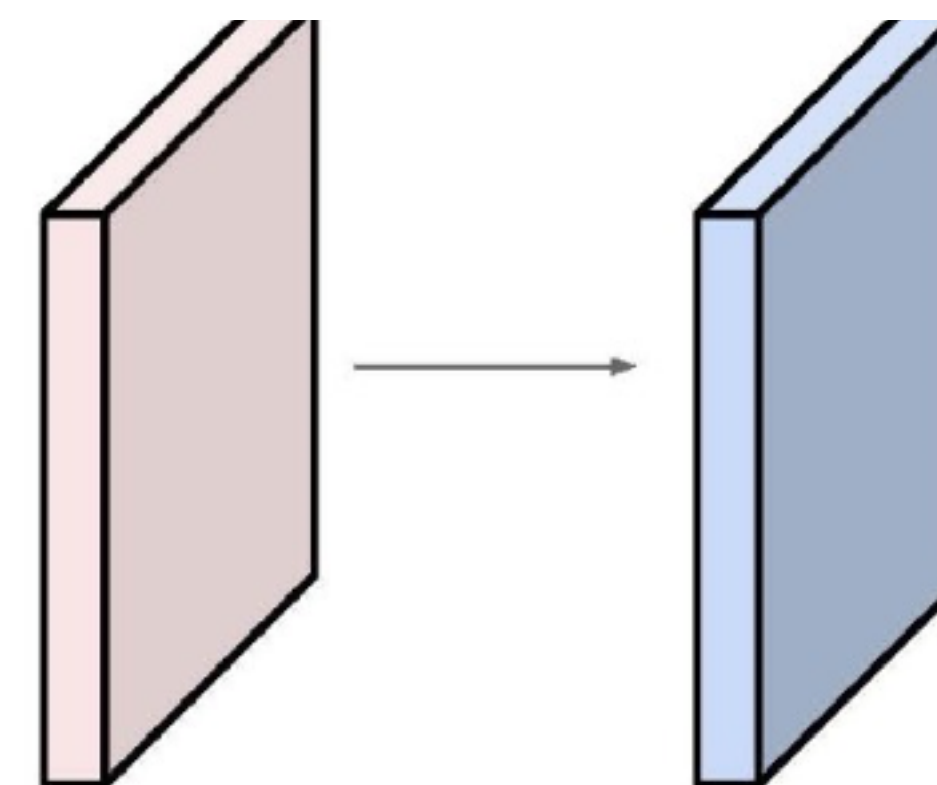
16x16x10



ConvNet

Examples time:

Input volume: **32x32x3**
10 **5x5** filters with stride **1**, pad **2**



Output volume size:
 $(32 + 2 * 2 - 5) / 1 + 1 = 32$ spatially, so
32x32x10

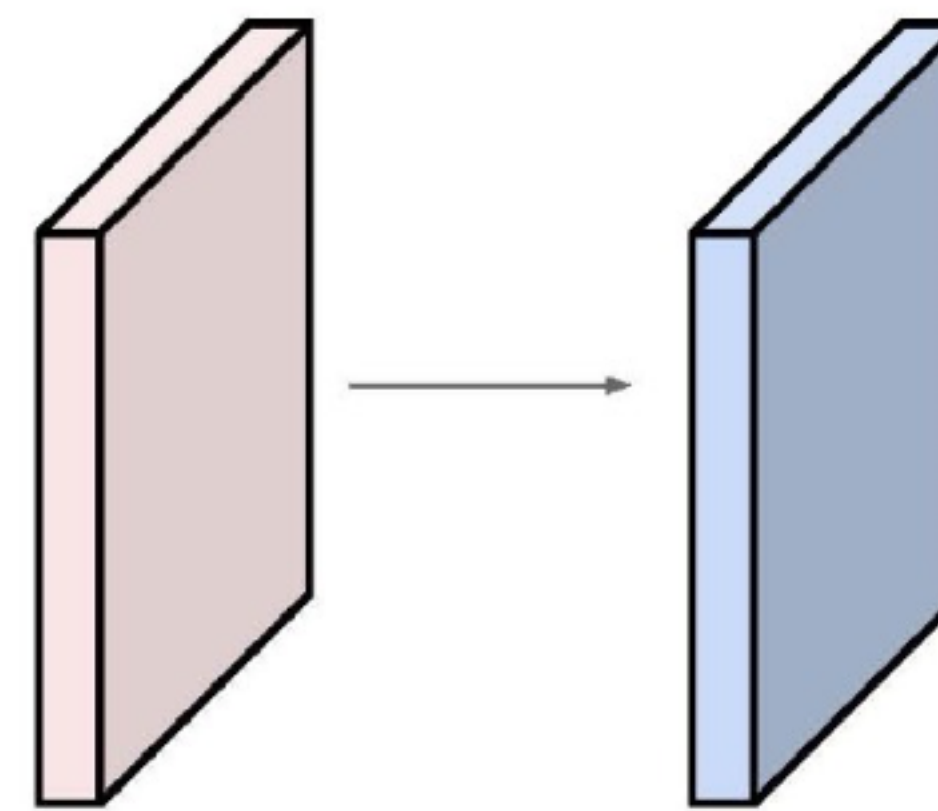
$$(W + 2 * P - F) / S + 1$$

ConvNet

Examples time:

Input volume: **32x32x3**

10 5x5 filters with stride 1, pad 2



Number of parameters in this layer?

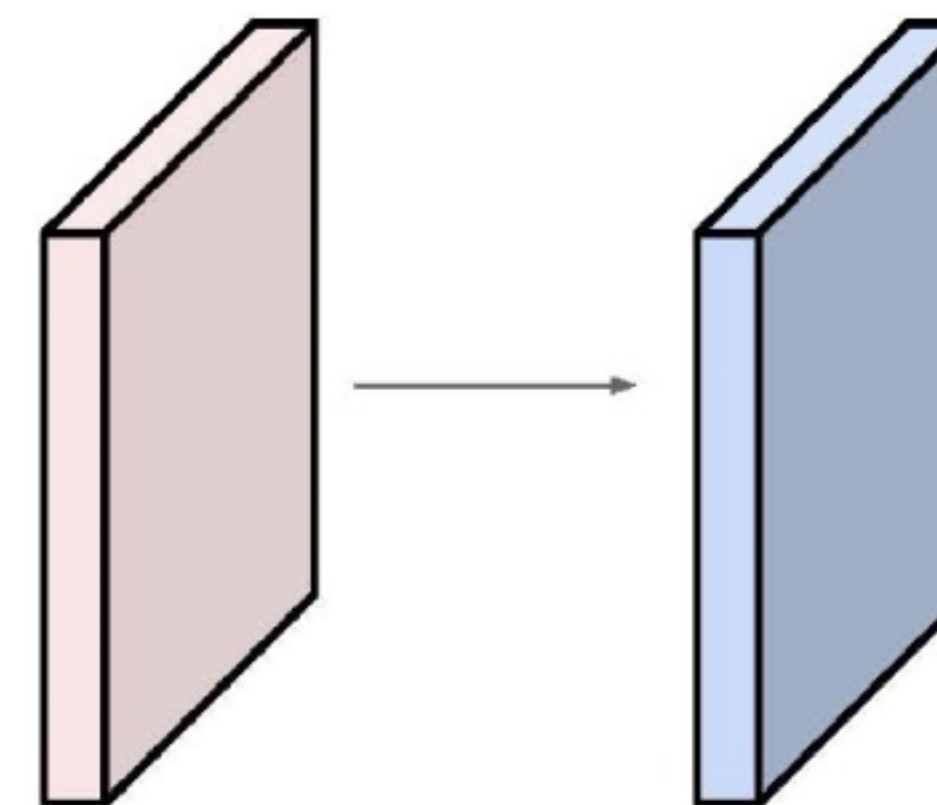
Poll

ConvNet

Examples time:

Input volume: **32x32x3**

10 **5x5** filters with stride 1, pad 2



Number of parameters in this layer?

each filter has $5*5*3 + 1 = 76$ params (+1 for bias)

=> $76*10 = 760$

$$(F \times F \times 3 + 1) \times K$$

Summary of Conv Layer

- ▶ Input: $W_1 \times H_1 \times D_1$
 - ▶ Output: $W_2 \times H_2 \times D_2$
 - ▶ $W_2 = (W_1 - F + 2P)/S + 1$
 - ▶ $H_2 = (H_1 - F + 2P)/S + 1$
 - ▶ $D_2 = K$
 - ▶ Parameters
 - ▶ No. of Filters: K
 - ▶ Their spatial extent: F
 - ▶ Stride: S
 - ▶ Padding: P
 - ▶ Parameters (with sharing)
 - ▶ Weights per filter $F \times F \times D_1$
 - ▶ For K filters and K biases $(F \times F \times D_1 + 1)K$
-
- ▶ Common Settings:
 - ▶ $K = \text{power of } 2$
 - ▶ e.g. 32, 64, 128, 512
 - ▶ $F = 3, S = 1, P = 1$
 - ▶ $F = 5, S = 1, P = 2$
 - ▶ $F = 5, S = 2, P = ?$ (whatever fits)
 - ▶ $F = 1, S = 1, P = 0$

Next

- More about CNN
 - Volumetric Filters
 - Layer and Filter dimensions etc
- Case Studies
 - AlexNet
 - VGG16
 - ResNet
 - Inception ...

